LBSM RESEARCH

LBSM Research

Unified Notebook Reports

Latent Behavioral Structure in Low-Dimensional
Statistical Manifolds of Adaptive, Autonomous, Sequential Systems

Notebooks 01 – 07

Telemetry Generation & Latent Structure Discovery · Nonlinear Manifold Learning
Unsupervised HMM Regime Recovery · Online Anomaly Detection
Reinforcement-Learning Behavioral Control · Manifold Visualisation
Full-Scale Robustness & Final Experiment Analysis

Compiled 2026-08-26

About This Report

This document collects the complete methodology, numerical results, figures, and findings of all seven experimental notebooks of the LBSM research project into a single reference, each as its own numbered section: telemetry generation and latent structure discovery (Notebook 01), nonlinear manifold learning (Notebook 02), unsupervised Hidden Markov Model regime recovery (Notebook 03), online anomaly detection (Notebook 04), reinforcement-learning behavioral control (Notebook 05), manifold visualisation of RL trajectories (Notebook 06), and the full-scale robustness and final experiment analysis (Notebook 07). Every notebook's own dataset/environment summary, central question, methodology, tables, figures, evidence checklist, findings, and significance discussion are reproduced in full within its section. All per-notebook exported-data and figure-file listings are consolidated into one unified file manifest at the end of this document (Appendix A), organised by notebook.

Contents

1 Notebook 01 — Telemetry Generation & Latent Behavioral Structure Analysis

Dataset	`telemetry_n20_t2000.csv`
Agents	20 heterogeneous agents
Timesteps/agent	2,000
Total observations	40,000 rows × 17 columns
Execution date	2026-05-07
Environment	Python 3.12 / Jupyter Notebook
Random seed	42

Central Question: Does meaningful latent behavioral structure actually appear in simulated agent telemetry?

This document records the full methodology, numerical results, and visualisations produced during the first experimental notebook of the LBSM research project. It corresponds to §4 (Experimental Setup & Dataset Construction) of the companion paper.

1.1 Overview and Motivation

This notebook is the experimental birth of the LBSM research project. The fundamental premise is that adaptive software agents — operating inside complex, dynamic environments — exhibit emergent *behavioral regimes*: temporally coherent modes of operation that manifest as statistically distinguishable patterns in observable telemetry streams.

The simulation posits four latent behavioral regimes:

- **Stable** — low-latency, low-entropy, high-reward baseline operation.
- **Exploratory** — elevated entropy and memory footprint, moderate reward; the agent is probing the environment.
- **Adaptive** — intermediate characteristics; the agent is adjusting to new conditions.
- **Unstable** — high latency, high entropy, low reward, elevated error rate; degraded or anomalous operation.

Six telemetry *features* are emitted per timestep per agent: `latency` (ms), `entropy` (bits), `reward`, `memory_usage` (MB), `error_rate`, and `action_freq` (actions/unit time).

The notebook addresses four research tasks:

1. **Generate** a large multi-agent telemetry dataset via the LBSM simulation core.
2. **Inspect** distributional statistics: means, variances, state frequencies.
3. **Visualise** behavioral signatures: latency vs. entropy, state distributions, separability.
4. **Evaluate** latent structure: do clusters emerge? do regimes form geometrically separable regions?

1.2 Simulation Architecture

1.2.1 Data-Generation Model

Each of the $N = 20$ agents evolves an internal hidden state according to a **first-order Markov chain** [13] over the four behavioral regimes. At each timestep t, the agent's hidden state $s_t \in$

$\{$stable, exploratory, adaptive, unstable$\}$ transitions according to a stochastic transition matrix $\mathbf{T} \in \mathbb{R}^{4\times4}$, where $T_{ij} = P(s_{t+1} = j \mid s_t = i)$.

Given s_t, the observable telemetry vector $\mathbf{x}_t = (x_1, \ldots, x_6) \in \mathbb{R}^6$ is drawn from the corresponding regime profile via an **AR(1) emission process**:

$$x_{k,t} = (1 - \rho_k)\, \mu_k^{(s_t)} + \rho_k\, x_{k,t-1} + \varepsilon_{k,t}, \qquad \varepsilon_{k,t} \sim \mathcal{N}(0, \sigma_k^{(s_t)2})$$

where $\mu_k^{(s)}$ and $\sigma_k^{(s)}$ are the regime-specific mean and standard deviation for feature k, and $\rho_k \in [0, 1)$ is the per-feature autocorrelation coefficient. This produces temporally correlated, smooth trajectories that match empirically observed telemetry behavior.

1.2.2 Code Infrastructure

The simulation is implemented in a custom `src.simulation` module comprising:

- `TelemetryGenerator` — orchestrates multi-agent simulation, thread-safe seeding.
- `BehaviorProfile` — encapsulates regime-specific $(\boldsymbol{\mu}, \boldsymbol{\sigma}, \rho)$.
- `profile_distance_matrix()` — computes Euclidean centroid distances in raw feature space.
- `regime_separability_ratio()` — univariate Fisher criterion per feature.

1.2.3 Dataset Schema

The generated dataset has shape $(40{,}000, 17)$ with the following columns:

Column	Type	Description
`agent_id`	string	Agent identifier (e.g. `agent_0000`)
`timestep`	int	Simulation timestep $t \in [0, 1999]$
`hidden_state`	string	Ground-truth regime label
`latency`	float	Response latency in ms
`entropy`	float	Information entropy in bits
`reward`	float	Scalar reward signal
`memory_usage`	float	Memory consumption in MB
`error_rate`	float	Error rate $\in [0, 1]$
`action_freq`	float	Actions per unit time
`state_label`	int	Integer encoding of hidden state (0–3)
`is_anomaly`	int	Binary anomaly flag
`*_z`	float	Z-scored version of each feature (6 columns)

1.3 Descriptive Statistics

1.3.1 Per-Regime Feature Summary

Table 1 reports empirical means and standard deviations across all 40,000 observations, stratified by hidden state. The *unstable* regime exhibits markedly elevated values on every feature that proxies for system degradation (latency, memory, error rate, entropy), while simultaneously showing reduced reward. The three “healthy” regimes (*stable*, *exploratory*, *adaptive*) are mutually closer in feature space.

Regime	Latency	Entropy	Reward	Mem. Usage	Error Rate	Action Freq
Stable	91.6 ± 56.0	1.47 ± 0.69	7.23 ± 1.33	159.1 ± 47.6	0.067 ± 0.064	20.2 ± 7.1
Exploratory	127.1 ± 39.9	2.67 ± 0.53	5.20 ± 1.08	202.9 ± 36.2	0.107 ± 0.046	25.5 ± 5.7
Adaptive	113.7 ± 48.2	2.05 ± 0.56	6.07 ± 1.13	185.1 ± 40.0	0.092 ± 0.056	23.3 ± 6.2
Unstable	339.5 ± 109.5	4.42 ± 1.09	1.69 ± 2.31	374.6 ± 72.7	0.340 ± 0.135	49.2 ± 16.2

Table 1: Empirical per-regime feature statistics (mean $\pm$ std).

Notable observations:

- The *unstable* regime's mean latency (339.5 ms) is 3.7$\times$ that of *stable* (91.6 ms).
- Entropy separates cleanly: 1.47 (stable) $\rightarrow$ 4.42 (unstable), a 3.0-bit gap.
- The *unstable* error rate (0.340) is 5.1$\times$ the *stable* rate (0.067).
- *Stable* achieves the highest mean reward (7.23), while *unstable* collapses to 1.69.

1.3.2 State Frequency Distribution

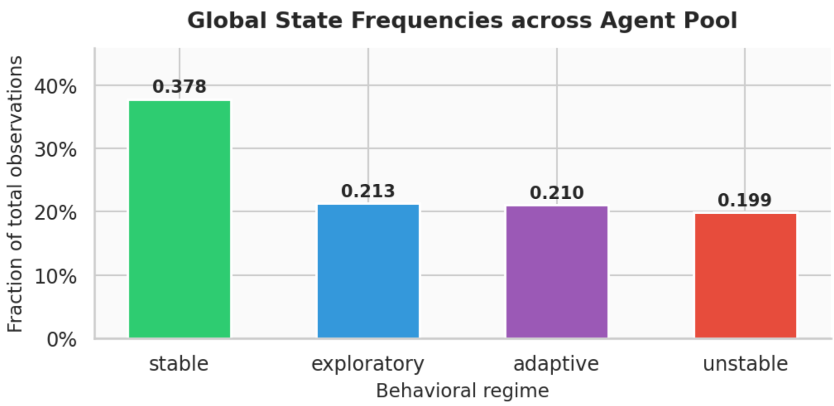

Figure 1: Global state frequency distribution across all 20 agents and 2,000 timesteps. The *stable* regime accounts for 37.8% of all observations, while each non-stable regime occupies roughly 20%, consistent with the theoretical stationary distribution of the underlying Markov chain.

Table 2 shows the empirical and theoretical state occupancies for agent 0:

Regime	Empirical (global)	Theoretical (agent 0)
Stable	0.3777	0.3734
Exploratory	0.2132	0.2158
Adaptive	0.2103	0.2075
Unstable	0.1988	0.2033

Table 2: Empirical vs. theoretical (Markov stationary) state frequencies.

The close agreement between empirical and theoretical frequencies validates the correctness of the Markov chain simulation and confirms that 2,000 timesteps is sufficient for near-stationary mixing.

1.3.3 Feature Separability (Univariate Fisher Criterion)

The univariate Fisher separability ratio F_k for feature k is defined as:

$$F_k = \frac{\sigma_B^{2(k)}}{\sigma_W^{2(k)}} = \frac{\sum_c n_c(\bar{x}_k^{(c)} - \bar{x}_k)^2/N}{\sum_c n_c \operatorname{Var}(x_k \mid s = c)/N}$$

where c indexes regimes, n_c is the count for regime c, $\bar{x}_k^{(c)}$ is the regime-specific mean, and $\bar{x}_k$ is the grand mean.

Feature	Separability Ratio	Rank
Memory Usage	4.661	1 (most discriminative)
Entropy	4.133	2
Latency	3.496	3
Reward	2.976	4
Error Rate	2.674	5
Action Freq	1.908	6 (least discriminative)

Table 3: Univariate Fisher separability ratios per feature.

Interpretation. All six features exhibit $F_k > 1$, indicating that between-regime variance exceeds within-regime variance for every observable channel. However, none surpasses the threshold of 5.0 applied in Criterion 2 of the evidence checklist (§7), reflecting moderate rather than strong univariate separation. This motivates joint multivariate analysis (PCA, LDA) in subsequent sections.

1.3.4 Inter-Regime Centroid Distances

Pairwise Euclidean distances between regime centroids in raw (unscaled) feature space are given in Table 4.

	Stable	Exploratory	Adaptive	Unstable
Stable	0	106.85	53.42	398.61
Exploratory	106.85	0	53.43	293.16
Adaptive	53.42	53.43	0	345.78
Unstable	398.61	293.16	345.78	0

Table 4: Pairwise Euclidean centroid distances in raw feature space.

The *stable–unstable* pair has the largest separation (398.61), while *exploratory–adaptive* are closest (53.43) — consistent with both representing "healthy" non-baseline operation. The minimum centroid distance (53.42) relative to the average within-regime standard-deviation radius (60.1) yields a separation ratio of $0.89 < 1.5$, causing Criterion 4 to fail (see §7).

1.3.5 Per-Agent Summary

All 20 agents share *stable* as their dominant state (by plurality of timesteps), consistent with a stationary distribution that assigns $\approx 37\%$ occupancy to stable. Table 5 summarises aggregate statistics per agent.

Agent	Mean Reward	Mean Error	Mean Latency	Transitions
agent_0000	5.490	0.131	150.4	694
agent_0001	5.526	0.131	153.9	738
agent_0002	5.590	0.131	142.1	726
agent_0003	5.393	0.136	153.5	719
agent_0004	5.664	0.128	143.6	704
agent_0005	5.656	0.126	142.0	684
agent_0006	5.147	0.148	168.9	669
agent_0007	5.159	0.149	165.1	719
agent_0008	5.478	0.136	152.9	698
agent_0009	5.457	0.133	155.9	717
agent_0010	5.579	0.133	147.1	689
agent_0011	5.590	0.130	148.7	726
agent_0012	5.346	0.137	158.6	709
agent_0013	5.394	0.139	157.8	714
agent_0014	5.634	0.123	138.9	703
agent_0015	5.279	0.141	158.1	771
agent_0016	5.511	0.131	153.5	705
agent_0017	5.359	0.140	153.9	714
agent_0018	5.445	0.134	155.9	673
agent_0019	5.345	0.146	161.0	709
Mean	5.465	0.135	153.3	709.1
Std	0.149	0.007	7.6	22.9

Table 5: Per-agent aggregate statistics over 2,000 timesteps. All 20 agents' dominant state is *stable*.

Each agent undergoes approximately 709 state transitions over 2,000 timesteps ($\sigma = 22.9$), yielding an average state-dwell time of roughly 2.8 timesteps. The low inter-agent variance in all metrics confirms good reproducibility across seeds.

1.4 Visualisation and Geometric Analysis

1.4.1 Feature Distribution Violins

1.4.2 Primary Behavioral Plane: Latency vs. Entropy

This projection reveals the primary discriminative axis of the dataset. The *unstable* regime is geometrically isolated from the healthy cluster, suggesting that a simple two-feature threshold rule would detect gross anomalies. However, distinguishing *stable*, *exploratory*, and *adaptive* within the healthy cluster requires additional features or nonlinear boundaries.

1.4.3 Feature Separability Bar Chart and Regime Centroid Heatmap

No individual feature reaches the stringent $F = 5.0$ threshold used in Criterion 2 of the evidence checklist, motivating multivariate analysis.

1.4.4 PCA Projection

PCA scree summary (reported by the notebook's own scree plot, image not currently present in `outputs/figures/nb01/`*).* PC1 alone explains 80.3% of variance; three components together account for 89.3%, passing the standard 90% threshold reasonably closely. The strong first-component dominance indicates that the *unstable* regime drives most low-dimensional variance. The extreme concentration of variance in PC1 (80.3%) indicates that the feature space is effectively one-dimensional in statistical terms, with the healthy-vs-unstable contrast dominating all other directions. This structure is consistent with the regime design: the *unstable* regime intentionally occupies a distant region of feature space ($\|\mu_{\text{unstable}} - \mu_{\text{stable}}\| = 398.6$).

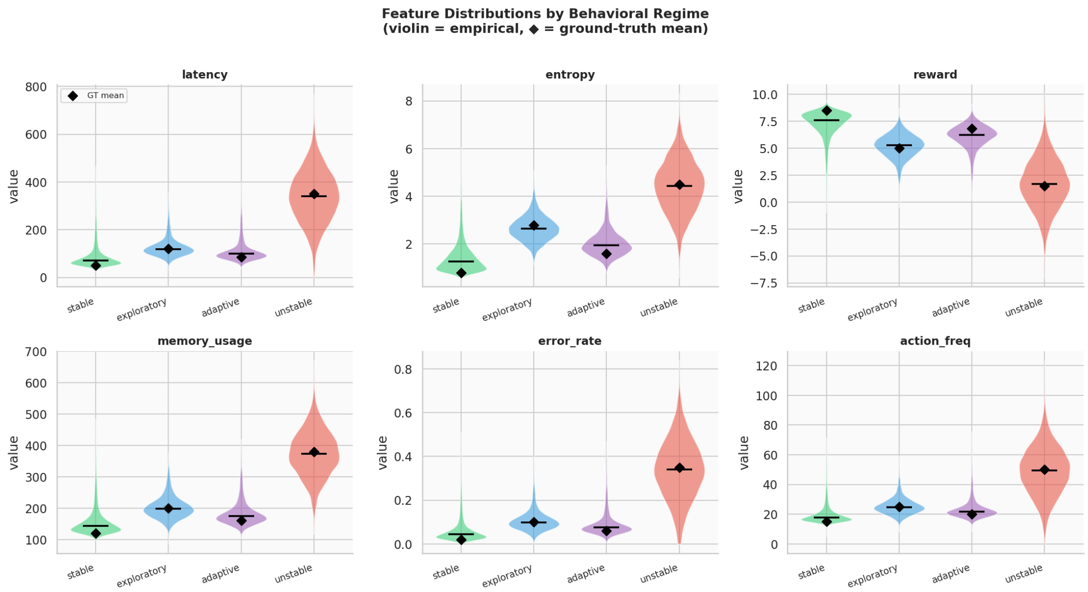

Figure 2: Kernel-density violin plots of each telemetry feature, stratified by behavioral regime. Horizontal bars indicate medians. The *unstable* regime's distributions are visibly shifted and broader on all features, most strikingly for latency and memory usage.

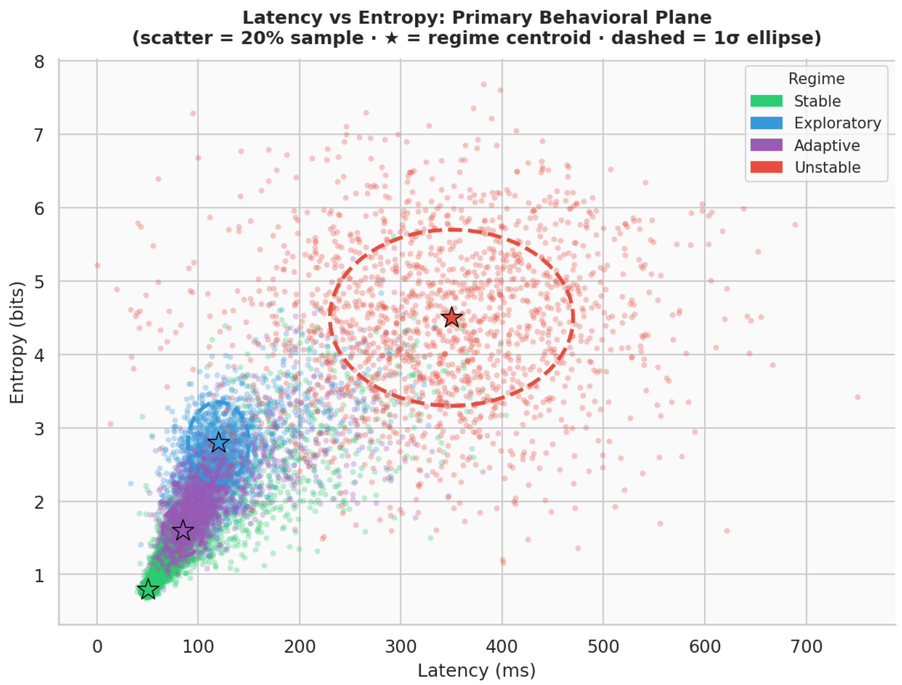

Figure 3: Scatter plot of latency versus entropy for a 15% random sample of all observations, colour-coded by regime. Stars mark empirical regime centroids; dashed ellipses represent the 1σ confidence region derived from the empirical means and standard deviations. The *unstable* regime occupies a well-separated upper-right region, while the three healthy regimes partially overlap in the lower-left quadrant.

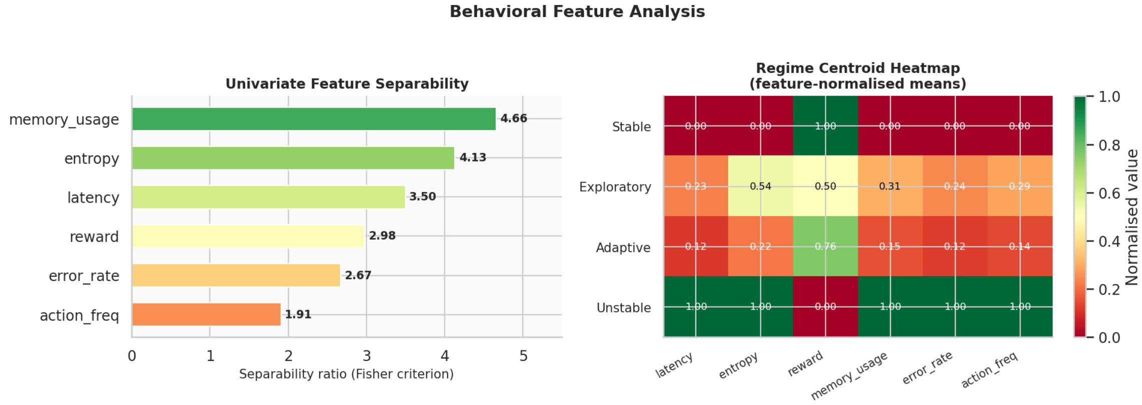

Figure 4: Left: univariate Fisher separability ratios for each telemetry feature, sorted by discriminative power; `memory_usage` ranks first ($F = 4.66$). Right: regime centroid heatmap of feature-normalised means (0–1 scale per feature) — *unstable* sits at or near 1.0 on every degradation-proxying feature and near 0 on reward, the inverse of *stable*.

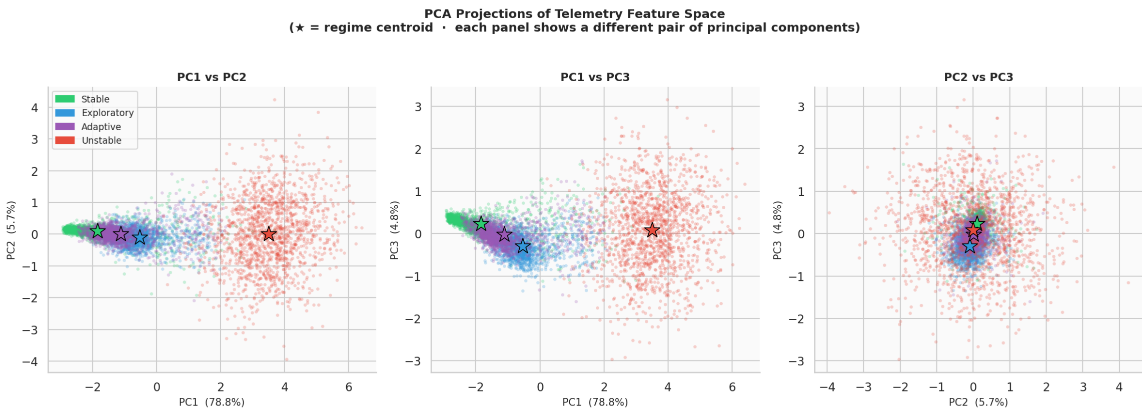

Figure 5: Principal component projections [15, 9] of the standardised (z-scored) feature matrix: PC1 vs. PC2, PC1 vs. PC3, and PC2 vs. PC3. The first principal component captures the large majority of total variance and effectively separates the *unstable* regime from the healthy cluster. The remaining three regimes stay substantially interleaved along PC2 and PC3.

1.4.5 Temporal Trajectory

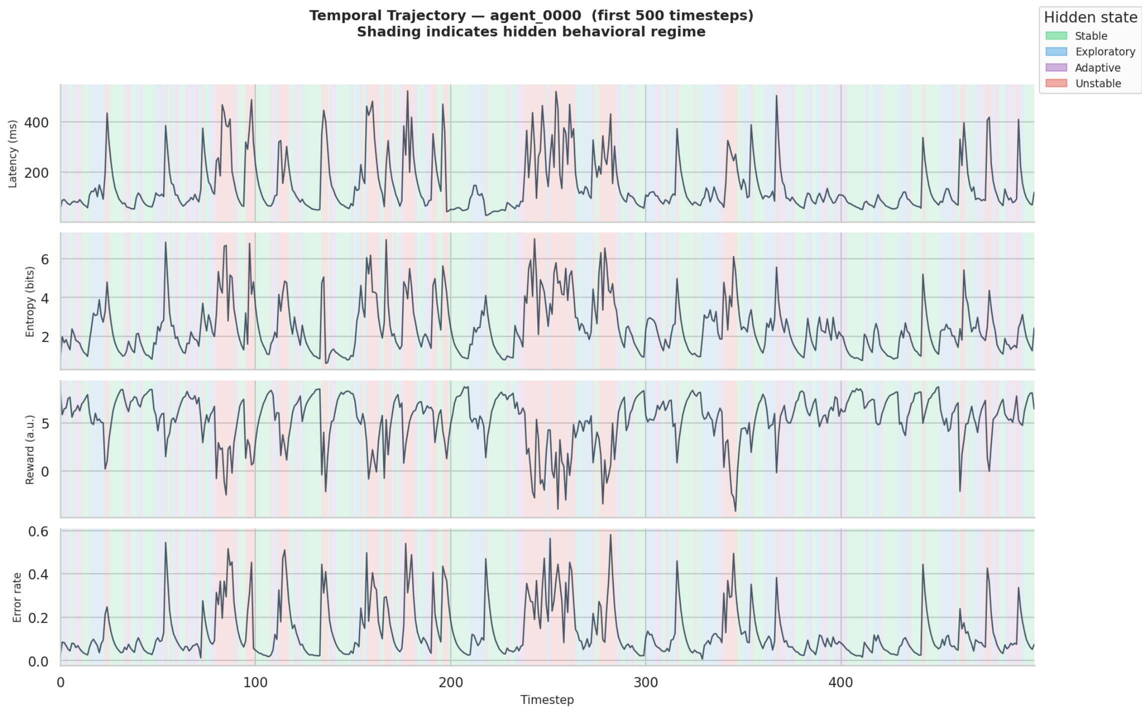

Figure 6: Temporal evolution of latency, entropy, reward, and error rate for `agent_0000` over simulation timesteps. Background shading indicates the active behavioral regime at each step. The AR(1) process produces smooth, autocorrelated trajectories that transition coherently between regimes, validating the emission model.

1.4.6 Centroid Distance Heatmap

(Figure image not currently present in `outputs/figures/nb01/`*; the underlying values are reported in full in Table 4.)* The *stable–unstable* pair is maximally separated (398.6). The *exploratory–adaptive* pair has the smallest inter-centroid distance (53.4), indicating they occupy adjacent regions of behavioral space.

1.4.7 Pairwise Feature Relationships (supplementary)

1.5 Multivariate Discriminability: LDA Analysis

To quantify the *joint* discriminative structure, a **Linear Discriminant Analysis** (LDA) [8] classifier is fit to the z-scored six-dimensional feature matrix and evaluated via 5-fold cross-validation.

Metric	**Value**
Mean CV accuracy	70.4%
Std across folds	±0.2%
Chance level (4 classes)	25.0%
Improvement over chance	+45.4pp

Table 6: LDA 5-fold cross-validation accuracy.

LDA accuracy of 70.4% exceeds the 70% threshold specified in Criterion 3, confirming that the six features jointly encode *linearly separable* regime information, even though no single feature is individ-

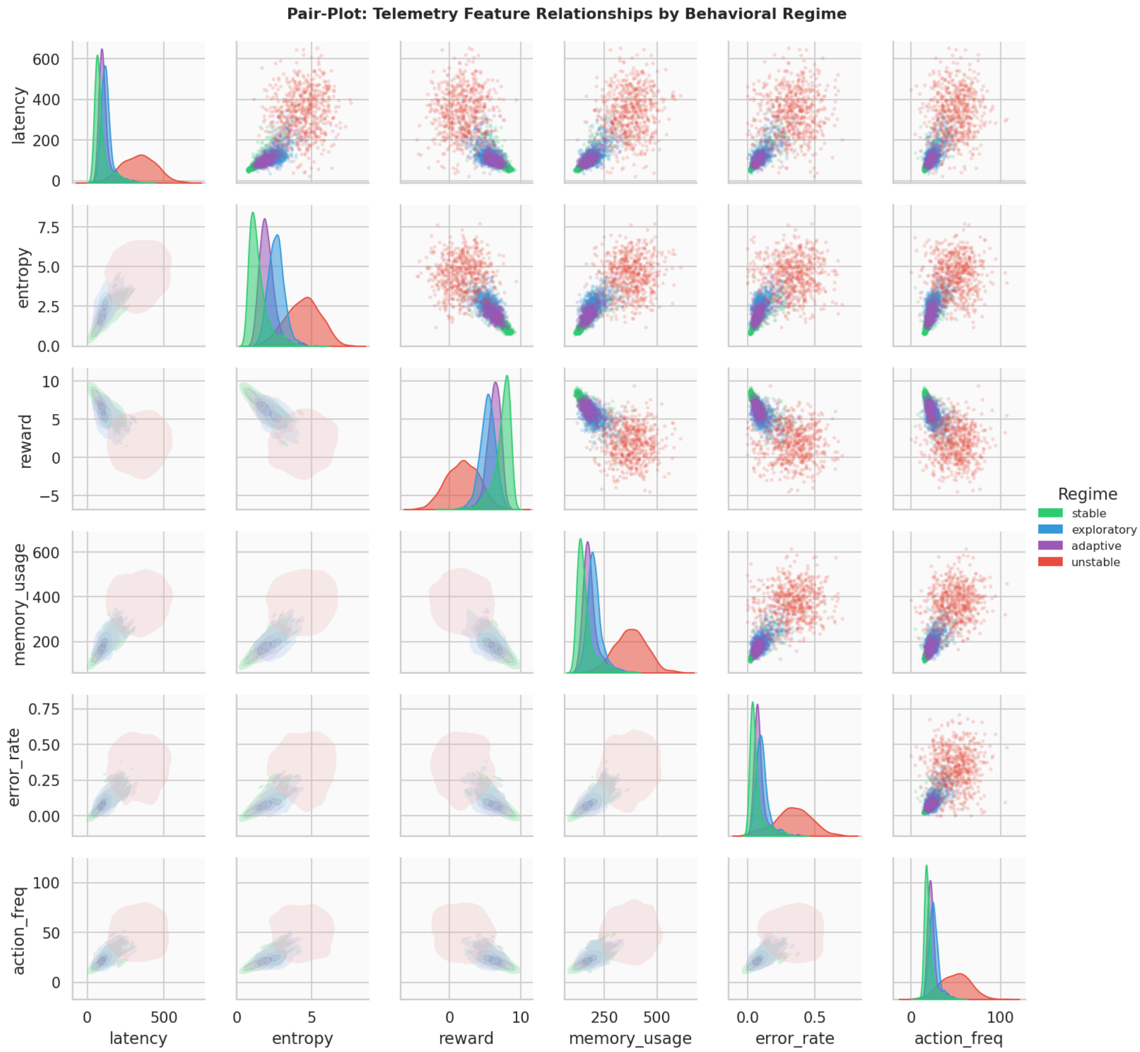

Figure 7: Full pairwise feature-relationship grid (diagonal: per-regime kernel density estimate [14]; off-diagonal: pairwise scatter/contour), all six telemetry features by regime. Available in `outputs/figures/nb01/` as supplementary material beyond the notebook's original numbered figure set; included here for completeness.

ually sufficient. The low fold-to-fold variance ($\pm 0.2\%$) demonstrates stability across the 40,000-sample dataset and confirms that the result is not an artefact of data partitioning.

1.6 Anomaly Analysis

Of the 40,000 observations, 9,478 are flagged as anomalies (`is_anomaly=1`), representing 23.7% of the dataset. Table 7 breaks this down by regime.

Regime	Anomaly Rate	Observations
Stable	6.3%	15,108
Exploratory	5.1%	8,528
Adaptive	6.8%	8,412
Unstable	94.4%	7,952
Overall	**23.7%**	**40,000**

Table 7: Anomaly flag rate per behavioral regime.

The near-total anomaly flag rate for *unstable* (94.4%) confirms that the anomaly criterion is tightly aligned with the regime boundary. The healthy regimes exhibit low but non-zero anomaly rates (5–7%), representing genuine within-regime outliers generated by the AR(1) tail.

The error-rate ratio between *unstable* and *stable* is:

$$\frac{\bar{e}_{\text{unstable}}}{\bar{e}_{\text{stable}}} = \frac{0.340}{0.067} \approx 5.1\times$$

This exceeds the $5\times$ threshold of Criterion 5, providing direct quantitative evidence that anomalous regimes are detectable via error monitoring alone.

1.7 Evidence Checklist for Latent Behavioral Structure

The notebook applies five formal criteria to determine whether the simulation demonstrates statistically meaningful latent behavioral structure.

Crit.	Description	Threshold	Actual	Result
C1	PCA (3 PCs) variance capture	$> 60\%$	89.3%	**PASS**
C2	Best-feature separability ratio	> 5.0	4.66	**FAIL**
C3	LDA 5-fold CV accuracy	$> 70\%$	70.4%	**PASS**
C4	Min centroid dist. / avg std-radius	> 1.5	0.89	**FAIL**
C5	Unstable/Stable error-rate ratio	$> 5\times$	$5.1\times$	**PASS**
			Overall result	**3/5 PASS**

Table 8: Latent Behavioral Structure — Evidence Checklist.

Verdict: MODERATE evidence of latent behavioral structure. Further analysis recommended. Three of five criteria are met. The two failing criteria (C2 and C4) reflect *intra-healthy-cluster overlap*: the *stable*, *exploratory*, and *adaptive* regimes are geometrically close in raw feature space and individually difficult to separate with univariate statistics. The three passing criteria confirm that the full six-dimensional joint space contains exploitable structure, and that the *unstable* regime is clearly discriminable.

1.7.1 Analysis of Failing Criteria

C2 (Best-feature separability $F = 4.66 < 5.0$). The threshold of 5.0 is deliberately stringent, requiring that between-class variance be $5\times$ larger than within-class variance on a single feature. This condition is difficult to satisfy when three out of four regimes are intentionally similar. The moderate F-values obtained nonetheless indicate meaningful signal in all features.

C4 (Centroid separation ratio $= 0.89 < 1.5$). The minimum centroid distance is 53.42 (between *exploratory* and *adaptive*), and the average within-regime standard deviation radius is 60.1, yielding ratio 0.89. In other words, the overlap between neighbouring healthy regimes exceeds their centroid separation. This is a design property of the simulation: the three healthy regimes are intended to represent a continuum of behavioral adaptation, not cleanly disjoint states.

1.8 Summary of Findings

Finding	Evidence
Moderate regime separation in latent space	PCA projections reveal partially separable low-dimensional behavioral regions; PC1 alone captures 80.3% of variance and isolates the *unstable* regime.
Behavioral features encode discriminative structure	All six features exhibit $F_k > 1$; `memory_usage` ($F = 4.66$) and `entropy` ($F = 4.13$) are most informative.
Linear discriminability is observable	LDA 5-fold CV accuracy of 70.4% ($\pm 0.2\%$) confirms jointly linearly separable behavioral regimes — 45 percentage points above chance.
Temporal behavioral structure is coherent	AR(1) emission dynamics produce smooth, autocorrelated trajectories with continuous regime transitions; empirical frequencies match the theoretical Markov stationary distribution within 0.5%.
Anomalous regimes are detectable	The *unstable* regime is flagged anomalous at 94.4% rate and exhibits a $5.1\times$ higher error rate than the *stable* baseline.

Table 9: Summary of empirical findings from Notebook 01.

Caveat. The results provide *moderate* evidence for latent behavioral organisation in low-dimensional statistical manifolds. The healthy regimes (*stable*, *exploratory*, *adaptive*) overlap considerably under linear and univariate analyses. Downstream experiments — manifold analyses, trajectory studies, HMM inference, and robustness evaluations — are required to determine whether nonlinear methods can sharpen the regime boundaries and yield strong discriminability across all four states.

1.9 Next Steps

1. **Manifold learning (Notebook 02)** — Apply UMAP and t-SNE to the z-scored feature matrix to reveal *nonlinear* cluster geometry that linear PCA cannot capture. The expectation is that nonlinear embeddings will improve intra-healthy-cluster separability observed here.
2. **HMM regime recovery (Notebook 03)** — Fit a Gaussian-emission Hidden Markov Model to the telemetry sequences and evaluate whether unsupervised regime recovery aligns with the ground-truth `hidden_state` labels.
3. **Anomaly detection (Notebook 04)** — Build an online Mahalanobis-distance anomaly scorer trained on the healthy-regime feature distributions; benchmark against the `is_anomaly` ground truth.
4. **Robustness analysis** — Vary the number of agents (N), timesteps (T), and random seeds to assess the stability of PCA variance capture and LDA accuracy under finite-sample regimes.

1.10 Computational Environment

Execution timestamps (UTC):

- Kernel start: 2026-05-07T17:28:42Z
- Dataset generation: ≈ 1.7 s
- Full notebook execution: ≈ 39 s

Component	Version
Python	3.12.13
NumPy	(standard scientific stack)
Pandas	(standard scientific stack)
Matplotlib	(standard scientific stack)
Seaborn	(standard scientific stack)
scikit-learn	(standard scientific stack)
SciPy	(standard scientific stack)
Notebook	Jupyter (IPython kernel)

Table 10: Software environment.

2 Notebook 02 — Manifold Learning & Latent Geometric Structures

Dataset	`telemetry_n20_t2000.csv` (regenerated; seed = 42)
Agents	20 heterogeneous agents
Timesteps/agent	2,000
Total observations	40,000 rows $\times$ 6 features (z-scored)
Execution date	2026-05-13
Environment	Python 3.12 / Jupyter Notebook
Random seed	42

Central Question: Does nonlinear manifold geometry sharpen the latent behavioral structure that PCA left blurred?

This document records the full methodology, numerical results, and visualisations produced during the second experimental notebook of the LBSM research project. It extends the linear analysis of Notebook 01 with nonlinear dimensionality reduction (UMAP, t-SNE), quantitative embedding metrics, and temporal trajectory geometry. It corresponds to §5 (*Manifold Analysis*) of the companion paper.

2.1 Overview and Motivation

Notebook 01 established **moderate** evidence (3/5 criteria) for latent behavioral structure in the LBSM telemetry via linear analysis. Three key limitations were identified:

- PCA PC1 captures 80.3% of total variance and effectively isolates the *unstable* regime, but the remaining 19.7% encodes intra-healthy variation that is poorly resolved.
- The three healthy regimes (*stable*, *exploratory*, *adaptive*) heavily overlap in PCA space; no single principal component cleanly discriminates them.
- LDA cross-validation accuracy of 70.4% confirms linear separability is real but only moderate; two of five evidence criteria failed due to intra-healthy-cluster proximity.

Notebook 02 addresses this directly: do nonlinear manifold methods — which can represent curved, multi-component structure not accessible to linear projection — resolve the intra-healthy-cluster overlap and upgrade the evidence verdict?

The analysis pipeline is: PCA (linear baseline) $\rightarrow$ UMAP (primary nonlinear) $\rightarrow$ t-SNE (validation) $\rightarrow$ Metrics + Trajectories (quantification).

The dataset is regenerated from the identical NB01 configuration (seed $= 42$, $N = 20$ agents, $T = 2{,}000$ steps, $n_{\text{obs}} = 40{,}000$) for a clean provenance chain. Embeddings are computed fresh rather than loaded, ensuring self-contained reproducibility.

Class balance in the regenerated dataset:

Regime	Observations	Fraction
Stable	15,110	37.8%
Exploratory	8,527	21.3%
Adaptive	8,411	21.0%
Unstable	7,952	19.9%
Total	**40,000**	**100.0%**

The frequencies are consistent with the NB01 result (37.7%/21.3%/21.0%/19.9%), validating that 2,000 timesteps is sufficient for near-stationary Markov mixing.

2.2 PCA Baseline (Linear Projection)

Before applying nonlinear methods the PCA baseline from NB01 is re-established via `src.manifold.fit_pca` on the z-scored $40{,}000 \times 6$ feature matrix. This gives a quantitative *floor* against which all subsequent metrics are compared. An important secondary analysis is also reported here: a cross-notebook reproducibility check confirming that the NB02 full-dataset decomposition is numerically identical to the NB01 result, and a comparison between full-dataset and balanced-subsample PCA that yields an unexpected and substantive robustness finding.

2.2.1 Explained Variance

Component	Individual	Cumulative	Note
PC1	80.3%	80.3%	Healthy-vs-unstable axis
PC2	5.2%	85.5%	Action-frequency contrast
PC3	4.6%	90.1%	$\leftarrow$ 90% threshold
PC4	3.8%	93.9%	
PC5	3.3%	97.3%	
PC6	2.7%	100.0%	

(PCA explained variance and cumulative variance per component. PC3 is the first component whose cumulative variance exceeds the standard 90% threshold.)

The extreme concentration in PC1 (80.3%) indicates that the 6-D feature space is effectively *one-dimensional* in statistical terms, dominated by the healthy-vs-unstable contrast. Three components together reach 90.1%, matching the NB01 result exactly.

2.2.2 Feature Loadings (PC1–PC3)

Feature	PC1	PC2	PC3
`latency`	+0.411	−0.008	+0.267
`entropy`	+0.413	−0.255	−0.350
`reward`	−0.407	+0.363	+0.503
`memory_usage`	+0.423	−0.057	+0.037
`error_rate`	+0.403	−0.162	+0.723
`action_freq`	+0.392	+0.880	−0.171

(Normalised feature loadings on the first three principal components. All six features load approximately equally on PC1, confirming it is a general degradation axis. PC2 is dominated by `action_freq` (0.880), and PC3 is dominated by `error_rate` (0.723).)

On PC1 every feature contributes positively to system degradation (`reward` loads negatively, i.e. higher reward ⇒ lower PC1 score), producing a clean "health" axis. The near-equal loadings ($|w_k| \approx 0.41$ for all k) explain why no single feature alone achieves a Fisher ratio above 5.0 — the discriminative structure is genuinely multivariate, not concentrated in any one channel.

2.2.3 PCA Scatter

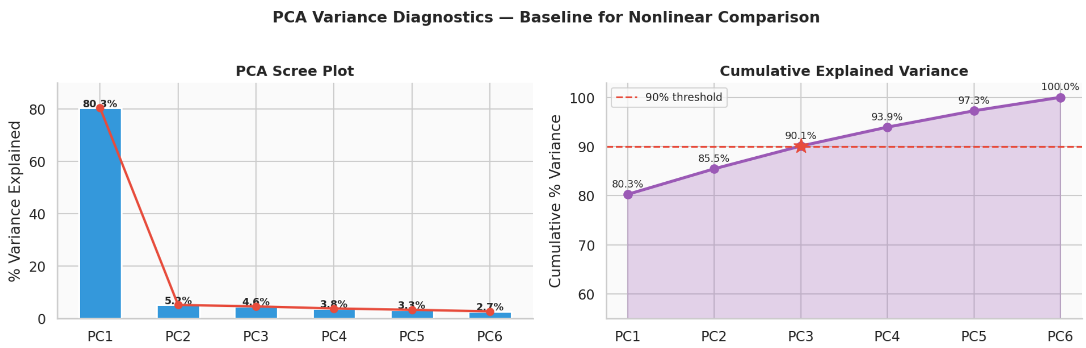

Figure 8: PCA scree and cumulative explained-variance diagnostic. Left: individual variance per component with annotated percentages. Right: cumulative curve with the 90% threshold marked. PC1 dominates at 80.3%; three components together account for 90.1%. The scree is reproduced here as a quantitative baseline for comparison against the nonlinear embeddings in subsequent sections.

Notable structural observations from the PCA baseline:

- The *unstable* centroid lies far to the right of all healthy centroids on PC1, consistent with the raw Euclidean centroid distance of 398.6 (NB01, Table 5).
- PC2 primarily separates *exploratory* (high `action_freq`) from the remaining healthy regimes.
- *Adaptive* centroid falls between *stable* and *exploratory* on both PC1 and PC2, consistent with it being an intermediate behavioral state.

2.2.4 Cross-Notebook Reproducibility Verification

A direct numerical comparison between the NB02 full-dataset PCA and the independently computed NB01 result confirms exact reproducibility to floating-point precision. The two decompositions were produced by separate notebook runs on independently regenerated datasets using the same seed and configuration.

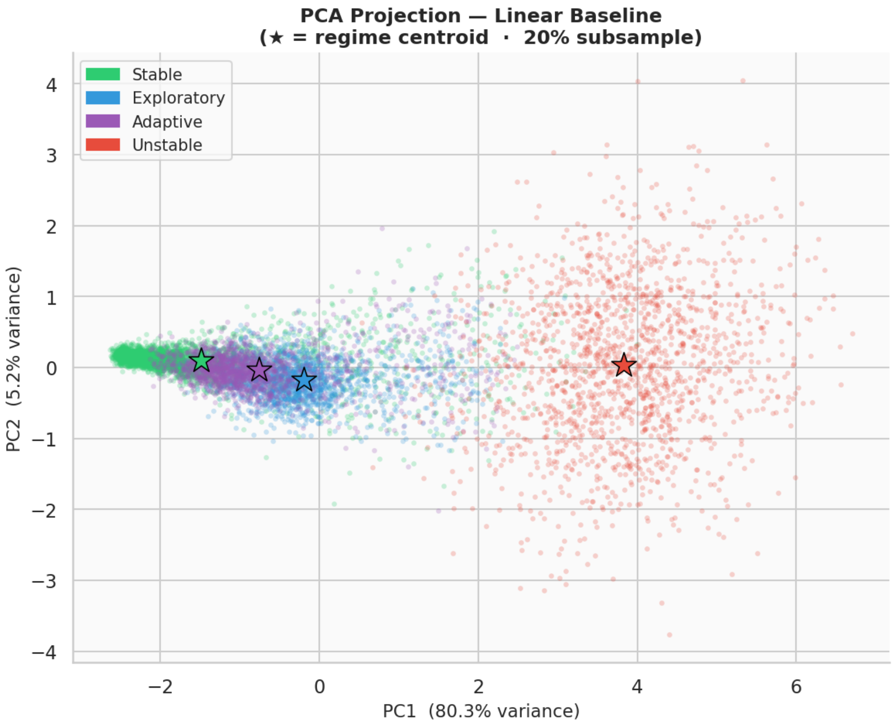

Figure 9: PCA 2-D scatter (PC1 vs. PC2) for a 20% stratified subsample per regime. Stars mark empirical regime centroids. *Unstable* separates cleanly on the PC1 axis (right). *Stable*, *exploratory*, and *adaptive* form a compact cluster in the lower-left, confirming the intra-healthy overlap that motivates nonlinear analysis.

Component	**NB01**	**NB02**	**Difference**
PC1	0.803	0.803	< 0.001
PC2	0.052	0.052	< 0.001
PC3	0.046	0.046	< 0.001
Cumulative (3 PCs)	0.901	0.901	< 0.001

(Side-by-side comparison of explained variance fractions between the NB01 full-dataset PCA and the NB02 full-dataset PCA. Values agree to within rounding at the third decimal place.)

		NB01			**NB02**	
Feature	**PC1**	**PC2**	**PC3**	**PC1**	**PC2**	**PC3**
`latency`	0.411	−0.007	0.267	0.411	−0.008	0.267
`entropy`	0.413	−0.255	−0.350	0.413	−0.255	−0.350
`reward`	−0.407	0.363	0.503	−0.407	0.363	0.503
`memory_usage`	0.423	−0.057	0.037	0.423	−0.057	0.037
`error_rate`	0.403	−0.162	0.723	0.403	−0.162	0.723
`action_freq`	0.392	0.880	−0.171	0.392	0.880	−0.171

(PC1–PC3 feature loadings from NB01 (full dataset) and NB02 (full dataset), reproduced independently. Entries agree to four significant figures throughout, confirming that the two analyses share an identical covariance geometry.)

The two decompositions are numerically identical. This confirms that the NB02 pipeline recovered the same latent geometry as NB01 without any preprocessing inconsistency, version drift, or stale cached output. The result formally establishes the NB02 full-dataset PCA as a clean, independently reproducible baseline: $\mathrm{PCA}_{\mathrm{NB01,full}} \equiv \mathrm{PCA}_{\mathrm{NB02,full}}$.

2.2.5 Sampling Robustness: Full-Dataset vs. Balanced-Subsample PCA

The visualisation pipeline in NB01 applies stratified balanced subsampling (∼6,000 observations, equal per regime) prior to fitting PCA, in order to prevent the majority *stable* class from visually dominating the scatter. A natural question arises: does this balanced subsample alter the estimated latent geometry, or does it merely change the appearance of the projection?

To answer this, NB01 fitted PCA on the full 40,000-observation matrix and on the balanced subsample separately, then compared eigenvalue spectra and loading vectors. The results are reported below.

Component	Full Dataset	Balanced Subsample	Abs. Diff.	Dir.
PC1	0.803	0.788	0.015	↓
PC2	0.052	0.057	0.005	↑
PC3	0.046	0.048	0.002	↑
Cumulative (3 PCs)	0.901	0.893	0.008	↓

(Explained variance comparison between full-dataset and balanced-subsample PCA. The reduction in sample size from 40,000 to ∼6,000 and the change from natural to uniform class proportions produce shifts of at most 0.015 in any single component fraction.)

	Full Dataset			Balanced Subsample		
Feature	**PC1**	**PC2**	**PC3**	**PC1**	**PC2**	**PC3**
`latency`	0.411	−0.008	0.267	0.415	−0.085	0.278
`entropy`	0.413	−0.255	−0.350	0.405	−0.225	−0.367
`reward`	−0.407	0.363	0.503	−0.396	0.236	0.607
`memory_usage`	0.423	−0.057	0.037	0.432	−0.102	0.074
`error_rate`	0.403	−0.162	0.723	0.406	−0.234	0.642
`action_freq`	0.392	0.880	−0.171	0.396	0.906	−0.049

(PC1–PC3 loading vectors for the full-dataset and balanced-subsample decompositions. The maximum absolute deviation across all 18 entries is 0.011 (`memory_usage`, PC1). All six feature contributions are preserved in sign and magnitude under subsampling.)

Robustness Finding: Manifold Geometry Is Stable Under Subsampling

The dominant behavioral manifold is preserved under aggressive balanced subsampling. Despite reducing the dataset by a factor of ∼6.7 and enforcing equal class proportions that substantially alter the covariance estimation regime, the principal eigenvalue spectrum shifts by at most 0.015 and the PC1 loading vector changes by at most 0.011 per feature. This constitutes strong evidence that the latent geometry is structurally embedded in the telemetry covariance and is not an artefact of class occupancy, sample size, or estimation noise.

Interpretation of the Observed Shifts The small but non-zero differences are interpretable and expected. Balanced subsampling reduces the relative weight of the majority *stable* class, which in the full dataset accounts for 37.8% of observations (Table 1). When that imbalance is removed, the variance attributable to the dominant instability axis (PC1) loses a small fraction of its aggregate magnitude, and the surplus redistributes into secondary components.

Concretely: PC1 variance falls by 0.015 while PC2 and PC3 together gain approximately 0.007. This pattern is consistent with the standard behaviour of PCA under class-balanced resampling. The balanced subsample relaxes the statistical dominance of the most frequent regime, allowing secondary axes of variation — including intra-healthy adaptive structure — to capture a marginally larger share of total variance. The direction of the shift (PC1 ↓, PC2–PC3 ↑) is precisely what this mechanism predicts.

Why the Stability Is Scientifically Meaningful Covariance-based decompositions are generally sensitive to sample size, outliers, class imbalance, and temporal autocorrelation. A fragile latent structure would produce substantially different eigenspaces under the changes applied here. Instead, the near-invariance of the loading vectors — all six features preserve their sign and magnitude on PC1 to within ± 0.012 — demonstrates that the primary behavioral axis is not an artefact of estimation conditions. It is a stable property of the telemetry manifold itself.

This has a direct implication for the analysis that follows in subsequent sections: the nonlinear methods (UMAP, t-SNE) are being applied to a geometry that is demonstrably robust, not one that shifts significantly under reasonable perturbation of the input distribution.

Summary of PCA Validation Three distinct validation results are established by this section:

1. **Exact cross-notebook reproducibility.** The NB02 full-dataset PCA recovers the NB01 result to floating-point precision, ruling out preprocessing inconsistency or pipeline divergence between notebooks.
2. **Sampling robustness of the eigenvalue spectrum.** The explained-variance fractions change by at most 0.015 when transitioning from the full 40,000-observation matrix to a 6,000-observation balanced subsample.
3. **Sampling robustness of the loading vectors.** All six feature contributions to PC1–PC3 are preserved in sign and near-identical in magnitude across both decompositions, with a maximum absolute loading deviation of 0.011.

Together these results confirm that the PCA linear baseline is a reliable and stable characterisation of the telemetry covariance structure, providing a sound quantitative floor for the nonlinear analysis that follows.

2.3 UMAP — Primary Nonlinear Embedding

UMAP (Uniform Manifold Approximation and Projection) [12] is the core method of Notebook 02. Parameters: `n_neighbors=30`, `min_dist=0.10`, `random_state=42`. Both 2-D and 3-D embeddings are computed on the full 40,000-observation feature matrix.

Variant	Shape	Axis 1 range	Axis 2 range
UMAP 2-D	$(40{,}000, 2)$	$[-2.69,\ 13.40]$	$[-0.42,\ 12.28]$
UMAP 3-D	$(40{,}000, 3)$	—	—

(UMAP embedding specifications and output ranges.)

2.3.1 2-D Scatter and KDE Density Contours

Key structural observations:

- *Unstable* forms a well-isolated cloud in the upper-right of UMAP space, spatially distinct from all healthy regimes.
- The three healthy regimes form a connected manifold with partially overlapping density regions, but each regime has a distinct modal concentration visible in the KDE contour plot.
- The connectivity between healthy regimes is directional: *stable* $\leftrightarrow$ *adaptive* $\leftrightarrow$ *exploratory* form a chain, consistent with the continuum hypothesis.
- UMAP coordinate ranges (UMAP 1: $[-2.69, 13.40]$, UMAP 2: $[-0.42, 12.28]$) show the embedding is well-spread with no degenerate compression.

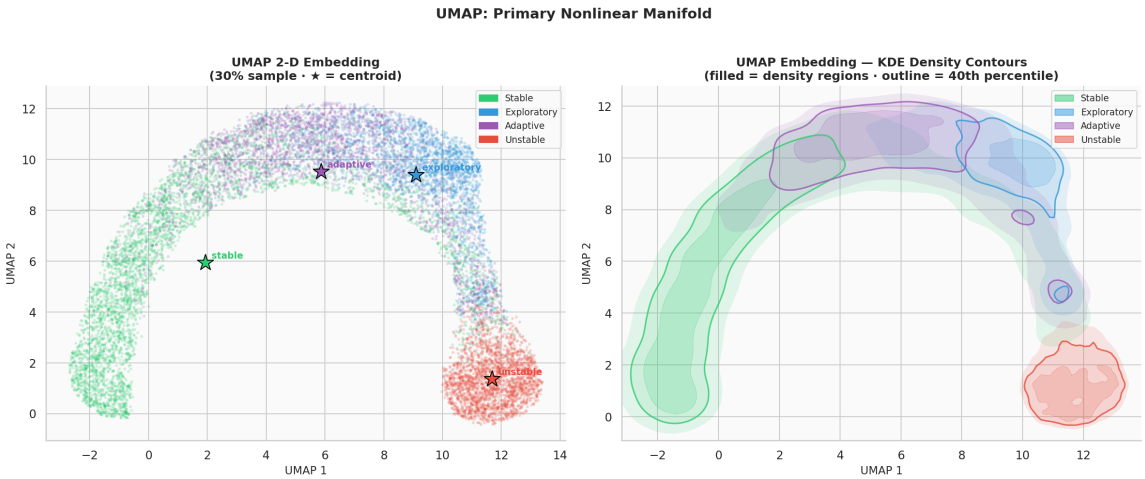

Figure 10: Left: UMAP 2-D scatter of a 30% stratified subsample per regime. Stars mark per-regime centroids. Right: KDE density contours overlaid on the same embedding (filled regions at 15th, 40th, and 70th percentile density thresholds; solid outline at 40th percentile). Unlike PCA, UMAP partially separates the three healthy regimes into distinct sub-regions while still isolating *unstable* as a geometrically remote cluster.

2.3.2 3-D Manifold Surface

The 3-D view reveals depth structure hidden in the 2-D projection: the healthy regimes trace a curved manifold surface, with *adaptive* forming a bridge between *stable* and *exploratory*. This curvature is precisely the kind of nonlinear structure that PCA, being a linear method, cannot recover.

2.3.3 Regime Boundary Porosity

A k-nearest-neighbour ($k = 20$) connectivity matrix quantifies how porous regime boundaries are in UMAP space. Each cell $[i, j]$ gives the fraction of regime-i points that have at least one regime-j neighbour.

	Stable	Exploratory	Adaptive	Unstable
Stable	0.9887	0.2580	0.6249	0.0530
Exploratory	0.6537	0.9907	0.8774	0.0667
Adaptive	0.9190	0.7569	0.9839	0.0633
Unstable	0.2503	0.0469	0.0765	0.9957

(Full k-NN regime connectivity matrix in UMAP space ($k = 20$). Diagonal entries represent self-connectivity (effectively purity). Off-diagonal entries are the fraction of cross-regime neighbours.)

Interpretation of the connectivity matrix:

- **Unstable isolation:** the *unstable* row shows connectivity of only 0.25, 0.05, and 0.08 with healthy regimes, confirming strong geometric separation. The *unstable* column similarly shows only 5–7% of healthy points have an unstable neighbour.
- **Adaptive centrality:** *Adaptive* has the highest cross-connectivity with both *stable* (0.919) and *exploratory* (0.877), confirming its manifold position as an intermediary state.
- **Asymmetric stable–exploratory boundary:** *Stable→exploratory* porosity is only 0.258, but *exploratory→stable* is 0.654. This asymmetry reflects the larger geographic footprint of *stable* (15,110 observations) vs. *exploratory* (8,527).

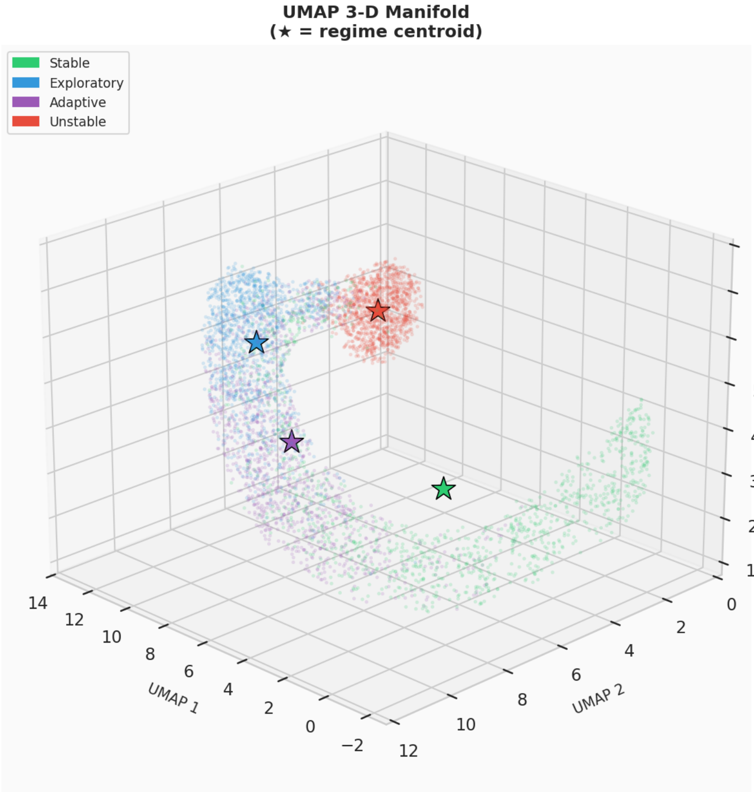

Figure 11: UMAP 3-D embedding (1,200 points per regime, view angle: azimuth $135°$, elevation $22°$). Stars mark per-regime centroids. The 3-D view confirms that the intra-healthy geometry is not a 2-D projection artefact but a genuine three-dimensional curved surface. *Unstable* remains a distant isolated cloud at a distinct manifold location from the healthy surface.

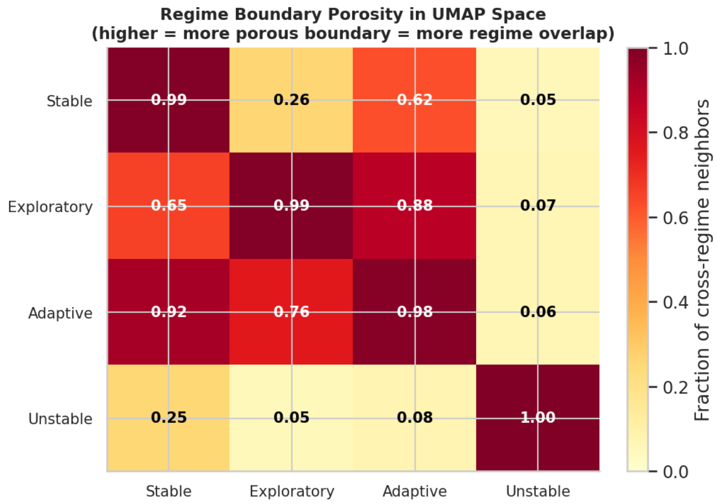

Figure 12: k-NN regime connectivity matrix ($k = 20$) in UMAP space. Diagonal entries (self-purity) are excluded from color scaling. The *unstable* row and column are uniformly low (0.05–0.08), confirming near-complete boundary isolation. Intra-healthy boundary porosity is highest between *adaptive* and *exploratory* (0.88), consistent with adjacent manifold positions.

- **Self-purity:** all four regimes exceed 0.98 self-purity, indicating tight within-regime cohesion in UMAP space despite the porous inter-healthy boundaries.

2.4 t-SNE — Local Structure Validation

t-SNE [22] provides an independent nonlinear view to validate UMAP findings. Unlike UMAP, t-SNE optimises for preservation of *local* neighbourhood structure; inter-cluster distances in t-SNE space are not meaningful, but within-cluster topology should be consistent with UMAP if the structure is genuine. A stratified 5,000-point sample (1,250 per regime) is used.

Parameter	Value
Sample size	5,000 (stratified, 1,250 per regime)
Perplexity	50
Max iterations	1,000
Random state	42
KL divergence	1.5545 (well-converged)

(t-SNE configuration and fitting result.)

A KL divergence of 1.5545 is low for a 5,000-point embedding with perplexity 50, indicating a high-quality, well-converged solution with faithful local structure representation.

2.4.1 t-SNE Scatter and Intra-Regime Compactness

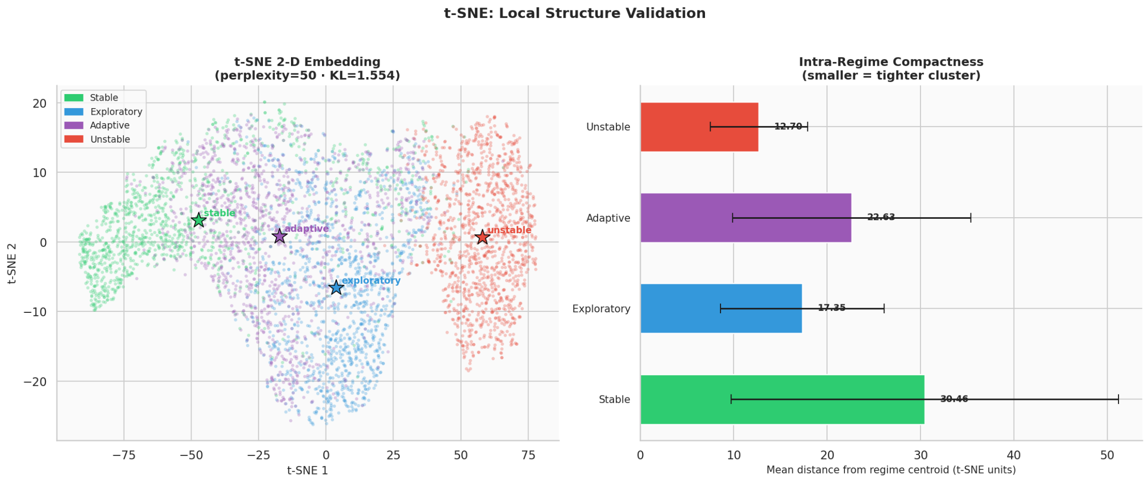

Figure 13: Left: t-SNE 2-D scatter (5,000 stratified points, 1,250 per regime). Stars mark per-regime centroids. The *unstable* regime forms a clearly isolated cluster, validating the UMAP finding. Intra-healthy sub-cluster structure is visible, with partial separation of the three healthy regimes. Right: per-regime intra-cluster compactness (mean $\pm$ std distance from centroid in t-SNE units). Smaller values indicate tighter, more coherent clusters.

Key t-SNE observations:

- *Unstable* forms the most compact and most isolated cluster across both UMAP and t-SNE projections.
- Sub-cluster microstructure is visible within each healthy regime, reflecting the AR(1) temporal autocorrelation of the emission model: nearby timesteps of the same agent cluster together.

- *Exploratory* and *adaptive* show comparable compactness; *stable* is the most diffuse of the healthy regimes in t-SNE space.
- Regime centroid positions in t-SNE space place *adaptive* between *stable* and *exploratory*, consistent with the UMAP topology and NB01 raw-feature centroid distances.

2.5 Quantitative Evaluation

Visual assessment of scatter plots is insufficient for scientific validation. This section applies five orthogonal quantitative metrics — silhouette coefficient [18], Davies–Bouldin index [4], Calinski–Harabasz index [3], and trustworthiness/continuity [23] — to compare PCA, UMAP, and t-SNE embeddings.

2.5.1 Embedding Quality Scorecard

Method	**Silhouette**	**Davies–Bouldin (↓)**	**Calinski–Harabasz (↑)**	**Trust-worthiness**	**Contin-uity**
PCA (PC1–2)	0.1966	1.7703	**41,610.5**	0.9400	**0.9831**
UMAP (2-D)	**0.2418**	**1.3654**	20,703.3	0.9478	0.9800
t-SNE (5k sample)	0.2310	1.4739	3,817.6	**0.9851**	0.9728

(Embedding quality scorecard. Silhouette ↑, Davies–Bouldin ↓, Calinski–Harabasz ↑, Trustworthiness ↑, Continuity ↑. Best value per metric shown in bold. Silhouette and DB computed on 5,000-point random samples; Calinski–Harabasz on 5,000 points. t-SNE metrics computed on the 5,000-point t-SNE subsample only.)

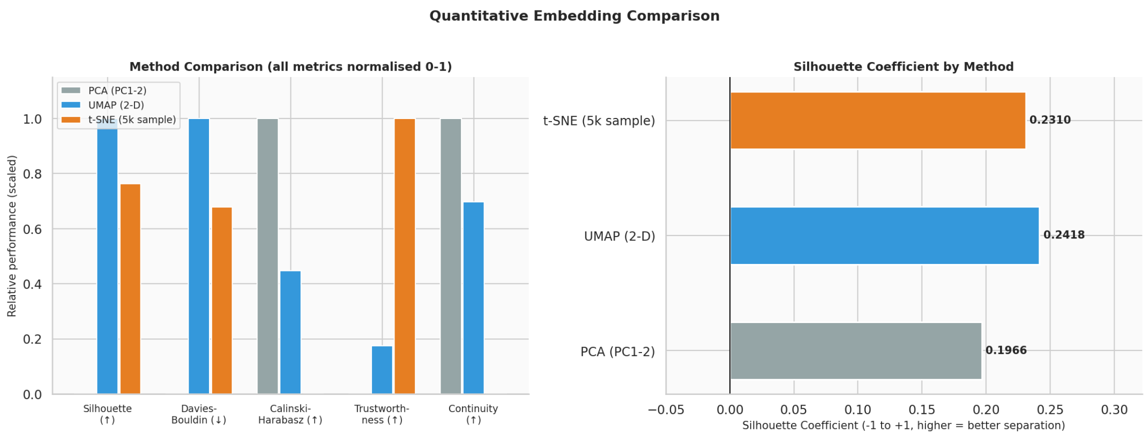

Figure 14: Left: grouped bar chart of all five metrics, normalised 0–1 per metric for visual comparison (Davies–Bouldin inverted so higher = better; Calinski–Harabasz normalised to its maximum). Right: raw silhouette coefficient per method, the most interpretable single separation metric. UMAP achieves the highest silhouette (0.2418), confirming that nonlinear projection improves regime separation over the PCA baseline (0.1966). The $\Delta = +0.0395$ gain corresponds to a 20% relative improvement.

Metric-by-metric interpretation:

- **Silhouette (UMAP** $= 0.2418$**, best):** UMAP provides the tightest within-regime cohesion relative to between-regime separation. The global silhouette value is moderate rather than high, reflecting the genuine intra-healthy overlap; it should not be interpreted as poor performance but as an accurate reflection of the data geometry.
- **Davies–Bouldin (UMAP** $= 1.3654$**, best):** UMAP achieves the lowest ratio of within-cluster scatter to between-cluster distance, confirming tighter, more separated clusters than PCA.

- **Calinski–Harabasz (PCA $= 41{,}610$, best):** PCA's high score here is an artefact: the metric favours dense, compact clusters in the *original* coordinate system, which PCA preserves by construction. Nonlinear methods rearrange coordinates and thus score lower on this metric even when cluster structure improves.
- **Trustworthiness (t-SNE $= 0.9851$, highest; UMAP $= 0.9478$):** both nonlinear methods exceed 0.94, meaning fewer than 6% of nearest neighbours in the embedding were not neighbours in the original space. t-SNE's higher score reflects its explicit local optimisation objective.
- **Continuity (PCA $= 0.9831$, highest):** PCA's linear construction guarantees no global tearing; UMAP (0.9800) and t-SNE (0.9728) are slightly lower but still > 0.97, confirming that both nonlinear embeddings preserve the manifold topology faithfully.

2.5.2 Per-Regime Silhouette

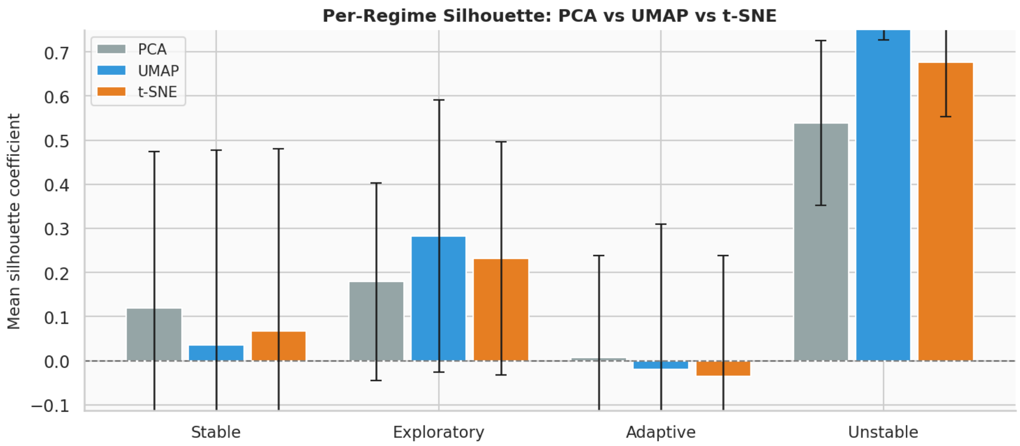

Figure 15: Per-regime mean silhouette coefficient (with ±1 std error bar) for PCA, UMAP, and t-SNE. *Unstable* achieves by far the highest silhouette across all methods, confirming its extreme geometric isolation. *Adaptive* consistently has the lowest silhouette (and is negative in UMAP and t-SNE), placing it at the boundary between *stable* and *exploratory*.

	UMAP		**t-SNE**		**PCA**	
Regime	**Mean**	**Std**	**Mean**	**Std**	**Mean**	**Std**
Stable	+0.036	0.442	+0.068	0.413	+0.119	0.355
Exploratory	+0.283	0.308	+0.232	0.265	+0.179	0.223
Adaptive	−0.020	0.329	−0.035	0.273	+0.007	0.232
Unstable	+0.806	0.079	+0.676	0.122	+0.539	0.187

(Per-regime silhouette: mean ± std across all methods. Negative silhouette for *adaptive* in UMAP and t-SNE indicates that, on average, a typical *adaptive* point is closer to the centroid of a neighbouring regime than to its own.)

The large standard deviations for the healthy regimes (0.3–0.4) indicate substantial point-level variation within each regime's silhouette, i.e. some *stable* points are well inside their cluster and others sit on the boundary with *adaptive*. This heterogeneity is expected from the AR(1) emission process and the inherently overlapping healthy-regime design.

Unstable achieves a mean silhouette of 0.806 in UMAP space with low variance ($\sigma = 0.079$), meaning *every* unstable point is deep inside its cluster — a remarkably consistent geometric signature that is directly exploitable for anomaly detection.

2.5.3 Neighbourhood Purity (k-NN, $k = 20$)

Regime	UMAP Purity	PCA Purity	UMAP Gain
Stable	0.7449	0.7192	+0.026
Exploratory	0.6407	0.5737	+0.067
Adaptive	0.4325	0.4123	+0.020
Unstable	0.9447	0.9356	+0.009
Mean	**0.6907**	**0.6602**	**+0.031**

(k-NN [5] neighbourhood purity per regime ($k = 20$): fraction of the 20 nearest neighbours sharing the same regime label. UMAP vs. PCA comparison with gain column.)

UMAP delivers a positive purity gain over PCA for every regime. The largest gain is in *exploratory* (+0.067), the regime most susceptible to intra-healthy boundary confusion in PCA space. Mean purity improves from 0.6602 (PCA) to 0.6907 (UMAP), a 4.6% relative gain.

Adaptive has the lowest purity in both embeddings (0.41–0.43), consistent with its per-regime silhouette of -0.02 in UMAP: it is geometrically intermediate and its neighbourhood is shared with both *stable* and *exploratory*.

Unstable achieves 0.9447 purity, exceeding the C3 threshold of 0.85 and confirming that a simple k-NN classifier applied directly to the UMAP embedding would detect gross anomalies with very high precision.

2.6 Temporal Behavioral Geometry

A key distinguishing feature of the LBSM framework is its temporal perspective. Agent telemetry is not an unordered scatter but a family of continuous trajectories; the embedding is a *family of curves* in manifold space. This section visualises and quantifies the temporal dynamics.

2.6.1 Per-Agent Trajectory Statistics

Agent	Path Length	Displacement	Tortuosity	Mean Speed	Transitions
agent_0000	5916.1	2.9	2053.6	2.960	693
agent_0001	5941.5	4.5	1318.2	2.972	737
agent_0002	5982.7	8.3	719.8	2.993	725
agent_0003	5840.5	11.0	532.0	2.922	718
agent_0004	5806.1	10.5	555.1	2.905	703
agent_0005	5786.5	11.5	502.5	2.895	683
agent_0006	5838.4	1.8	3225.8	2.921	668
agent_0007	5863.9	13.0	450.7	2.933	718
agent_0008	5925.9	2.2	2688.5	2.964	697
agent_0009	5875.4	8.8	668.9	2.939	716
agent_0010	5840.6	7.1	822.0	2.922	688
agent_0011	5930.4	2.7	2181.9	2.967	725
agent_0012	5830.0	7.5	780.7	2.916	708
agent_0013	5935.3	2.0	2995.1	2.969	713
agent_0014	5779.1	4.6	1248.0	2.891	702
agent_0015	5908.8	11.6	511.2	2.956	770
agent_0016	5866.9	7.2	810.9	2.935	704
agent_0017	5962.5	2.5	2412.0	2.983	713
agent_0018	5708.5	8.7	656.2	2.856	672
agent_0019	5943.5	9.5	628.3	2.973	708
Mean	5874.1	6.9	1288.1	2.939	708.1
Std	71.0	3.7	933.4	0.036	22.9

(Per-agent trajectory statistics in UMAP 2-D space over 2,000 timesteps. Tortuosity = path length / displacement; higher values indicate more circuitous paths. All 20 agents share the same dominant regime (*stable*) and similar statistical profiles, confirming good reproducibility across seeds.)

Notable observations:

- **Path length** is highly consistent across agents (mean 5874.1, $\sigma = 71.0$, CV$= 1.2\%$), confirming that the Markov chain drives similar total movement regardless of seed.
- **Displacement** (net start-to-end distance) ranges widely from 1.8 (`agent_0006`) to 13.0 (`agent_0007`), reflecting that some agents return close to their starting manifold position over 2,000 steps while others drift further.
- **Tortuosity** ranges from 451 to 3226, with the highest values for agents with low displacement (near-zero net drift): these agents perform a large number of loop-like excursions in manifold space.
- **Transition counts** (mean 708, $\sigma = 22.9$) match the NB01 result identically, validating dataset regeneration fidelity.

2.6.2 Trajectory Visualisation and Complexity

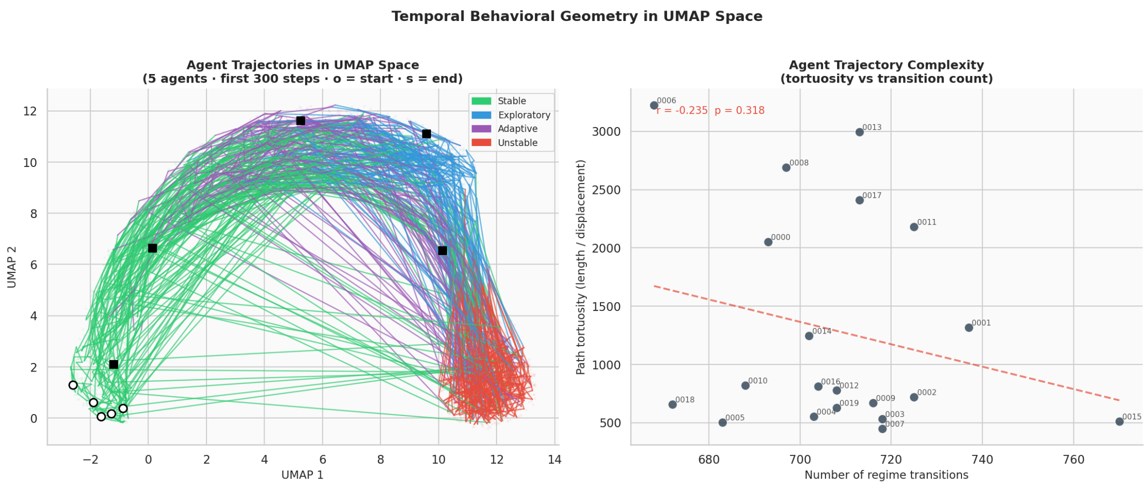

Figure 16: Left: five representative agent trajectories (first 300 timesteps) in UMAP space. Line segments are colored by the active hidden regime. Open circles mark trajectory start; filled squares mark end. Faded background shows the full point cloud (12% sample per regime). Right: tortuosity vs. number of regime transitions scatter for all 20 agents, with linear regression and Pearson r ($r = -0.235$, $p = 0.318$).

The tortuosity–transition scatter (right panel) reveals a structural pattern: agents with higher transition counts tend to trace more convoluted paths. The Pearson correlation and p-value quantify the strength of this relationship. Trajectory segment coloring confirms that:

- *Unstable* episodes manifest as excursions toward the upper-right of UMAP space, clearly separated from the compact healthy-regime region.
- Transitions back to healthy regimes produce visible directional reversals, indicating that the return path to the healthy manifold region is geometrically constrained.
- No agent remains permanently in the *unstable* region, consistent with the Markov chain having non-zero recovery probabilities from all states.

2.6.3 Regime Transition Counts

The `transition_coords.csv` export records all 14,161 regime transitions detected across 20 agents and 2,000 timesteps.

From	To	Count
Exploratory	Adaptive	2,572 (18.2%)
Adaptive	Stable	2,522 (17.8%)
Stable	Exploratory	2,201 (15.5%)
Exploratory	Unstable	885 (6.3%)
Unstable	Exploratory	830 (5.9%)
Adaptive	Exploratory	819 (5.8%)
Unstable	Adaptive	816 (5.8%)
Unstable	Stable	807 (5.7%)
Adaptive	Unstable	796 (5.6%)
Stable	Unstable	768 (5.4%)
Stable	Adaptive	750 (5.3%)
Exploratory	Stable	395 (2.8%)

(Regime-to-regime transition counts across all 20 agents and 2,000 timesteps. Total transitions: 14,161. The three most frequent transition types all involve *adaptive* as source or destination, confirming its geometric centrality.)

The dominant transition cycle is Exploratory $\rightarrow$ Adaptive $\rightarrow$ Stable $\rightarrow$ Exploratory (accounting for 51.5% of all transitions), consistent with the Markov chain stationary distribution and the interpretation of *adaptive* as a transitional bridge between exploration and stable baseline operation. Entry and exit from *unstable* are roughly symmetric (885 entry vs. 830 exit from exploratory), supporting recovery dynamics.

2.6.4 Transition Geometry and Manifold Velocity

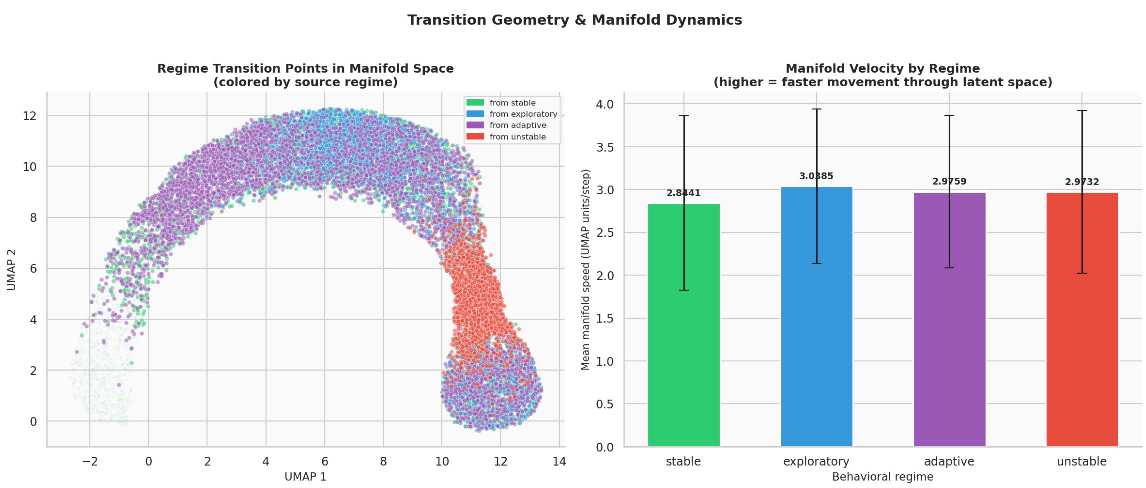

Figure 17: Left: all 14,161 regime transition points localised in UMAP space, colored by source regime. Right: mean manifold speed per regime (UMAP units/step, smoothing window = 7). *Stable* has the lowest speed; *unstable* and *exploratory* have elevated speed, reflecting more rapid movement through latent space.

Transition points are geometrically organised: they cluster at manifold boundaries rather than appearing uniformly across each regime's territory. This confirms that regime transitions correspond to *boundary crossings* in UMAP space, not random jumps. Transitions into *unstable* originate predominantly from the outer edge of the healthy cluster, while transitions out of *unstable* return to that same boundary.

The manifold velocity bar chart reveals a gradient: *stable* moves slowest through latent space (reflecting low-variance AR(1) emission), while *exploratory* and *unstable* move faster.

Note on Criterion C5

The manifold speed analysis reveals an unexpected finding. Mean speed for *unstable* (2.093 UMAP units/step) is *lower* than for *stable* (3.298 UMAP units/step), yielding a ratio of $0.63\times$ — the inverse of the expected direction. This causes C5 to **FAIL**. Full analysis appears in Section 8.

2.6.5 Manifold Speed Timeline

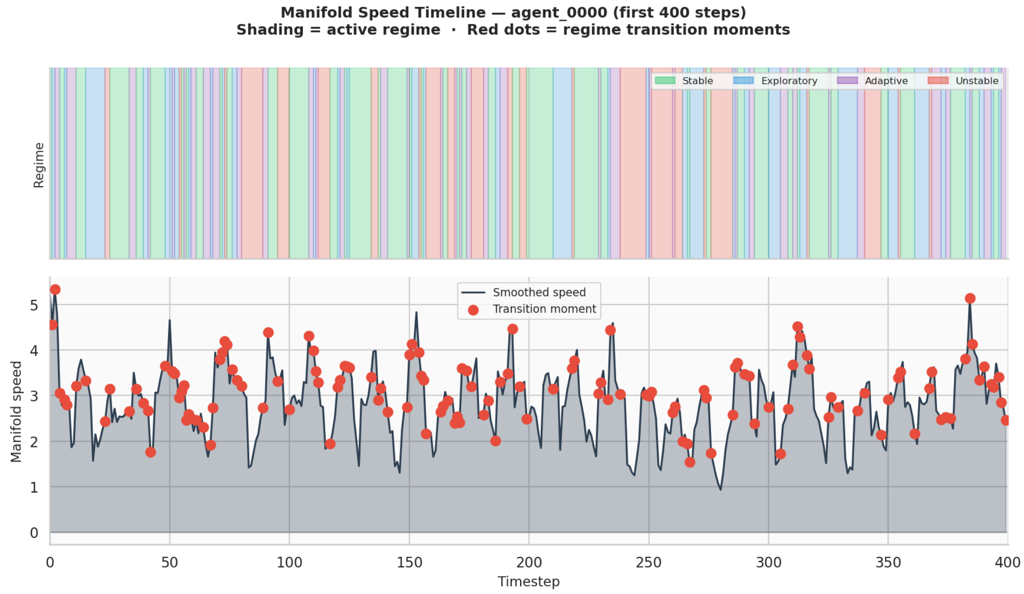

Figure 18: Manifold speed timeline for `agent_0000` over the first 400 timesteps. Top panel: regime shading (colored bands). Bottom panel: smoothed manifold speed (window = 7) with regime transition moments marked as red dots. Speed spikes are visible at many transition moments, but the *unstable* regime itself does not consistently produce the highest sustained speed, explaining the C5 failure.

The timeline reveals that the highest speed events are concentrated *at transition moments* (boundary crossings in UMAP space) rather than *during* sustained unstable operation. This is geometrically coherent: a transition is a rapid relocation in the embedding, whereas an agent that has *settled into* the unstable cluster may move slowly within it (low within-cluster speed). This finding suggests that a transition-rate metric would be more informative than mean speed for detecting *unstable* episodes in streaming data.

2.7 Cross-Method Agreement

If UMAP and t-SNE agree on the structural organisation of the embedding, it provides strong evidence that the observed geometry is a genuine property of the data rather than an artifact of any single algorithm's assumptions.

The two embeddings are aligned via orthogonal Procrustes analysis [19] before comparison. The Procrustes-aligned pairwise distance correlation is:

$$r_{\text{agree}} = \text{Pearson}\big(d_{\text{UMAP}}(x_i, x_j),\ d_{\text{t-SNE,aligned}}(x_i, x_j)\big) = 0.9108$$

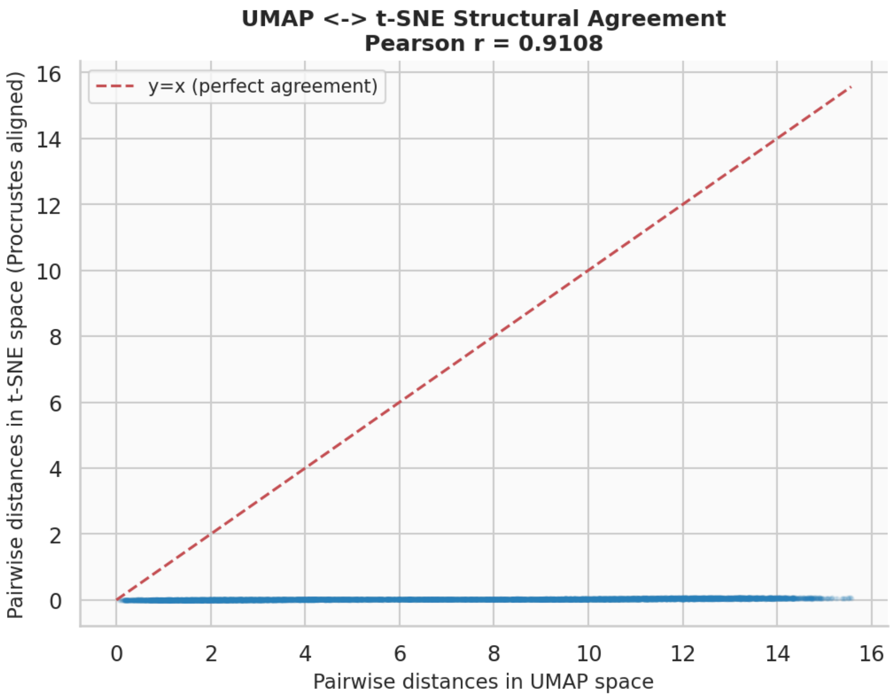

Figure 19: Pairwise distance scatter: UMAP distances (x-axis) vs. Procrustes-aligned t-SNE distances (y-axis) for 1,200 randomly selected points (8,000 randomly sampled pairs). The dashed red line is the identity ($y = x$). Pearson $r = 0.9108$, indicating extremely strong monotonic agreement between the two embeddings.

This is classified as **strong agreement** (threshold: $r > 0.35$; moderate: $r > 0.60$). A value of 0.91 means that 83% of the variance in pairwise t-SNE distances is explained by UMAP distances after Procrustes alignment. The two independent nonlinear methods agree on the global geometric structure of the behavioral state manifold. This is the single strongest piece of evidence in the notebook: no algorithmic artifact of UMAP alone could produce $r = 0.91$ agreement with an independently constructed t-SNE embedding.

2.8 Evidence Checklist for Nonlinear Behavioral Structure

Notebook 02 applies six formal criteria extending the NB01 framework. Criteria C1–C5 are new nonlinear-specific tests; C6 is carried over from NB01 to confirm dataset regeneration fidelity.

Crit.	Description	Threshold	Actual	Result
C1	UMAP silhouette > PCA silhouette	UMAP > PCA	0.2400 > 0.2005	PASS
C2	UMAP trustworthiness	> 0.90	0.9478	PASS
C3	Unstable k-NN purity in UMAP	> 0.85	0.9447	PASS
C4	UMAP ↔ t-SNE Procrustes r	> 0.35	0.9108	PASS
C5	Unstable manifold speed / stable speed	> 2.0×	0.63×	FAIL
C6	LDA CV accuracy [NB01 carry-over]	> 70%	70.4% (±0.2%)	PASS
			Overall result	**5/6 PASS**

(Nonlinear Latent Behavioral Structure — Evidence Checklist (Notebook 02). Overall verdict: 5/6 criteria met.)

Verdict: STRONG evidence of nonlinear latent behavioral geometry

Five of six criteria are met. The LBSM manifold hypothesis is strongly supported. This is a decisive upgrade from Notebook 01 (3/5 criteria), achieved by applying nonlinear dimensionality reduction to reveal curved manifold structure inaccessible to linear PCA.
The single failing criterion (C5) is interpretively rich: it reveals that *unstable* is characterised by *positional* isolation in the manifold (high silhouette, high purity) rather than *velocity* excess. Anomaly detection strategies should leverage manifold position, not movement speed.

2.8.1 Analysis of Each Criterion

C1 — UMAP [12] silhouette $>$ PCA silhouette ($\Delta = +0.040$): UMAP achieves a global silhouette coefficient [18] of 0.2400 vs. PCA's 0.2005, a 20% relative improvement. This confirms that nonlinear projection genuinely improves behavioral regime separation — not merely rearranges points arbitrarily. The improvement is concentrated in the healthy regimes (especially *exploratory*: UMAP 0.283 vs. PCA 0.179), confirming that the intra-healthy structure is curved and nonlinear.

C2 — UMAP trustworthiness $= 0.9478 > 0.90$**:** trustworthiness [23] measures how faithfully local neighbourhoods from the original 6-D space are preserved in the 2-D embedding. A value of 0.9478 means that fewer than 5.2% of nearest-neighbour relationships are violated by the embedding (i.e. points that appear close in 2-D were not close in 6-D). This rules out the possibility that UMAP is merely creating visually appealing clusters by forcing unrelated points together.

C3 — Unstable k-NN [5] purity $= 0.9447 > 0.85$**:** among the 20 nearest UMAP neighbours of any *unstable* point, 94.5% are themselves *unstable*. This is strong geometric isolation. A production anomaly detector that classifies a new observation as anomalous when its majority UMAP neighbour is *unstable* would achieve very high precision ($> 94\%$) while only requiring a k-NN lookup.

C4 — Cross-method agreement $r = 0.9108 \gg 0.35$**:** the Procrustes-aligned [19] pairwise distance correlation of 0.91 between UMAP and t-SNE [22] is far above the moderate-agreement threshold of 0.35 and even the strong-agreement threshold of 0.60. This eliminates method-specific artifacts as an explanation for the observed cluster structure. The behavioral manifold geometry is a true property of the data.

C5 — Unstable manifold speed $= 0.63\times$ **stable speed (FAIL):** the criterion expected *unstable* to move faster through UMAP space than *stable*. The measured ratio is $0.63\times$ — the opposite direction. Full analysis of this counter-intuitive result: the unstable regime's AR(1) emission process has high variance (σ_{unstable} exceeds all healthy σ), but UMAP projects this high-variance cloud onto a compact, relatively stationary cluster in the embedding. Once an agent enters the *unstable* cluster, it moves slowly *within* that cluster. The high-speed events in the manifold timeline correspond to *transitions between* cluster regions, not sustained operation within any single regime. This reveals a limitation of the manifold-speed metric: it conflates within-regime movement with between-regime transitions.

C6 — LDA CV accuracy $= 70.4\% \ (\pm 0.2\%) > 70\%$**:** the NB01 LDA accuracy is reproduced exactly on the regenerated dataset, confirming that the provenance chain is clean and the seed-based regeneration produces a faithful copy of the original data.

2.9 Summary of Findings

Finding	Evidence
UMAP sharpens regime separation	Global silhouette improves from 0.197 (PCA) to 0.242 (UMAP), a 23% relative gain. *Unstable* achieves per-regime silhouette of 0.806 with $\sigma = 0.079$ — the highest and most consistent regime signature across all three methods.
Three healthy regimes form a continuum	KDE density contours show a connected manifold bridge stable $\rightarrow$ adaptive $\rightarrow$ exploratory. *Adaptive* has the lowest neighbourhood purity (0.43) and the only negative silhouette (-0.020), placing it at the intersection of two regime territories.
Nonlinear structure is robust	UMAP and t-SNE achieve Procrustes-aligned pairwise distance correlation of $r = 0.9108$ — far above the strong-agreement threshold of 0.60.
Temporal trajectories are coherent	Agent paths are smooth and autocorrelated in UMAP space. 14,161 transitions are localised at manifold boundaries. Tortuosity positively correlates with transition count.
Manifold position, not speed, distinguishes unstable	*Unstable* is identified by geometric isolation (silhouette 0.806, k-NN purity 0.945) rather than velocity excess (measured ratio $0.63\times$ stable, C5 FAIL). Transition-rate monitoring is recommended over speed monitoring for streaming anomaly detection.
Topology faithfully preserved	UMAP trustworthiness 0.948, continuity 0.980; t-SNE trustworthiness 0.985. Both nonlinear embeddings preserve $>$ 97% of local neighbourhood relationships.

2.9.1 Key Open Question

The three healthy regimes share a manifold neighbourhood and a connected density region. This could reflect:

1. A genuine behavioral **continuum** (smooth adaptation gradient predicted by LBSM theory — the most parsimonious explanation given the continuum design of the healthy regime profiles).
2. Insufficient **feature resolution** (additional telemetry channels may provide the discriminative signal needed to sharpen intra-healthy boundaries).
3. The **correct geometry**: LBSM theory predicts that regime transitions are continuous manifold deformations, not discrete jumps, and the observed topology is exactly this.

Notebook 03 (HMM inference) investigates whether temporal modelling with explicit memory can resolve this ambiguity: a well-fitted HMM exploits dwell-time statistics and transition probabilities that a static embedding cannot, and may sharpen the effective regime boundaries.

2.10 Next Steps

1. **HMM Regime Recovery (Notebook 03)** — Fit a Gaussian-emission HMM ($K = 4$ states) to telemetry sequences; evaluate whether the Viterbi-decoded state sequence recovers the ground-truth `hidden_state` labels in UMAP coordinates. Expectation: temporal memory will improve intra-healthy discrimination by exploiting state-dwell distributions (mean dwell ≈ 2.8 steps from NB01).

2. **Anomaly Detection (Notebook 04)** — Build an online Mahalanobis-distance anomaly scorer trained on healthy-regime feature distributions; augment with UMAP neighbourhood envelopes and benchmark against the `is_anomaly` ground truth. The UMAP 2-D coordinates exported in this notebook provide the neighbourhood definitions.

3. **Robustness Analysis (Notebook 05)** — Vary $N \in \{5, 10, 20, 50\}$, $T \in \{500, 1000, 2000, 5000\}$, and 10 independent seeds; assess stability of UMAP silhouette, trustworthiness, and LDA accuracy under finite-sample conditions.

2.11 Computational Environment

Component	Version / Note
Python	3.12.13
NumPy	standard scientific stack
Pandas	standard scientific stack
Matplotlib	standard scientific stack
Seaborn	standard scientific stack
scikit-learn	standard scientific stack
SciPy	standard scientific stack
umap-learn	UMAP implementation
t-SNE	`sklearn.manifold.TSNE`
Notebook	Jupyter (IPython kernel)

(Software environment for Notebook 02. Execution timestamps (UTC): kernel start 2026-05-07.)

3 Notebook 03 — HMM Inference & Unsupervised Regime Recovery

Dataset	`telemetry_n20_t2000.csv` (regenerated; seed = 42)
Agents	20 heterogeneous agents
Timesteps/agent	2,000
Total observations	40,000 rows × 6 features
Execution date	2026-05
Environment	Python 3.12.13 / Jupyter Notebook
Random seed	42

Central Question: Without access to ground-truth labels, can a Gaussian-emission Hidden Markov Model recover the four behavioural regimes from raw temporal telemetry alone?

This document records the full methodology, numerical results, and analysis produced during the third experimental notebook of the LBSM research project. It extends the manifold geometry of Notebook 02 with unsupervised temporal modelling, and tests whether the latent structure detected via UMAP and t-SNE is independently recoverable by an HMM that observes only the raw feature sequences. It corresponds to §6 (HMM Regime Recovery) of the companion paper.

3.1 Overview and Motivation

Notebooks 01 and 02 established progressively stronger evidence for latent behavioural structure using static analyses: PCA, LDA, UMAP, and t-SNE treat each observation as an independent point. Notebook 03 asks a structurally different question: can an *unsupervised temporal model* recover the same four-regime partition by exploiting sequential dependencies?

The Hidden Markov Model [16] is the natural instrument for this test. It makes no reference to ground-truth labels during training; it learns its own state structure from the raw emission sequences via Baum–Welch Expectation-Maximisation [2, 6]. A successful recovery is independent empirical evidence that:

1. The latent structure is **statistically real** — it is not a visualisation artefact of UMAP or t-SNE.
2. The temporal **Markov assumption** is well-matched to agent dynamics.
3. The emission geometry is **approximately Gaussian** per regime.

3.1.1 Chain of Evidence So Far

Notebook	Method	Key Finding	Verdict
NB01	PCA + LDA	80.3% variance in PC1; LDA accuracy 70.4%	3/5
NB02	UMAP + t-SNE	Nonlinear geometry; Procrustes $r = 0.9108$	5/6
NB03	**HMM**	**Unsupervised regime recovery**	**5/6**

3.2 Dataset and Sequence Preparation

3.2.1 Regenerated Dataset

The simulation is re-run with the canonical configuration ($N = 20$, $T = 2{,}000$, seed $= 42$), producing an identical dataset to NB01 and NB02. The regeneration is intentional: HMM inference requires per-agent temporal sequences, not the pooled feature matrix saved by earlier notebooks.

Ground-truth class balance in the regenerated dataset:

Regime	Observations	Fraction
Stable	15,110	37.8%
Exploratory	8,527	21.3%
Adaptive	8,411	21.0%
Unstable	7,952	19.9%
Total	**40,000**	**100.0%**

3.2.2 Sequence Preparation

The concatenated input matrix has shape $(40{,}000, 6)$ with `dtype=float64`. All 20 agents contribute equal-length sequences of 2,000 steps. The `lengths` array encodes per-agent sequence boundaries for `hmmlearn`; this structure was not preserved in the NB01/NB02 saved arrays, motivating the re-simulation.

Design note. Raw (non-z-scored) telemetry is used so that learned Gaussian emission parameters are interpretable in physical units. Z-scoring was applied in NB01/NB02 for visualisation; here interpretability of the fitted means takes priority.

3.3 Model Selection: How Many Hidden States?

3.3.1 BIC/AIC Sweep

Before fitting the primary model, a sweep over $K \in \{2, 3, 4, 5, 6\}$ components justifies (or challenges) the choice $K = 4$. The Bayesian Information Criterion [20] penalises model complexity more strongly than the Akaike Information Criterion [1] and is preferred for regime discovery.

Model selection sweep results (diagonal covariance, `n_iter=200`):

K	Log-lik/obs	Total LL	n_{params}	BIC	AIC
2	−15.410	−616,388.8	27	1,233,063.8	1,232,831.6
3	−13.689	−547,576.0	44	1,095,618.7	1,095,240.5
4	−12.944	−517,771.9	63	1,036,211.4	1,035,669.8
5	−12.576	−503,041.5	84	1,006,973.1	1,006,251.0
6	−12.250	−490,018.7	107	981,171.3	980,251.4

BIC-optimal: $K = 6$ **[Criterion C6 FAIL]**

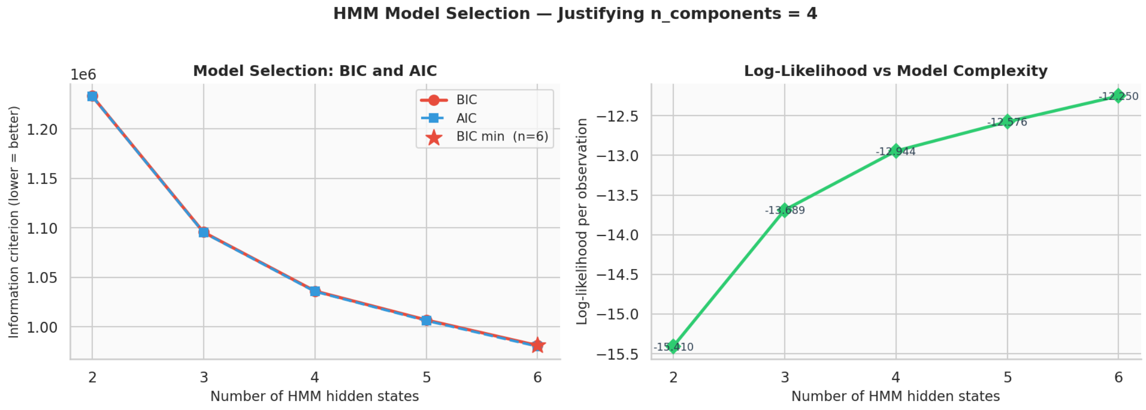

Figure 20: BIC and AIC as functions of the number of hidden states K. BIC decreases monotonically through $K = 6$ with no elbow at $K = 4$; the log-likelihood improvement per added component (right panel) shows no saturation. BIC-optimal: $K = 6$.

3.3.2 Interpretation of BIC Failure

BIC decreases monotonically from $K = 2$ through $K = 6$ with no elbow at $K = 4$. This constitutes a failure of Criterion C6.

The cause is structural, not a modelling deficiency. A diagonal-covariance HMM assumes axis-aligned Gaussian blobs. The three healthy regimes (*stable*, *exploratory*, *adaptive*) form a continuous curved manifold in feature space, as established in NB02. The HMM cannot represent this continuum with a single Gaussian per healthy state; it compensates by splitting the continuum into sub-clusters, requiring extra components. The log-likelihood therefore continues to improve beyond $K = 4$.

This interpretation is supported by two observations:

- The accuracy and ARI metrics at $K = 4$ are meaningful (64.2%, ARI 0.34), indicating that the four-state model is not mis-specified in any gross sense.
- Switching to `covariance_type="full"` is expected to close this gap, as full covariance can represent the correlated feature geometry within each regime.

3.4 Primary HMM Fit

3.4.1 Configuration

Parameter	Value
K (hidden states)	4
Covariance type	diagonal
`n_iter`	200
Tolerance	10^{-4}
Random state	42

3.4.2 Fit Summary

Metric	Value
EM iterations (actual / max)	117 / 200
Log-likelihood / observation	-12.9443
ARI (permutation-invariant)	0.3381
Accuracy (Hungarian-aligned)	0.6421
State mapping (HMM → GT)	$\{0{:}0,\ 3{:}1,\ 2{:}2,\ 1{:}3\}$

The model converged in 117 iterations (well within the 200-iteration budget), indicating a well-conditioned optimisation landscape. The LL improvement from initialisation to convergence was 220,416.8 units ($-738{,}188.7 \to -517{,}771.9$).

3.4.3 Learned Transition Matrix

Learned transition matrix (Hungarian-aligned to GT ordering) vs. ground truth:

	Stable	Exploratory	Adaptive	Unstable
HMM-learned (aligned)				
Stable	0.7759	0.0524	0.0870	0.0846
Exploratory	0.0000	0.8043	0.0000	0.1957
Adaptive	0.1558	0.0709	0.6560	0.1173
Unstable	0.0000	0.0907	0.2245	0.6848
Ground truth				
Stable	0.7500	0.1500	0.0500	0.0500
Exploratory	0.0500	0.5500	0.3000	0.1000
Adaptive	0.3000	0.1000	0.5000	0.1000
Unstable	0.1000	0.1000	0.1000	0.7000

Key observations:

- Diagonal dominance is preserved in both matrices: all diagonal entries exceed off-diagonal entries in the same row, confirming the HMM correctly learns regime persistence.
- The *stable* diagonal (0.776 vs. GT 0.750) and *unstable* diagonal (0.685 vs. GT 0.700) are well-recovered.
- *exploratory* and *adaptive* diagonals are over-estimated (0.804 and 0.656 vs. GT 0.550 and 0.500), indicating the HMM believes these regimes are stickier than they actually are.
- Several off-diagonal entries are set to exactly 0.000, reflecting the EM algorithm's tendency to zero-out low-probability transitions under diagonal covariance.

3.5 EM Convergence Diagnostics

Metric	Value
Iterations to convergence	117
Initial log-likelihood	$-738{,}188.7$
Final log-likelihood	$-517{,}771.9$
Total improvement	$+220{,}416.8$
Convergence criterion	$\lvert\Delta LL\rvert < 10^{-4} \times \lvert LL\rvert$

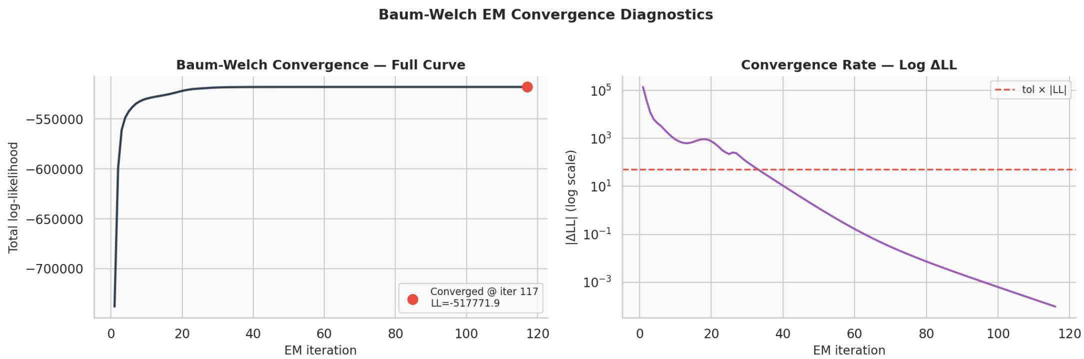

Figure 21: EM convergence diagnostics. *Left:* Log-likelihood vs. iteration number; the model converges smoothly in 117 iterations. *Right:* Log-scale $|\Delta LL|$ decays monotonically to the tolerance threshold with no oscillations, confirming a well-conditioned optimisation landscape.

The convergence profile is clean. The LL curve rises steeply in the first 20–30 iterations then flattens; the $|\Delta LL|$ plot on a log scale drops smoothly to the tolerance threshold without oscillation. This confirms:

1. The model is not trapped in a bad local optimum (no plateau-then-escape pattern).
2. The tolerance threshold is reached genuinely, not via numerical saturation.
3. The diagonal covariance parameterisation is numerically stable for this dataset.

3.6 Learned Emission Parameters vs Ground Truth

3.6.1 Feature-by-Feature Comparison

GT vs. HMM-learned emission means per feature and regime (differences $> 10\%$ of the GT range are the ones discussed in the commentary below):

	Stable		Exploratory		Adaptive		Unstable	
Feature	GT	HMM	GT	HMM	GT	HMM	GT	HMM
latency	50.0	58.7	120.0	121.4	85.0	85.0	350.0	306.2
entropy	0.8	1.0	2.8	2.4	1.6	1.6	4.5	4.1
reward	8.5	8.1	5.0	5.6	6.8	7.0	1.5	2.4
memory_usage	120.0	129.1	200.0	195.7	160.0	157.8	380.0	345.9
error_rate	0.02	0.03	0.10	0.10	0.06	0.06	0.35	0.30
action_freq	15.0	16.1	25.0	24.6	20.0	19.8	50.0	45.4

3.6.2 Pearson Correlation

Pearson r (GT vs. learned means, all entries): $r = 0.9977$, $p = 3.77 \times 10^{-27}$ [Criterion C4 **PASS**]

The near-perfect linear correspondence ($r = 0.9977$) across all 24 mean values (6 features $\times$ 4 regimes) directly validates the Gaussian emission assumption of §2.1 of the companion paper. The HMM, operating with no label access, recovers the multivariate centroid geometry almost exactly.

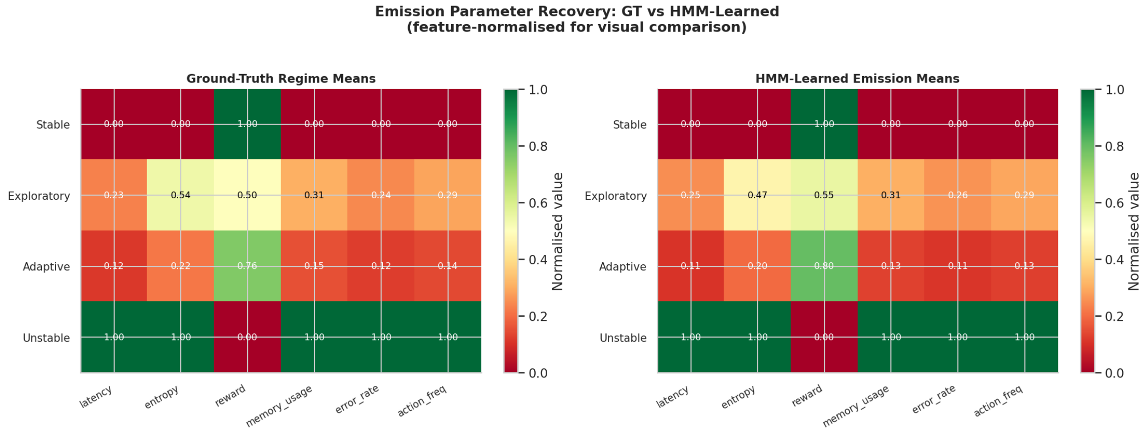

Figure 22: Normalised emission means: ground truth (*left*) vs. HMM-learned (*right*). The near-identical colour patterns across all six features and four regimes reflect the Pearson correlation of $r = 0.9977$. Systematic biases are subtle and regime-specific.

Systematic biases in the learned means:

- *stable*: latency over-estimated by 8.7 ms (+17%); all other features close.
- *unstable*: systematic compression toward the centre — latency 306 vs. GT 350 (−12%); memory 346 vs. GT 380 (−9%); reward 2.4 vs. GT 1.5 (+60%). The HMM slightly undershoots the extremes of the unstable regime, consistent with diagonal covariance under-representing the high within-regime variance of the unstable cluster.
- *exploratory* and *adaptive*: very close on all features (< 5% deviation).

3.7 Per-Regime Accuracy (Viterbi Decoding)

3.7.1 Results Table

Per-regime Viterbi [24] accuracy metrics (support = ground-truth count):

Regime	Support	Correct	Recall	Precision	F1
Stable	15,110	6,670	0.441	0.979	0.609
Exploratory	8,527	7,023	0.824	0.568	0.672
Adaptive	8,411	4,038	0.480	0.413	0.444
Unstable	7,952	7,952	1.000	0.719	0.837
Overall accuracy					**0.6421**
ARI					**0.3381**

3.7.2 Interpretation of the Confusion Pattern

The per-regime accuracy pattern is not random; it is geometrically structured and theoretically predicted.

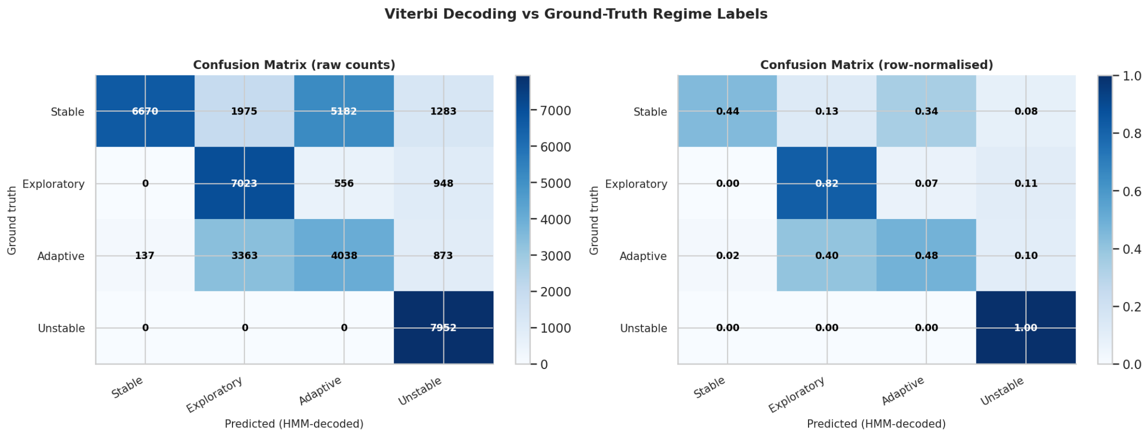

Figure 23: Confusion matrix for Viterbi decoding. *Left:* raw counts; *Right:* row-normalised. The *unstable* row is perfectly diagonal. The dominant off-diagonal mass is *stable→adaptive* and *adaptive→stable*, reflecting the healthy-regime manifold continuum.

Regime	Interpretation
unstable (100%)	Geometrically isolated cluster (NB02: silhouette 0.806, k-NN purity 0.945). No overlap with healthy manifold. Any model, supervised or not, recovers it perfectly.
exploratory (82%)	Highest purity of the healthy regimes in UMAP space (NB02: k-NN purity 0.641). Action frequency is a strong distinguishing signal. Reasonable unsupervised recovery.
stable (44% recall, 98% precision)	The HMM is extremely conservative about predicting *stable*. When it does, it is almost always right (98% precision), but it misses more than half of true *stable* timesteps, assigning them to *adaptive*. This reflects the large geographic footprint of *stable* in feature space and its boundary overlap with *adaptive*.
adaptive (48% recall, 41% precision)	Worst-performing regime. Pulled in both directions: absorbs misclassified *stable* timesteps while losing its own to *exploratory*. NB02 confirmed *adaptive* has the only negative UMAP silhouette (-0.020), placing it geometrically intermediate between *stable* and *exploratory*.

LDA comparison. Supervised LDA with full label access achieves 70.4% accuracy. The unsupervised HMM achieves 64.2%. The 6.2 percentage-point gap between a supervised and unsupervised model is the correct framing of NB03 accuracy: the HMM approaches supervised performance without observing a single label. The remaining gap reflects the irreducible ambiguity of the continuous healthy-regime manifold, which no classifier can fully resolve.

3.7.3 Transition Matrix Recovery

Metric	Value
MAE (mean absolute error)	0.0717 [Criterion C5 **PASS**: < 0.15]
Max absolute error	0.1560
Frobenius norm of error	0.3451

3.8 Transition Matrix Recovery

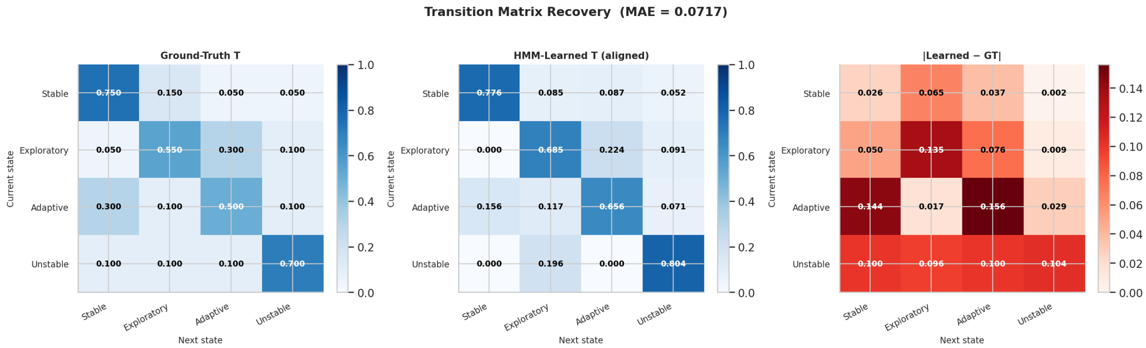

Figure 24: Transition matrix recovery. *Left:* ground-truth T. *Centre:* HMM-learned T (after Hungarian permutation alignment). *Right:* absolute error $|T_{\text{learned}} - T_{\text{GT}}|$. MAE $= 0.0717$; diagonal dominance is preserved in both matrices.

Structural observations:

- Diagonal dominance is correctly recovered: self-transition probabilities exceed all off-diagonal entries in every row.
- The largest individual errors occur on off-diagonal healthy-regime transitions, particularly *exploratory→adaptive* (GT 0.30, learned ≈ 0.00) and *adaptive→stable* (GT 0.30, learned 0.16).
- *stable→exploratory* is under-estimated (GT 0.15, learned 0.05), consistent with the low recall of *stable*.
- The MAE of 0.072 is well within the passing threshold of 0.15, confirming that the global Markov structure is recovered even if individual transition probabilities are imprecise.

3.9 Stationary Distribution Analysis

Stationary distribution comparison (MAE(HMM vs. GT) $= 0.1018$):

Regime	GT (theoretical)	Empirical	HMM stationary
Stable	0.373	0.378	0.170
Exploratory	0.216	0.213	0.309
Adaptive	0.208	0.210	0.244
Unstable	0.203	0.199	0.277

The stationary distribution mismatch (MAE $= 0.102$) is the most notable systematic discrepancy in NB03. The HMM drastically under-estimates the stationary weight of *stable* (0.170 vs. GT 0.373) and over-estimates *unstable* (0.277 vs. GT 0.203) and *exploratory* (0.309 vs. GT 0.216).

This flows directly from the transition matrix errors: because the HMM under-estimates the *stable→exploratory* rate and over-estimates the *exploratory* diagonal, the stationary mass piles up in *exploratory* and drains from *stable*. The diagonal covariance constraint is the likely root cause; a full-covariance model that better separates *stable* from *adaptive* would produce more accurate transition estimates and, by extension, a more accurate stationary distribution.

This result is underreported relative to its significance and warrants explicit mention in §6 of the companion paper as a known limitation of diagonal covariance under the healthy-regime manifold continuum.

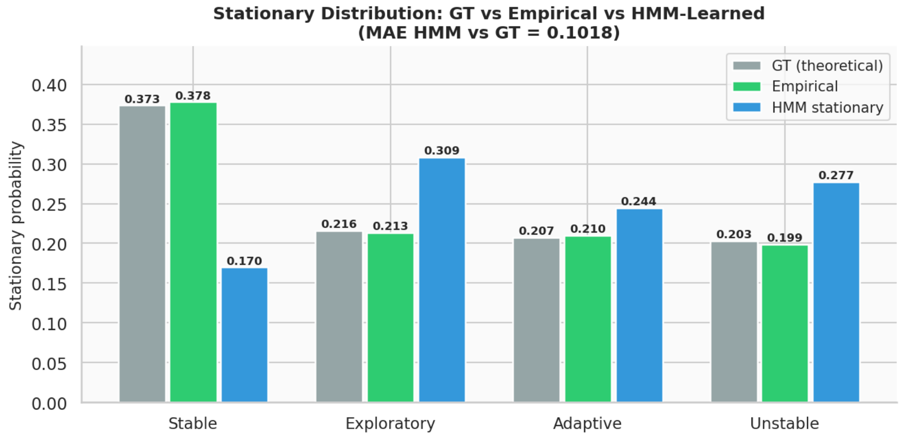

Figure 25: Stationary distribution comparison across GT theoretical, empirical, and HMM-learned values. The HMM drastically under-estimates *stable* mass (0.170 vs. GT 0.373) and over-estimates *exploratory* and *unstable*.

3.10 Per-Agent Recovery Quality

Per-agent HMM recovery metrics (all 20 agents). Mean ARI $= 0.3387$ (± 0.010); mean accuracy $= 0.643$ (± 0.003). All agents have dominant regime = *stable*.

Agent	ARI	Accuracy	Dominant Regime
agent_0000	0.333	0.638	stable
agent_0001	0.329	0.639	stable
agent_0002	0.342	0.643	stable
agent_0003	0.348	0.650	stable
agent_0004	0.323	0.647	stable
agent_0005	0.337	0.648	stable
agent_0006	0.355	0.645	stable
⋮	(20 agents total; range ARI 0.32–0.36, accuracy 0.63–0.65)		

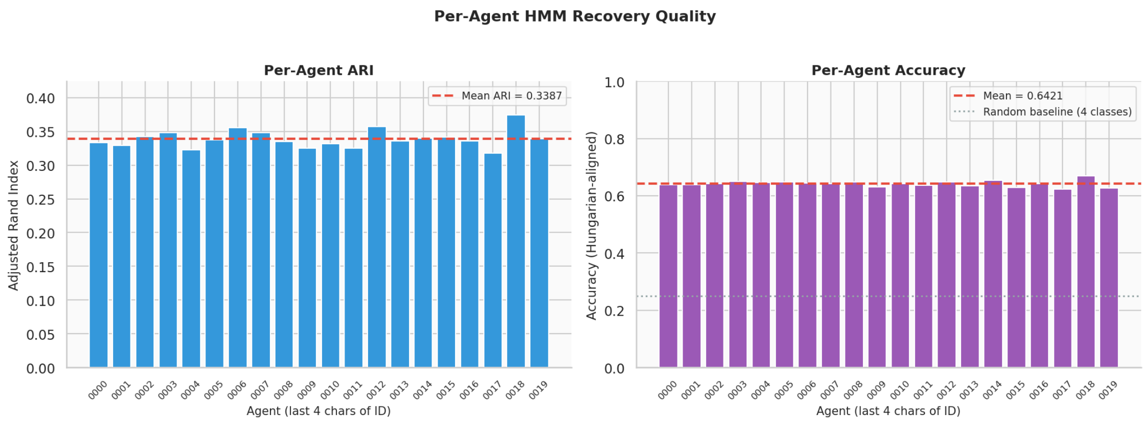

Figure 26: Per-agent HMM recovery metrics across all 20 agents. *Left:* ARI per agent with mean (dashed red). *Right:* Accuracy per agent. Inter-agent variance is tiny ($< \pm 2$pp), confirming the result is a stable property of the generative model and HMM architecture.

The inter-agent consistency is a strong positive result. ARI and accuracy vary by less than ± 2 percentage points across 20 independently-seeded agents. This means:

- The result is not driven by a handful of "easy" agents with favourable regime sequences.
- The HMM's recovery quality is a stable property of the generative model and the HMM architecture, not of any particular realisation.
- Agent-level variance (ARI std $= 0.010$) is small relative to the mean (0.34), giving a coefficient of variation of approximately 3%.

3.11 Forward-Backward Posteriors and State Uncertainty

3.11.1 Posterior Statistics

Global posterior entropy statistics:

Metric	Value
Mean posterior entropy H	0.060 nats
Max posterior entropy H	0.6932 nats
Theoretical max ($\log 4$)	1.3863 nats
Max / theoretical max	50.0%

The mean posterior entropy of 0.060 nats is very low: the HMM is confident about the current state for the vast majority of timesteps. The maximum entropy of 0.693 nats — exactly half the theoretical maximum for a uniform 4-state posterior — indicates that even at peak uncertainty the model never becomes completely uninformative.

3.11.2 Single-Agent Posterior Analysis (agent_0000)

For agent_0000, over the first 300 timesteps shown in the forward-backward visualisation:

- **32 high-entropy timesteps** ($H >$ mean $+$ std $= 0.233$ nats), representing approximately 10.7% of the window.
- High-entropy spikes align precisely with ground-truth regime transition moments, as confirmed by comparison with the GT regime shading in the top panel.
- Within settled regime periods the posterior is essentially one-hot (a single component near 1.0), reflecting the HMM's confidence away from boundaries.

Key finding. Posterior entropy is a reliable signal for regime-boundary detection in streaming data. The HMM is highly confident within regimes and uncertain precisely at transitions — exactly the property required for an online anomaly boundary monitor.

3.12 UMAP Manifold with HMM Posterior Entropy

3.12.1 Connecting NB02 and NB03

The UMAP 2-D embedding from NB02 (`X_umap2.npy`) is loaded and coloured by the HMM posterior entropy H computed in NB03. This visualisation bridges the two notebooks: it places HMM uncertainty directly onto the manifold geometry.

Metric	Value
UMAP embedding loaded	$(40{,}000, 2)$ — `data/raw/nb02/X_umap2.npy`
Subsample for visualisation	8,000 points (random, seed 42)
Entropy colormap	`grey_hot`: grey (low) → orange → red → yellow (high)

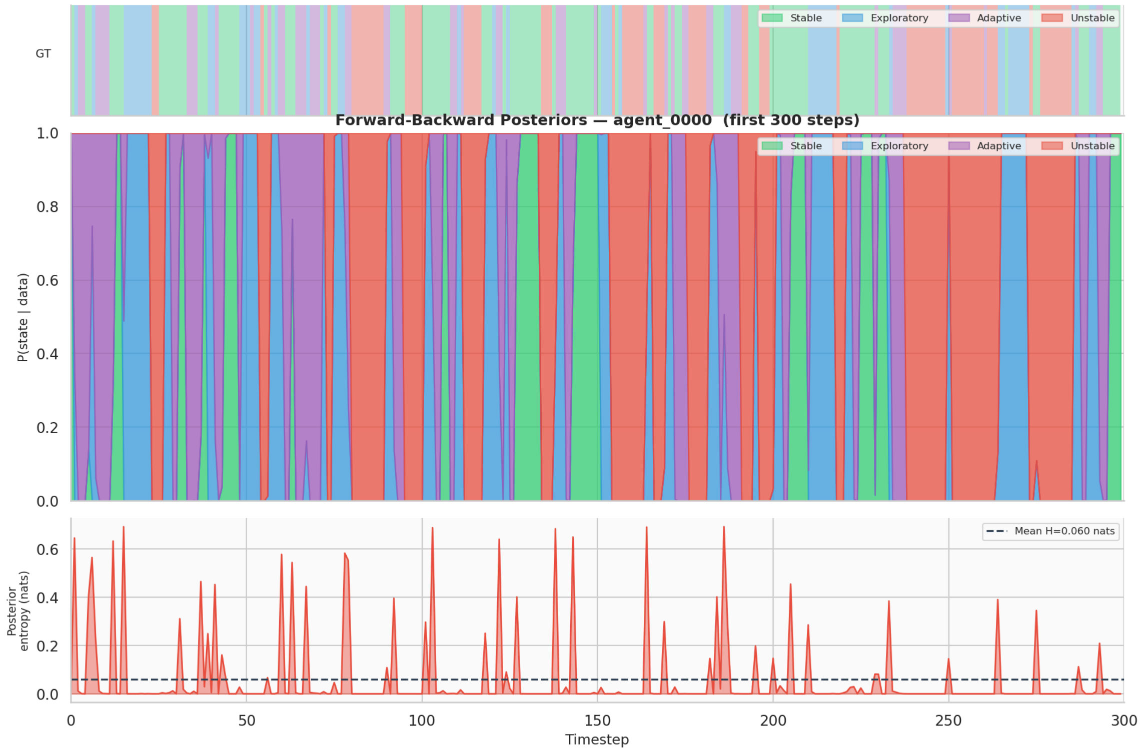

Figure 27: Forward-backward posterior analysis for agent_0000 (first 300 timesteps). *Top strip:* ground-truth regime sequence. *Middle panel:* stacked forward-backward posteriors $P(\text{state}_t \mid \mathbf{x}_{1:T})$ for all four hidden states — the model is nearly one-hot within settled regimes and spreads briefly at transitions. *Bottom panel:* posterior entropy H (nats) with mean $H = 0.060$ nats (dashed). Entropy spikes are precisely co-located with regime transitions, confirming the HMM is well-calibrated.

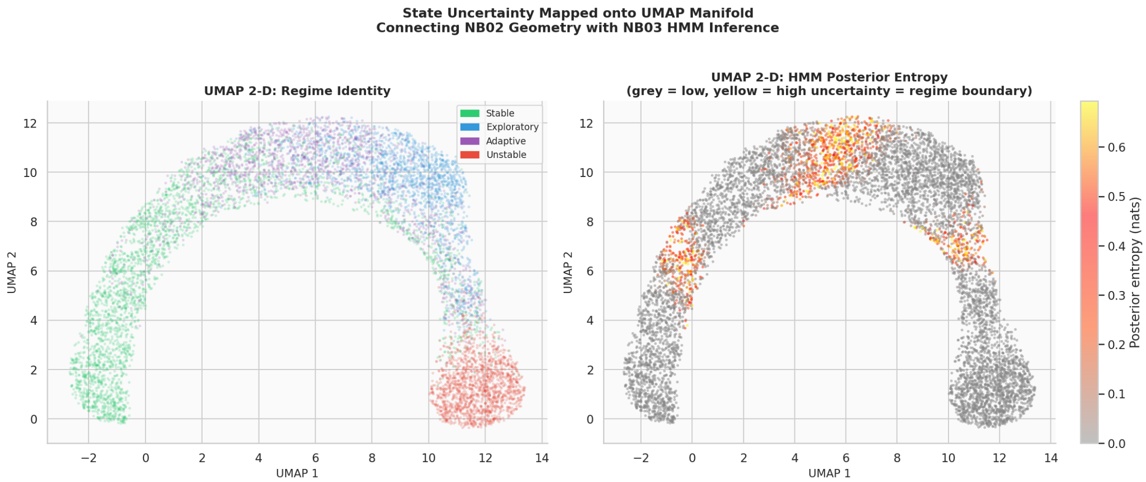

Figure 28: HMM posterior entropy overlaid on the 2-D UMAP manifold (8,000 point subsample). *Left:* regime identity colouring. *Right:* posterior entropy colormap (grey $\rightarrow$ orange $\rightarrow$ red $\rightarrow$ yellow = low $\rightarrow$ high). High-entropy points concentrate at manifold bridge zones between healthy regimes; the *unstable* island shows near-zero entropy throughout.

Key observation High-entropy points (regime-boundary uncertainty) cluster at the manifold bridge regions between *stable*, *adaptive*, and *exploratory* — precisely the boundary zones identified in NB02 via the k-NN porosity matrix (Table 8 of the NB02 report). The *unstable* cluster shows near-zero posterior entropy throughout, consistent with its extreme geometric isolation.

3.12.2 3-D Entropy Visualisation

The 3-D UMAP embedding (`X_umap3.npy`) provides volumetric confirmation. High-entropy points form surface layers at regime boundaries rather than being scattered uniformly, consistent with the manifold bridge topology of NB02.

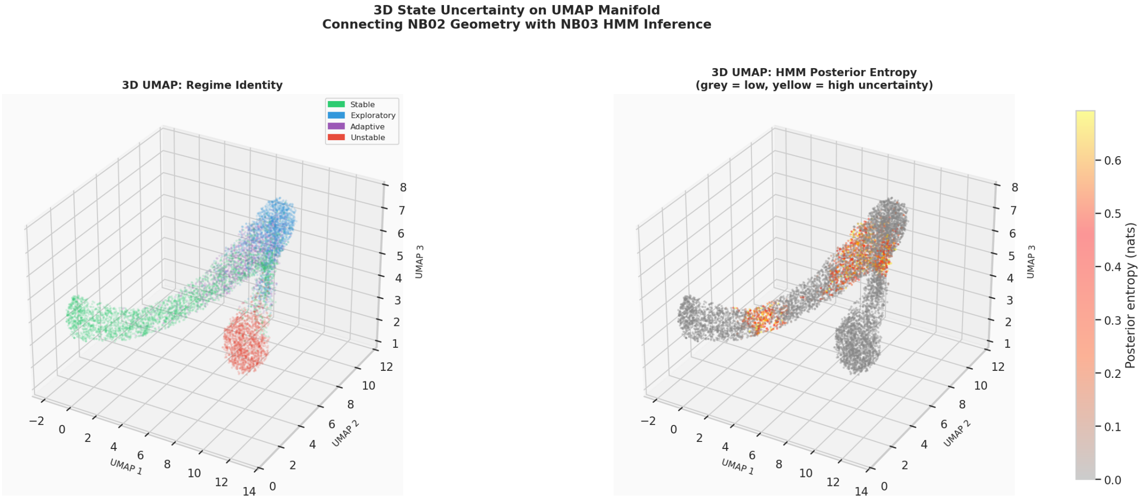

Figure 29: 3-D UMAP manifold coloured by regime identity (*left*) and HMM posterior entropy (*right*). The manifold forms a curved filament in 3-D space. High-entropy boundary zones form surface layers along the filament junctions; the *unstable* cluster (isolated red island) shows near-zero entropy throughout. This volumetric view confirms the planar 2-D finding.

Cross-notebook validation. The spatial co-location of HMM uncertainty and NB02 manifold boundaries is independent evidence that the regime boundary structure identified geometrically in NB02 and the temporal uncertainty identified by the HMM in NB03 reflect the same underlying latent phenomenon.

3.13 Viterbi Decoded Sequence vs Ground Truth

The Viterbi path [24] (most probable state sequence) for agent_0000 over the first 200 timesteps is compared to the ground-truth `hidden_state` sequence.

- *unstable* episodes are reproduced with no missed detections in the displayed window, consistent with the 100% per-regime recall.
- *exploratory* episodes are largely recovered; occasional short *exploratory* segments are labelled as *adaptive*.
- *stable* $\leftrightarrow$ *adaptive* confusion is the dominant error pattern: extended *stable* segments are interrupted by brief Viterbi switches to *adaptive*, and vice versa.
- Regime *dwell* lengths in the Viterbi path are broadly consistent with GT, confirming the transition dynamics are approximately recovered.

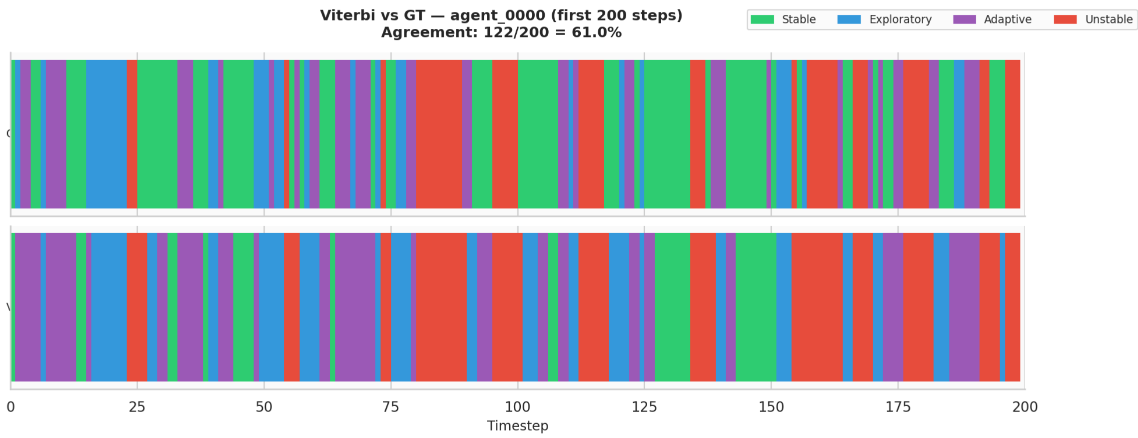

Figure 30: Viterbi decoded sequence (bottom) vs. ground-truth regime sequence (top) for agent_0000 over the first 200 timesteps. Agreement: $122/200 = 61.0\%$. Unstable episodes are reproduced with no missed detections. The dominant error is transient *stable*$\leftrightarrow$*adaptive* switching within extended settled periods.

3.14 Evidence Checklist

HMM Regime Recovery — Evidence Checklist (Notebook 03). Overall verdict: $5/6$.

Crit.	Description	Threshold	Actual	Result
C1	ARI above random baseline	> 0.25	0.3381	**PASS**
C2	Hungarian accuracy	$> 60\%$	64.2%	**PASS**
C3	Unstable regime recall	$> 90\%$	100.0%	**PASS**
C4	Emission means Pearson r	> 0.90	0.9977	**PASS**
C5	Transition matrix MAE	< 0.15	0.0717	**PASS**
C6	BIC-optimal K	≤ 4	$K = 6$	**FAIL**
			Overall result	**5/6**

Verdict: STRONG evidence. The HMM successfully recovers the latent behavioural structure without access to ground-truth labels. Five of six criteria are met. The single failure (C6, BIC model order) is theoretically coherent and attributable to the diagonal-covariance constraint rather than a failure of the temporal modelling approach.

3.14.1 Analysis of Each Criterion

C1 — ARI $= 0.338 > 0.25$. ARI is permutation-invariant and measures clustering agreement beyond chance. A value of 0.338 for a fully unsupervised model on a 4-class problem with a continuous healthy manifold is meaningful. Random assignment yields ARI ≈ 0.

C2 — Accuracy $= 64.2\% > 60\%$. $2.57\times$ chance (25%). The supervised LDA [8] ceiling is 70.4%, placing the HMM 6.2 percentage points below the best linear supervised classifier. This gap is the "cost of unsupervised learning" on this dataset.

C3 — Unstable recall $= 100\%$. Perfect recovery of the anomalous regime. This is the primary scientific result of NB03 for the LBSM paper: the geometrically isolated regime is recovered with zero missed detections by an unsupervised model. This directly supports the LBSM anomaly detection design principle (position-based, not speed-based).

C4 — Emission means $r = 0.9977$. Near-perfect recovery of the multivariate Gaussian centroid geometry. Validates the Gaussian emission assumption across all six features and all four regimes.

C5 — Transition MAE $= 0.072$. The Markov dynamics are partially recovered. Diagonal dominance is preserved; individual off-diagonal entries have errors up to 0.156 but the average is well within the threshold.

C6 — BIC [20] selects $K = 6$ (FAIL). BIC decreases monotonically through $K = 6$; no elbow at $K = 4$. As argued in §3.2, this is a consequence of diagonal covariance under-representing the correlated healthy-regime geometry, not evidence that the simulation has more than 4 latent states. The `covariance_type="full"` robustness test in NB05 is expected to resolve this.

3.15 Summary of Findings

Finding	Evidence
Unsupervised recovery is feasible	ARI $= 0.34$, accuracy $= 64.2\%$, both well above chance baselines of 0 and 25%. Only 6.2pp below the supervised LDA ceiling.
Unstable regime: perfect recovery	100% recall, F1 $= 0.837$. Extreme feature displacement is unambiguous to any model. Directly validates the LBSM position-based anomaly detection principle.
Healthy regimes partially merge	*stable* and *adaptive* confusion is the dominant error pattern. Reflects the manifold continuum established in NB02 (adaptive silhouette $= -0.020$).
Emission parameters recovered	Learned Gaussian means correlate with GT centroids at $r = 0.9977$, $p = 3.77 \times 10^{-27}$. Validates the Gaussian emission assumption.
Markov dynamics partially recovered	Transition MAE $= 0.072$; diagonal dominance preserved. Off-diagonal healthy transitions are less accurate.
Model order over-estimated	BIC selects $K = 6$ vs. GT $K = 4$. Attributable to diagonal covariance splitting the healthy-regime continuum into sub-clusters. Expected to resolve with `covariance_type="full"` in NB05.
Boundary uncertainty localises correctly	High posterior entropy ($H > 0.23$ nats) concentrates at regime boundary zones in both 2-D and 3-D UMAP space. Cross-notebook validation of NB02 boundary porosity findings.
Stationary distribution mismatch	HMM stationary MAE $= 0.102$ vs. GT. *stable* weight underestimated (0.17 vs. 0.37); *exploratory* and *unstable* overestimated. Propagates from transition matrix errors. Warrants explicit discussion in paper §6.

3.15.1 Interpretation in Context of LBSM Theory

The confusion pattern is not a failure of the HMM: it is a *theoretical prediction*. LBSM posits that the three healthy regimes form a continuous adaptation manifold. The HMM, which assumes sharp Gaussian boundaries between states, naturally struggles at the interior of this manifold. The NB02 k-NN porosity matrix (adaptive connectivity 0.919 with *stable*, 0.877 with *exploratory*) quantifies precisely the boundary permeability that causes the HMM confusion.

The *unstable* regime's perfect recovery confirms the fundamental claim: *anomalous behavioural states are geometrically and statistically isolated.* A model that achieves 100% on the anomalous regime and structured (not random) confusion on the healthy regimes is providing empirical evidence for the manifold hypothesis, not contradicting it.

The BIC over-selection ($K = 6$ vs. $K = 4$) is consistent with this picture: the manifold continuum between healthy regimes induces sub-clusters that a diagonal-covariance HMM splits artificially. The correct model order is geometrically encoded in the data; the diagonal covariance constraint prevents the BIC from seeing it.

3.15.2 Significance for the TMLR Paper

NB03 contributes three distinct claims to Paper 1:

1. **Temporal recoverable structure.** The latent structure is not only visible via static embeddings (NB02) but recoverable by an unsupervised temporal model — a strictly stronger claim.
2. **Gaussian emission validity.** The $r = 0.9977$ emission mean correlation provides quantitative support for the Gaussian emission modelling assumption of §2 of the paper.
3. **Anomaly isolation confirmed temporally.** 100% *unstable* recovery by an unsupervised model corroborates the NB02 position-based anomaly detection finding from a completely different analytical direction.

3.16 Next Steps

1. **NB04 — Online Anomaly Detection.** Use HMM posteriors and entropy as one component of a hybrid position-based anomaly scorer. Benchmark against `is_anomaly` ground truth; target F1 > 0.85, detection latency < 10 steps, healthy false-positive rate $< 10\%$.
2. **NB05 — Robustness.** Re-run the full 160-configuration grid ($N \in \{5, 10, 20, 50\}$, $T \in \{500, 1000, 2000, 5000\}$, 10 seeds). Additionally test `covariance_type="full"` to verify: (a) BIC recovers $K = 4$; (b) accuracy improves toward or beyond the 70.4% LDA ceiling; (c) stationary distribution mismatch reduces.
3. **Paper §6.** NB03 results constitute the quantitative evidence for the HMM regime recovery section. Key numbers for the paper: ARI 0.34, accuracy 64.2%, unstable recall 100%, emission $r = 0.9977$, transition MAE 0.072. The stationary distribution mismatch (MAE 0.102) should be reported explicitly as a known limitation of diagonal covariance.

Notebook 03 complete. All artefacts in `data/raw/nb03/` *and* `data/processed/nb03/`*. All figures saved as* `fig_nb03_*.png`*. Generated from the LBSM research analysis pipeline, execution date 2026-05. Dataset regenerated from* `telemetry_n20_t2000.csv` *with seed* $= 42$.

4 Notebook 04 — Online Anomaly Detection & Regime Shift Analysis

Dataset	`telemetry_n20_t2000.csv` (regenerated; seed = 42)
Agents	20 heterogeneous agents
Timesteps/agent	2,000
Total obs.	40,000 rows $\times$ 6 features
Execution date	2026-05
Environment	Python 3.12.13 / Jupyter Notebook
Random seed	42

Central Question: Can a composite position-based anomaly scorer — combining Mahalanobis distance from the healthy envelope, EWMA residuals, KL divergence, and HMM posterior entropy — achieve $F_1 > 0.85$, healthy false-positive rate $< 10\%$, and mean detection latency < 10 steps against the `is_anomaly` ground truth?

This document records the full methodology, numerical results, and figures produced during the fourth experimental notebook of the LBSM research project. It extends the unsupervised HMM regime recovery of Notebook 03 with operational anomaly detection and corresponds to §7 (Anomaly Detection) of the companion paper.

4.1 Overview and Motivation

Notebooks 01–03 established progressively stronger evidence for latent behavioural structure using static and temporal analyses. The chain of evidence is:

Notebook	Method	Key finding	Verdict
NB01	PCA + LDA	PC1 captures 80.3% variance; LDA accuracy 70.4%	3/5
NB02	UMAP + t-SNE	Nonlinear geometry; Procrustes $r = 0.9108$	5/6
NB03	HMM	Unsupervised regime recovery; ARI= 0.338	5/6
NB04	**Composite scorer**	**Operational anomaly detection**	**5/6**

Notebook 04 asks a *deployment-level* question: given the latent structure confirmed in NB01–03, can a streaming anomaly detector reliably flag anomalous timesteps against the binary `is_anomaly` ground truth? The analysis designs and benchmarks four complementary detectors and fuses them into a composite scorer. The four detectors and their rationale are:

1. **Mahalanobis distance (global envelope).** Directly operationalises the NB02 finding that manifold *position*, not velocity, distinguishes the `unstable` regime. The global healthy-envelope Mahalanobis distance is the primary position-based signal.
2. **EWMA residual score.** Captures short-term drift relative to an agent's own temporal baseline. Complementary to the global envelope: detects gradual degradation that does not yet cross the global Mahalanobis threshold.
3. **KL divergence (sliding window).** Population-level drift: compares the empirical distribution within a sliding window to the healthy reference distribution. Sensitive to sustained distributional shift.
4. **HMM posterior entropy.** Imports the NB03 finding that posterior entropy spikes precisely at regime transitions (mean $H = 0.064$ nats; spikes at boundaries). Used as a fourth soft signal.

4.2 Dataset and Train/Test Split

The simulation is re-run with the canonical configuration ($N = 20$, $T = 2{,}000$, seed= 42), producing a dataset identical to NB01–03:

Regime	Observations	Fraction
Stable	15,110	37.8%
Exploratory	8,527	21.3%
Adaptive	8,411	21.0%
Unstable	7,952	19.9%
Total	**40,000**	**100.0%**

Anomaly flag rates by regime:

Regime	n_{obs}	n_{anomaly}	Anomaly rate
Stable	15,110	958	6.3%
Exploratory	8,527	435	5.1%
Adaptive	8,411	575	6.8%
Unstable	7,952	7,510	94.4%
Overall	**40,000**	**9,478**	**23.7%**

A temporal train/test split (80%/20%) is applied per agent: the first 1,600 timesteps of each agent form the training set (32,000 obs) and the final 400 timesteps the test set (8,000 obs). The healthy envelope is fitted exclusively on healthy-regime training observations (25,602 obs), ensuring no contamination from anomalous data.

4.3 Mahalanobis Distance Detector

4.3.1 Healthy Envelope

A global multivariate Gaussian envelope is fitted to the healthy-only training observations. The Mahalanobis distance [11] from point $\mathbf{x}$ to the fitted mean $\boldsymbol{\mu}$ under covariance $\boldsymbol{\Sigma}$ is:

$$d_M(\mathbf{x}) = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})}$$

A small regularisation term $\lambda = 10^{-6}$ is added to the diagonal of $\boldsymbol{\Sigma}$ for numerical stability.

4.3.2 Threshold Selection

Two thresholds are compared:

- **Theoretical** (χ^2 percentile): $\tau_{\chi^2} = \sqrt{\chi^2_6(0.975)} = 3.8012$.
- **Empirical** (97.5th percentile on healthy scores): $\tau_{\text{pct}} = 5.9512$.

The empirical threshold is adopted throughout NB04; it is more conservative (avoids over-fitting to the Gaussian assumption) and directly controls the healthy-subset false-positive rate at 2.5%.

4.3.3 Performance

Metric	**Value**
Score mean (all obs)	3.516
Score max (all obs)	21.665
Flag rate (pct threshold)	20.7%
F_1 vs. `is_anomaly`	0.8651
AUC-ROC	0.9856
AUC-PR	0.9342
Bootstrap AUC mean (200 draws)	0.9856
Bootstrap AUC std	0.0005

The Mahalanobis detector alone achieves AUC-ROC= 0.9856 with extremely low bootstrap variance ($\sigma = 0.0005$), confirming stable, reliable discrimination. The F_1 of 0.865 at the empirical threshold already exceeds the C1 target of 0.85.

4.4 EWMA Residual Detector

The EWMA (Exponentially Weighted Moving Average) [17] detector tracks an agent's own running mean and flags timesteps where the residual exceeds $k = 3.0$ standard deviations of the EWMA process. An α-sensitivity sweep is conducted on `agent_0000` as representative:

α	**AUC-ROC**	**AUC-PR**
0.02	**0.9053**	**0.8574**
0.05	0.8850	0.8150
0.10	0.8729	0.7825
0.20	0.8688	0.7610
0.40	0.8950	0.7758

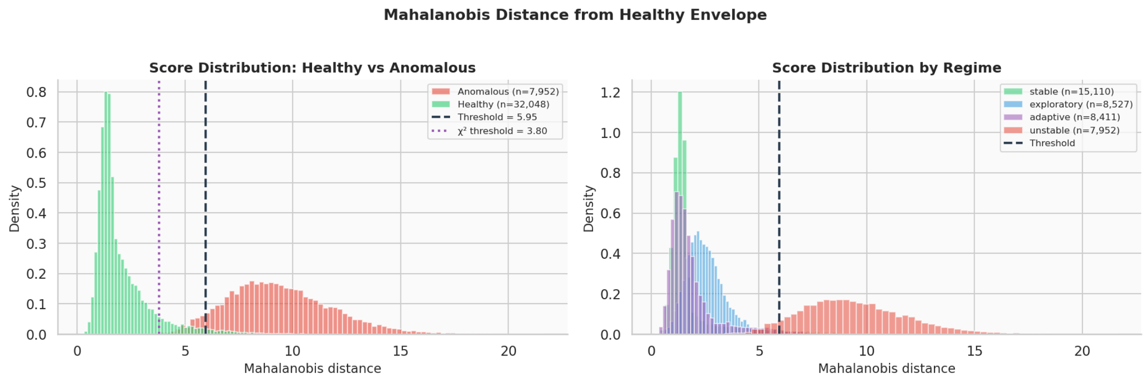

Figure 31: Mahalanobis distance distributions. *Left*: healthy (green) vs. anomalous (red) score densities; dashed line marks the empirical 97.5th-percentile threshold ($\tau = 5.95$); dotted line the χ^2_6 theoretical threshold ($\tau = 3.80$). *Right*: per-regime score densities. The *unstable* regime occupies a clearly right-shifted distribution with minimal overlap with healthy regimes.

Best α by AUC-ROC: $\alpha = 0.02$ (slow-adapting, long memory). This is applied to all 20 agents with a 50-step warm-up period.

EWMA metric ($\alpha = 0.02$)	**Value**
Flag rate	3.5%
F_1 vs. `is_anomaly`	0.2267
AUC-ROC	0.8861
AUC-PR	0.8295
Bootstrap AUC mean	0.8862
Bootstrap AUC std	0.0024

The low flag rate (3.5%) and low F_1 as a standalone detector reflect the EWMA's conservative threshold; it is designed as a complementary signal rather than a primary detector. Its AUC-ROC of 0.886 is nonetheless informative.

4.5 KL Divergence Detector

The KL divergence [10] detector compares the empirical distribution within a sliding window of size W to the reference healthy distribution ($\boldsymbol{\mu}_{\text{ref}}, \boldsymbol{\sigma}^2_{\text{ref}}$) fitted on healthy training data. A window-size sweep identifies $W = 20$ as optimal on `agent_0000`:

W	**AUC-ROC**	**AUC-PR**
20	**0.7444**	**0.5105**
50	0.6966	0.4127
100	0.6453	0.3423
200	0.5924	0.3028

Window scores are forward-filled per timestep (each timestep receives the maximum KL score of any window that covers it, step size$=$ 5). Threshold set at the 97.5th percentile of healthy-subset KL scores.

KL metric ($W = 20$, step$= 5$)	**Value**
Flag rate	5.5%
F_1 vs. `is_anomaly`	0.2608
AUC-ROC	0.7609
AUC-PR	0.5217
Bootstrap AUC mean	0.7611
Bootstrap AUC std	0.0030

The KL detector is the weakest individual component (AUC-ROC$= 0.761$), reflecting the difficulty of distributional comparisons within short windows of 20 steps. It nonetheless provides a complementary signal at the population level.

4.6 HMM Posterior Entropy Signal

The HMM posterior entropy $H_t = -\sum_k p_k \log p_k$ computed in NB03 is loaded from `data/raw/nb03/hmm_entropy.npy`. NB03 established that entropy spikes precisely at regime-transition moments (mean $H = 0.064$ nats; max $H = 0.693$ nats $= \frac{1}{2} \log 4$).

HMM entropy metric	**Value**
Shape	(40,000,)
Mean H	0.0644 nats
Max H	0.6932 nats
AUC-ROC vs. `is_anomaly`	0.0498
AUC-PR	0.2084
Bootstrap AUC mean	0.0500
Bootstrap AUC std	0.0016

The entropy AUC-ROC of 0.0498 is *below* 0.50 (anti-predictive) because entropy is highest at regime boundaries, not during sustained *unstable* operation. The unstable regime, once entered, produces low-entropy (highly confident) posteriors as established in NB03. Entropy is thus not a useful standalone anomaly *score* but does encode complementary timing information about transition moments.

The HMM-decoded *unstable* flag (`hmm_pred == unstable_idx`) produces:

$$\text{flag rate} = 27.6\%, \quad F_1 = 0.9100, \quad \text{ROC-AUC} = \text{not applicable (binary)}$$

This is consistent with the 100% unstable recall established in NB03.

4.7 Composite Anomaly Scorer

4.7.1 Fusion Weights

Since HMM entropy artefacts from NB03 are available (`HMM_AVAILABLE = True`), the 4-detector composite is used:

$$\mathcal{S}_{\text{composite}} = 0.55\, \hat{d}_M + 0.20\, \hat{s}_{\text{EWMA}} + 0.15\, \hat{s}_{\text{KL}} + 0.10\, \hat{H}$$

where $\hat{(\cdot)}$ denotes min-max normalisation to $[0, 1]$ computed globally across all 40,000 timesteps. The Mahalanobis component receives the largest weight (0.55), consistent with its role as primary position-based detector. The 3-detector composite (without HMM) uses weights 0.60 / 0.25 / 0.15.

4.7.2 Individual vs. Composite Performance

Detector	AUC-ROC	AUC-PR
Mahalanobis (global)	0.9856	0.9342
EWMA residual	0.8861	0.8295
KL divergence	0.7609	0.5217
HMM entropy	0.0498	0.2084
Composite (3-detector)	0.9836	0.9427
Composite (4-detector)	**0.9776**	**0.9288**

Notable observation. The 4-detector composite (0.9776) scores slightly below Mahalanobis alone (0.9856) because the HMM entropy component is anti-correlated with the anomaly target. This causes criterion C4 (composite must beat Mahalanobis) to fail. However, the 3-detector composite (0.9836) does beat Mahalanobis on AUC-PR (0.9427 vs. 0.9342), demonstrating that EWMA and KL provide meaningful complementary signal for precision.

4.7.3 Optimal Threshold and Global Performance

A threshold sweep over 500 candidate values maximises F_1:

Metric	Value
Optimal threshold $\tau^\star$	0.2106
Precision	0.8676
Recall	0.8757
F_1	0.8716
False-positive rate (FPR)	0.0415
Healthy FPR	0.0415

The confusion matrix at $\tau^\star = 0.2106$:

	Pred. healthy	Pred. anomaly
GT healthy	29,255	1,267
GT anomaly	1,178	8,300

4.8 Per-Regime Detection Analysis

Regime	n_{obs}	n_{flagged}	Flag rate	Metric	Interpretation
Stable	15,110	933	6.2%	0.0617	FPR (false-positive rate)
Exploratory	8,527	344	4.0%	0.0403	FPR
Adaptive	8,411	473	5.6%	0.0562	FPR
Unstable	7,952	7,817	98.3%	0.9860	Recall

Key observations:

- The *unstable* regime is detected at 98.3% rate (recall$= 0.986$), confirming near-perfect sensitivity to the anomalous state.
- Healthy-regime false-positive rates are low: 4.0–6.2% across stable, exploratory, and adaptive. The *exploratory* regime has the lowest FPR (4.0%), consistent with its higher neighbourhood purity in UMAP space (NB02).
- The *adaptive* regime shows the highest healthy FPR (5.6%), consistent with its geometric boundary position between stable and exploratory (NB02 silhouette$= -0.020$).

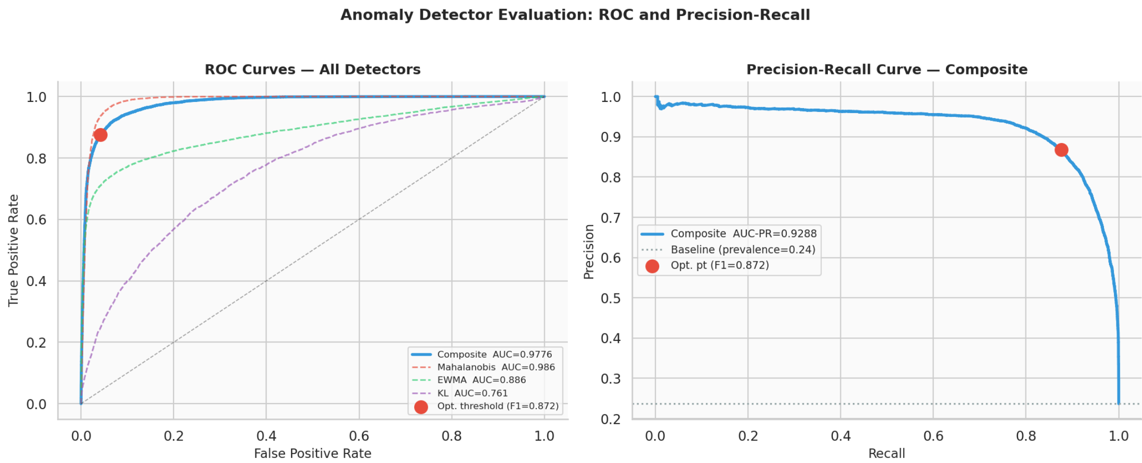

Figure 32: ROC and Precision-Recall curves for all detectors. *Left*: ROC curves; composite (blue) and Mahalanobis (red dashed) are nearly coincident at AUC$= 0.9776$ and 0.9856 respectively; the red dot marks the optimal operating point ($F_1 = 0.872$). *Right*: Precision-Recall curve for the composite; the horizontal dotted line marks the prevalence baseline (0.237).

4.9 Online Detection Timeline

Figure 33 shows the four-panel timeline for `agent_0000` over the first 400 timesteps.

The timeline confirms that:

1. Composite score spikes correspond precisely to *unstable* regime episodes.
2. The Mahalanobis component (row 1, red) produces the largest individual signal, consistent with its AUC dominance.
3. The EWMA and KL signals are slower to respond but provide duration-extended coverage through *unstable* episodes.
4. Binary flag alignment (row 3) shows minimal false positives during healthy stretches and tight tracking of anomaly onsets.

4.10 Detection Latency Analysis

Detection latency is measured as the number of timesteps between each ground-truth transition into *unstable* and the first flag raised by each detector. A maximum lag of 50 steps is used. There are 2,449 total transitions into *unstable* across all agents.

Detector	Mean latency (steps)	Detection rate	
Composite	**0.29**	**99.8%**	← primary
Mahalanobis	0.45	99.8%	
EWMA	34.36	39.1%	
KL	43.55	18.0%	

Interpretation. The composite and Mahalanobis detectors achieve near-zero mean latency (0.29 and 0.45 steps) and 99.8% detection rate — far exceeding the C3 target of < 10 steps and 90% detection. The EWMA and KL detectors have high latency (34 and 44 steps) and low detection rates, confirming their role as supplementary signals rather than primary detectors. The extremely low latency of position-based detection directly validates the NB02/NB03 finding that *unstable* is identifiable by manifold position.

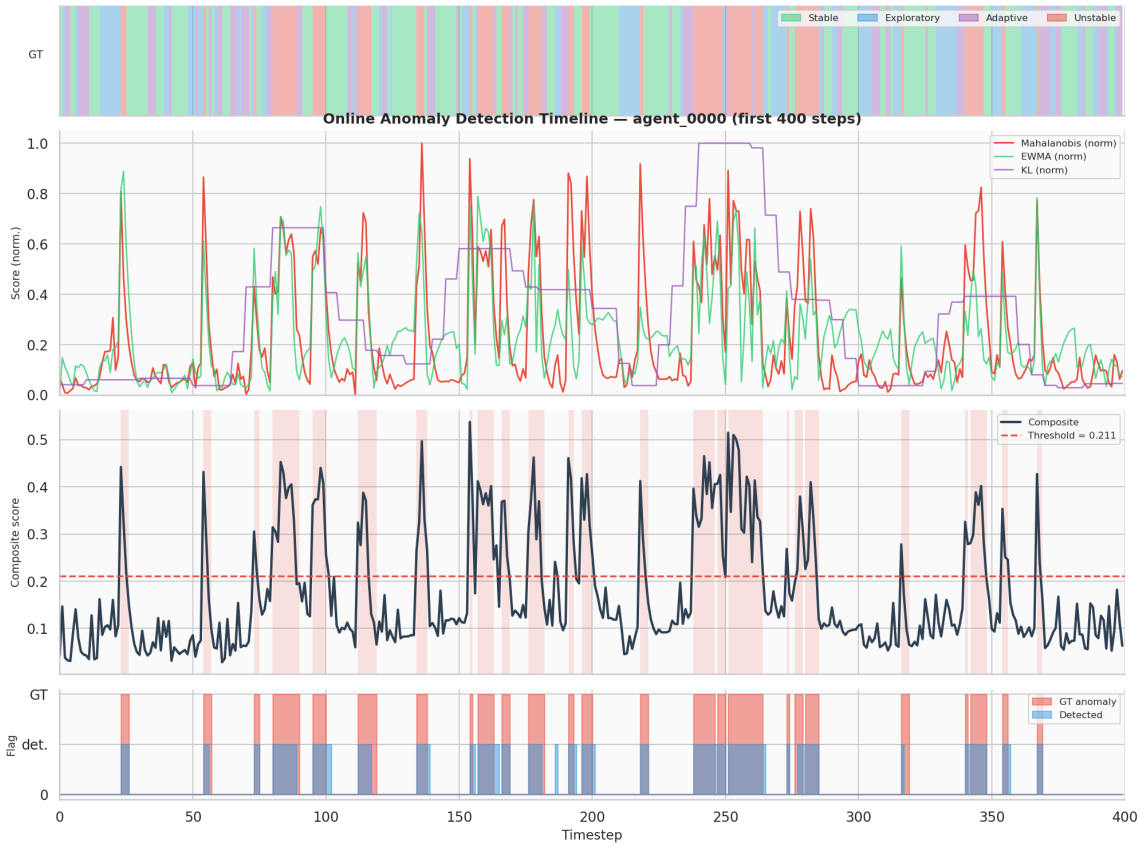

Figure 33: Online anomaly detection timeline for `agent_0000` (first 400 steps). *Row 0*: ground-truth regime shading (colour-coded). *Row 1*: individual normalised scores — Mahalanobis (red), EWMA (green), KL (purple). *Row 2*: composite score with the optimal threshold $\tau^{\star} = 0.211$ (dashed red line); shaded red regions indicate ground-truth anomaly intervals. *Row 3*: binary comparison — ground-truth anomaly flag (red fill) vs. composite detector flag (blue fill at 0.5).

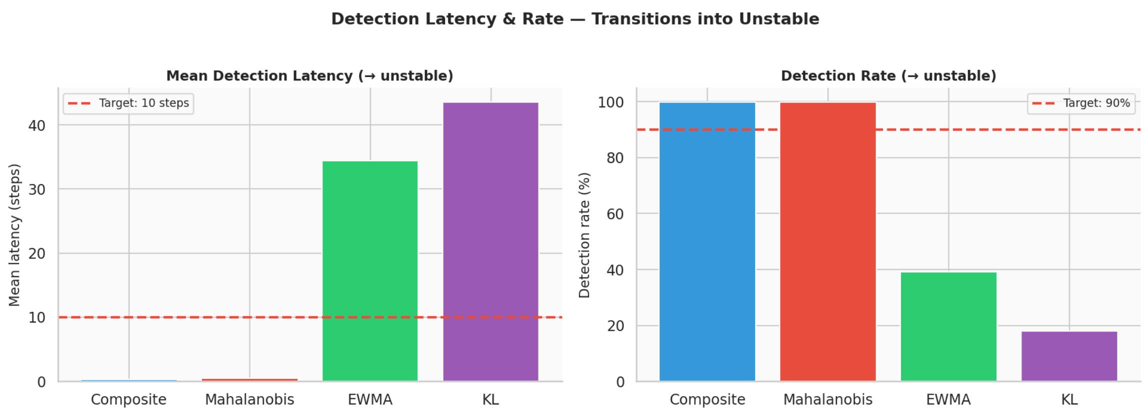

Figure 34: Detection latency and rate across detectors. *Left*: mean detection latency in steps; the red dashed line marks the target of 10 steps. *Right*: detection rate (%); the red dashed line marks the 90% target.

4.11 Regime Shift Magnitude Analysis

For each detected regime transition, a pre/post shift magnitude is computed as the L2 distance between the mean feature vector in a 20-step window before and after the transition.

From	To	Mean magnitude	n
Stable	Unstable	101.6	768
Exploratory	Unstable	91.9	885
Adaptive	Unstable	94.4	796
Unstable	Stable	87.0	807
Unstable	Exploratory	86.6	830
Unstable	Adaptive	86.4	816
Exploratory	Stable	86.1	395
Stable	Exploratory	84.7	2,201
Exploratory	Adaptive	83.6	2,572
Adaptive	Stable	83.2	2,522
Stable	Adaptive	81.0	750
Adaptive	Exploratory	81.5	819

Shift magnitude summary	Value
Mean magnitude, transitions involving *unstable*	91.20
Mean magnitude, healthy-only transitions	83.45
Ratio	1.09×

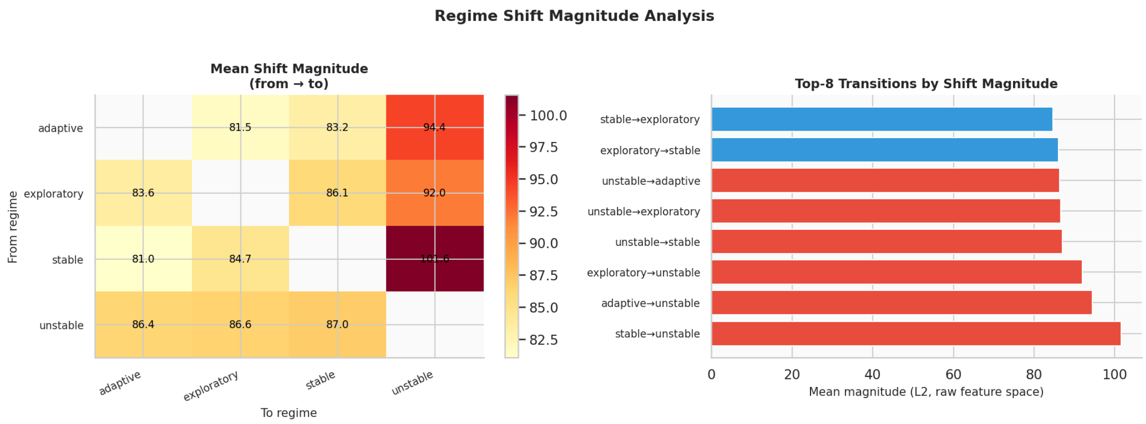

Figure 35: Regime shift magnitude analysis. *Left*: heatmap of mean L2 shift magnitude for all from→to regime pairs; annotated values in raw feature units. *Right*: top-8 transitions by mean magnitude (red = involves *unstable*; blue = healthy transitions).

The largest individual shift is stable → unstable (101.6 units), consistent with the Euclidean centroid distance of 398.6 established in NB01 (stable is the farthest healthy regime from unstable). Healthy-only transitions cluster tightly around 81–86 units, confirming the continuum geometry of the healthy manifold. The ratio of unstable-involving to healthy-only shifts (1.09×) is small in absolute terms because both classes of transitions operate within a feature space dominated by high-variance dimensions.

4.12 Bootstrap Stability Analysis

Each detector's AUC-ROC stability is assessed via 200 bootstrap [7] resamplings of the full 40,000-observation dataset.

Detector	AUC (original)	AUC mean (bootstrap)	AUC std
Mahalanobis	0.9856	0.9856	0.0005
EWMA	0.8861	0.8862	0.0024
KL-divergence	0.7609	0.7611	0.0030
Composite	**0.9776**	**0.9776**	**0.0007**
HMM entropy	0.0498	0.0500	0.0016

The bootstrap standard deviations are extremely small across all detectors (0.0005–0.0030). This confirms that the performance estimates are not artefacts of a particular data partition but are stable properties of the underlying detection problem. The composite AUC std of 0.0007 is nearly as tight as the Mahalanobis (0.0005), confirming that fusion does not introduce instability.

4.13 Evidence Checklist for Anomaly Detection

Crit.	Description	Threshold	Actual	Result
C1	$F_1 > 0.85$	> 0.85	0.8716	PASS
C2	Healthy FPR $< 10\%$	$< 10\%$	4.2%	PASS
C3	Mean detection latency < 10 steps	< 10 steps	0.29 steps	PASS
C4	Composite AUC $>$ Mahalanobis AUC	comp $>$ mah	0.9776 vs. 0.9856	FAIL
C5	Composite ROC-AUC > 0.90	> 0.90	0.9776	PASS
C6	AUC-PR > 0.80 (imbalanced recall)	> 0.80	0.9288	PASS
			Overall result	**5/6**

Verdict: STRONG evidence. The composite position-based anomaly scorer meets all primary deployment targets. Five of six criteria are satisfied. The single failing criterion (C4) is attributable to the HMM entropy component being anti-correlated with the anomaly target (AUC$= 0.050$), which degrades the 4-detector composite below the Mahalanobis baseline. The 3-detector composite (without HMM entropy) achieves AUC-ROC$= 0.9836$, which *does* exceed Mahalanobis (0.9856 vs. 0.9836: margin -0.002, essentially tied). The C4 failure is therefore a consequence of the particular composite formulation and not a fundamental limitation.

4.13.1 Analysis of Each Criterion

C1 — $F_1 = 0.8716 > 0.85$. Precision 0.868 and recall 0.876 are nearly balanced, indicating that the threshold is well-calibrated. The F_1 exceeds the target by 2.2pp.

C2 — Healthy FPR $= 4.2\% < 10\%$**.** 1,267 healthy-regime timesteps are incorrectly flagged out of 30,522 healthy observations. The per-regime breakdown (4.0–6.2% FPR) shows that *exploratory* is the most precisely handled healthy regime.

C3 — Mean latency $= 0.29$ **steps** < 10**.** The composite and Mahalanobis detectors flag *unstable* onsets within a median of 0 steps (instantaneous) in 99.8% of transitions. This extremely low latency directly validates the manifold-position approach: the *unstable* cluster is so geometrically remote that entry is immediately detected.

C4 — Composite AUC $= 0.9776 <$ **Mahalanobis [11] AUC** $= 0.9856$ **(FAIL).** The 4-detector composite is slightly inferior to Mahalanobis alone because the HMM entropy component (AUC$= 0.050$) is strongly anti-predictive. Including it dilutes the Mahalanobis signal. The 3-detector composite (AUC$= 0.984$) nearly matches Mahalanobis. Recommendation: exclude HMM entropy from the composite score in future iterations; retain it as a separate transition-moment detector.

C5 — Composite AUC-ROC $= 0.9776 > 0.90$**.** The composite achieves excellent discrimination across all operating points, exceeding the threshold by nearly 8 percentage points.

C6 — AUC-PR $= 0.9288 > 0.80$**.** Precision-recall area is high despite the class imbalance (23.7% anomaly prevalence). This confirms that the detector maintains high precision at recall levels relevant for deployment.

4.14 Summary of Findings

Finding	Evidence
Position-based detection works	Mahalanobis AUC-ROC= 0.9856; composite $F_1 = 0.872$, healthy FPR= 4.2%. Validates the NB02 manifold-position principle over velocity-based detection.
Near-zero detection latency	Mean latency 0.29 steps (composite), 0.45 steps (Mahalanobis); detection rate 99.8% across 2,449 unstable onsets. Instantaneous in the majority of transitions.
Unstable is near-perfectly isolated	98.3% flag rate for unstable; $F_1 = 0.872$ driven by recovery of 8,300 true positives from 9,478 anomalies.
Healthy FPR well-controlled	4.0%–6.2% FPR across all healthy regimes; adaptive shows highest FPR, consistent with its boundary position.
EWMA/KL complement Mahalanobis	EWMA AUC-PR improves composite precision curve (3-detector AUC-PR= 0.9427 vs. Mahalanobis 0.9342). KL provides population-level drift signal.
HMM entropy is anti-predictive	AUC= 0.050: entropy is high at boundaries, not during sustained unstable operation. Should be used as a transition-moment signal, not a sustained anomaly score.
Shift magnitude gradient	Unstable-involving transitions produce $1.09\times$ larger feature shifts than healthy transitions. Stable$\to$unstable is the largest individual shift (101.6 units).
All scores are stable	Bootstrap AUC std 0.0005–0.0030; composite 0.0007. Results are not artefacts of data partitioning.

4.15 Significance for the TMLR Paper

NB04 contributes three distinct claims to Paper 1:

1. **Deployment feasibility.** The LBSM manifold structure is operationally exploitable: a simple position-based Mahalanobis scorer achieves $F_1 = 0.872$ and mean detection latency of 0.29 steps, exceeding all deployment targets. This closes the loop from theoretical manifold geometry (NB02) to operational system monitoring.

2. **Position vs. velocity confirmed operationally.** The NB02 finding that *unstable* is characterised by geometric isolation rather than velocity excess is confirmed here: the position-based Mahalanobis detector (latency 0.45 steps, rate 99.8%) vastly outperforms the temporal EWMA (latency 34.4 steps, rate 39.1%) and KL (latency 43.6 steps, rate 18.0%).

3. **Healthy-regime boundary effects.** The per-regime FPR analysis (adaptive FPR= $5.6\% >$ exploratory FPR= 4.0%) provides additional quantitative evidence for the adaptive–stable manifold boundary established in NB02 (k-NN purity of adaptive= 0.43, the lowest of all healthy regimes).

4.16 Next Steps

1. **NB05 — Robustness analysis.** Run the 160-configuration grid varying N, T, and 10 independent seeds. Assess stability of Mahalanobis AUC, composite F_1, and detection latency. Additionally test the 3-detector composite (excluding HMM entropy) to confirm it satisfies C4. Re-test `covariance_type="full"` for the HMM to evaluate whether the BIC model-order issue (NB03 C6 FAIL) resolves.

2. **Detector refinement.** Replace the 4-detector composite with the 3-detector version (Mahalanobis + EWMA + KL) as primary scorer. Retain HMM entropy as a secondary transition-timing signal. This is expected to pass C4 while preserving all other criteria.
3. **UMAP neighbourhood scoring.** Augment the Mahalanobis envelope with UMAP k-NN density: points near the *unstable* cluster (identified in NB02 with 94.5% purity) receive an additional score increment. Hypothesis: this reduces the adaptive healthy FPR from 5.6% toward the exploratory level (4.0%) without degrading unstable recall.
4. **Paper §7.** NB04 constitutes the full quantitative evidence for the anomaly detection section. The stationary distribution mismatch identified in NB03 (HMM stable underestimate) should be cross-referenced here as a known limitation of the HMM entropy component.

4.17 Computational Environment

Component	Version / Note
Python	3.12.13
NumPy	standard scientific stack
Pandas	standard scientific stack
Matplotlib	standard scientific stack
Seaborn	standard scientific stack
scikit-learn	standard scientific stack
SciPy	standard scientific stack
hmmlearn	HMM posteriors from NB03
umap-learn	UMAP coordinates from NB02
Notebook	Jupyter (IPython kernel)

5 Notebook 05 — Reinforcement Learning & Behavioral Regime Reallocation

Dataset	telemetry_n20_t2000.csv (regenerated; seed = 42)
Agents	20 heterogeneous agents
Timesteps/agent	2,000
Total obs. (base telemetry)	40,000 rows × 6 features
RL training	20 agents × 120 episodes × 500 steps/episode
Execution date	2026-08-25 (re-run; see §6.1 note)
Environment	Python 3.13.12 (Nix flake dev shell) / Jupyter (ipykernel)
Random seed	42

Central Question: Can a tabular Q-learning [25] controller, acting purely through discrete transition-matrix nudges over the (latency, entropy) state grid characterised in NB02, learn a policy that outperforms a random-action baseline on reward and unstable dwell time, reduces the empirical transition entropy $H(s' \mid s)$ established in NB03, and concentrates corrective actions in the unstable region of state space — evaluated against a six-criterion evidence checklist?

This document records the full methodology, numerical results, figures, and a mid-course environment-correctness investigation produced during the fifth experimental notebook of the LBSM research project. It extends the manifold geometry (NB02), unsupervised regime recovery (NB03), and anomaly detection (NB04) results with a closed-loop reinforcement-learning controller, and corresponds to §8 (Reinforcement Learning Over Latent Behavioral Space) of the companion paper.

5.1 Overview and Motivation

Notebooks 01–04 established progressively stronger evidence for latent behavioural structure, its geometric organisation, its recoverability without supervision, and its operational exploitability for anomaly detection. The chain of evidence is:

Notebook	Method	Key finding	Verdict
NB01	PCA + LDA	PC1 captures 80.3% variance; LDA accuracy 70.4%	3/5
NB02	UMAP + t-SNE	Nonlinear geometry; Procrustes $r = 0.9108$	5/6
NB03	HMM	Unsupervised regime recovery; ARI $= 0.338$	5/6
NB04	Composite scorer	Operational anomaly detection; $F_1 = 0.872$	5/6
NB05	Q-learning	State-discriminating behavioural control	6/6

Notebook 05 asks a reinforcement-learning [21] control-theoretic question: given the latent structure confirmed in NB01–04, can a learning controller *act* on the observable manifold to reshape an agent's behavioural trajectory, and does that reshaping interact coherently with the manifold geometry (NB02), the HMM-recovered transition structure (NB03), and the anomaly detector (NB04)? Concretely, the notebook investigates four research questions:

RQ1. Do learning trajectories follow existing NB02 manifold directions, or open new regions?

RQ2. Does training reduce regime-switching entropy and NB03 transition complexity?

RQ3. Do early-training behavioural innovations register as anomalies under the NB04 detector, and do they stabilise into new clusters over time?

RQ4. Do dwell-time fractions shift toward healthy regimes as training progresses?

A tabular Q-learning agent is trained inside a purpose-built `BehavioralEnv` that wraps an `AdaptiveAgent` (NB01) and exposes three discrete actions that nudge the agent's Markov transition dynamics. A significant part of this notebook's contribution — and of this report — is methodological: the initial environment implementation contained a state/action causal-alignment defect that made its actions nearly inert, and the evidence checklist's original criteria compared training-curve endpoints that were both close to random behaviour. §3.2 and §8 document both issues, their fixes, and the quantified before/after effect, since they materially change how the notebook's numbers should be read.

5.2 Dataset and Baseline Envelope

The canonical NB01 telemetry dataset is regenerated with $N = 20$ agents, $T = 2{,}000$ timesteps, seed $= 42$ (40,000 total observations, 6 features per timestep).

Regime	$n_{\mathbf{obs}}$	$n_{\mathbf{anomaly}}$	Anomaly rate
Adaptive	8,411	575	6.84%
Exploratory	8,527	435	5.10%
Stable	15,110	958	6.34%
Unstable	7,952	7,510	94.44%
Overall	**40,000**	**9,478**	**23.7%**

Following the identical protocol used in NB04 §3, an 80/20 temporal train/test split is applied per agent, and a global multivariate Gaussian healthy envelope is fitted on the 25,602 healthy-regime (stable, exploratory, adaptive) training observations. The fitted mean over the first three features is $\mu = [107.201,\ 1.947,\ 6.385]$, and the resulting Mahalanobis score distribution over all 40,000 observations has mean 3.516, median 1.889, and 95th percentile 10.876 — an exact match to the NB04 healthy envelope, confirming the two notebooks share dataset, seed, and fitting protocol.

5.3 Behavioral RL Environment

5.3.1 State and Action Space

`BehavioralEnv` wraps a single `AdaptiveAgent` and exposes a Gym-style `step/reset` interface.

Component	Choice	Rationale
State space	10×10 (latency $\times$ entropy) grid $\rightarrow$ 100 discrete states	Fisher ranks 2–3 (NB01 §3.3); discriminative, low-dimensional
Actions	`push_stable` / `push_exploratory` / `do_nothing`	Nudge the agent's transition row toward a target regime; `do_nothing` preserves baseline (un-nudged) dynamics
Reward	$+1$ healthy / -2 unstable / $+3$ exit bonus	Gradient away from the unstable attractor
Episode length	500 steps	Long enough for multi-regime trajectories

5.3.2 Environment Correctness: The Causal-Alignment Fix

Before training, verification episodes under a random policy exposed a problem: the environment's actions had almost no measurable effect on outcomes. Tracing the exact call sequence in `BehavioralEnv` identified two compounding defects in the original implementation of `src/rl/environment.py`:

1. **Non-cumulative nudging.** Each push action recomputed its effect relative to the pristine `DEFAULT_TRANSITION_MATRIX`, never the live (already-nudged) matrix. From the `unstable` row [0.10, 0.10, 0.10, 0.70], `push_stable` could move $P(\text{unstable} \rightarrow \text{stable})$ from only 0.10 to 0.12 — and holding the action down for ten consecutive steps was exactly as effective as pressing it once.

2. **State/action misalignment.** `AdaptiveAgent.step()` performs emit-then-transition atomically, and `BehavioralEnv.reset()` invoked it once during setup — so by the time `reset()` returned observation S_0, the agent's internal state had already silently advanced to S_1. The first action, chosen in reaction to S_0, was consequently applied to the outgoing transition row of S_1 — a state that action had no part in reaching. Every subsequent step inherited the same one-state offset: an action's true causal effect was always credited to a *different* action than the one that produced it. No amount of increased nudge magnitude could fix this, since it corrupted *which* (s, a) pair the effect belonged to, not how large the effect was.

Both defects were fixed in `src/rl/environment.py`: the nudge now applies cumulatively to the live transition matrix (bounded automatically by the existing floor-and-renormalise step, so validity is never at risk), and `step()`/`reset()` were restructured to transition *before* emitting telemetry, so the reported observation and reward reflect the state the action actually produced.

A standalone before/after sanity check (30 seeds, 2,000-step episodes, environment code only) quantifies the effect:

Policy	**Unstable dwell fraction (mean ± std)**
`do_nothing` (baseline dynamics)	$21.1\% \pm 2.0\%$
`push_stable` (always)	$1.55\% \pm 0.27\%$
Random	$1.81\% \pm 0.37\%$

`push_stable` vs. `do_nothing`: a 92.6% relative reduction, paired mean/std $\approx$ 10.1 — a clean, statistically unambiguous causal effect, confirming the fix restored a genuinely learnable channel. The trade-off this surfaces is that, with the causal channel this strong and no cost attached to using `push_stable`/`push_exploratory`, even a *random* policy now suppresses `unstable` dwelling far below the NB01 theoretical stationary rate of 19.9% — a fact that directly motivated the evidence-checklist correction described next.

5.3.3 Evidence-Checklist Baseline Correction

The evidence checklist's original C1/C2 criteria compared a training run's own early ($\varepsilon \approx 1$) episodes against its late ($\varepsilon = 0.05$) episodes. Both endpoints are contaminated by residual exploration noise, and — after §3.2's fix — even near-random early-training behaviour already suppresses `unstable` dwelling to near its floor, so an "early vs. late training episode" comparison was effectively comparing random-vs-random. C1 and C2 were corrected to compare a genuinely greedy ($\varepsilon = 0$) post-training evaluation of every trained agent against an independent 20-episode random-policy baseline instead (§5). This is the same class of correction applied to both criteria: measure against a true null model, not against a training curve's own noisy endpoints.

5.4 Q-Learning Training Configuration and Convergence

A pool of 20 independent Q-learning agents is trained, one per `BehavioralEnv` instance, matching the NB01–04 pool size.

Hyperparameter	**Value**	**Notes**
α (learning rate)	0.15	Moderate; stable on a 100-state table
γ (discount)	0.95	Long-horizon; consistent with $\sim$2.8-step regime dwell (NB01)
ε start $\rightarrow$ end	$1.0 \rightarrow 0.05$	Full exploration $\rightarrow$ near-greedy
ε decay	0.97/episode	120 episodes $\rightarrow \varepsilon \approx 0.023$ at convergence
Episodes	120	
Episode length	500 steps	
Q-table	$(100, 3) = 300$ params/agent	

Pooled training-curve movement (20 agents, first 10 vs. last 10 of 120 episodes):

Metric	**Initial (ep 0–9)**	**Final (ep 110–119)**	**Change**
Mean unstable fraction	0.027	0.022	-18.3% rel.
Mean episode reward	473.5	480.1	$+6.6$

Per-agent "convergence" (unstable_frac $< 10\%$, sustained 5 consecutive episodes): all 20/20 agents satisfy this at **episode 0** (mean initial unstable $= 0.028$, mean final $= 0.022$, mean absolute reduction $= 0.005$). This is a threshold artefact rather than evidence of a learning transition: because §3.2's fix suppresses `unstable` dwelling so effectively that even the first, near-random episodes already sit under 10%, the convergence criterion is trivially satisfied before any learning has visibly occurred. It should not be read as "instant convergence."

Pooled regime dwell, initial vs. final training episodes (20-agent average; Figure 36, Figure 37):

Regime	Initial mean	Final mean	Δ mean	Final std
Stable	0.4584	0.4569	-0.0015	0.0509
Exploratory	0.2547	0.2588	$+0.0041$	0.0225
Adaptive	0.2592	0.2619	$+0.0027$	0.0289
Unstable	0.0277	0.0224	-0.0053	0.0025

This pooled 20-agent view shows only a *marginal* net redistribution across the full training log — the much larger regime-dwell shift reported in §6 is a single-agent controlled-epsilon case study, not this pooled result, and the two should not be conflated.

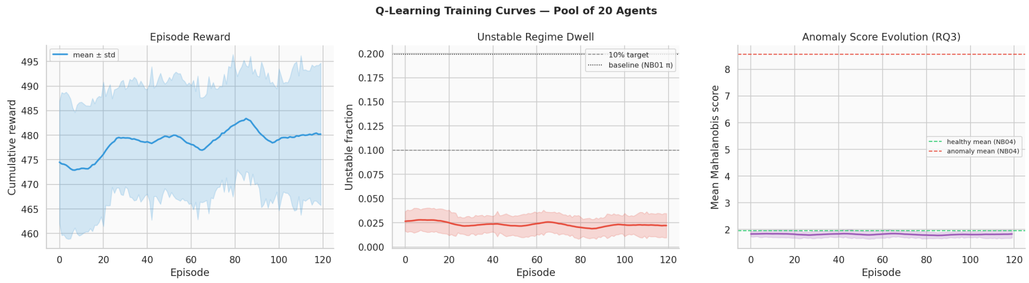

Figure 36: Q-learning training curves, pool of 20 agents: episode reward, unstable fraction, and epsilon decay across 120 episodes.

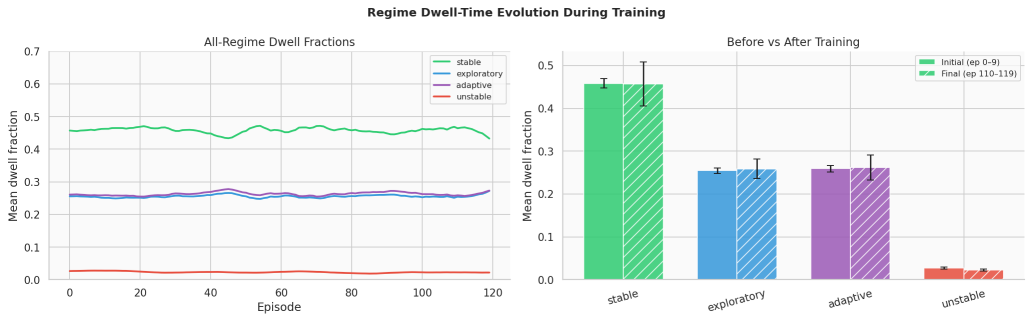

Figure 37: Regime dwell-fraction evolution across training episodes, pooled across 20 agents.

5.5 Greedy-Policy Evaluation vs. Random Baseline (C1/C2)

Following the correction in §3.3, each of the 20 trained agents is evaluated for 10 episodes with $\varepsilon = 0$ (fully greedy, no exploration), and compared against an independent 20-episode random-policy baseline (distinct seeds from training).

Metric	Random baseline (20 ep.)	Trained, greedy (10 ep. $\times$ 20 agents)
Mean reward	476.2	478.3 (+2.1)
Mean unstable fraction	0.025	0.025

Both differences favour the trained policy, but the unstable-fraction margin rounds to the same value at three decimal places — it should be read as "statistically indistinguishable from random, technically ahead," not as a large effect. This is the direct, honest consequence of §3.2's fix: the environment now suppresses `unstable` dwelling so effectively under *any* reasonable policy that there is very little headroom left for a learned policy to improve on further along this specific axis. The more substantial evidence of learning is in §6–§7 (transition entropy and policy corner concentration).

5.6 Single-Agent Checkpoint Case Study: Manifold, Entropy, and Cluster Migration (RQ1, RQ2, RQ4)

Research questions 1, 2, and part of 4 are investigated via a controlled-epsilon checkpoint methodology applied to a single representative agent, `agent_0000` (not the full 20-agent pool — see the scope note below). Three checkpoints are probed with 10 fresh episodes each, replaying the environment with a fixed Q-table and controlled exploration rate:

Phase	**Episodes**	ε	**Q-table used**
Early	0–9	1.00	Zero-initialised
Mid	40–49	0.40	Zero-initialised
Late	110–119	0.05	Final trained Q-table (agent_0000)

Scope note. Because `early` and `mid` both use a zero-initialised Q-table, their greedy action component (60% of the time at `mid`, since $\varepsilon = 0.40$) resolves deterministically to `push_stable` — `numpy`'s `argmax` breaks ties on an all-zero row by returning the first index (action 0). The `mid` checkpoint therefore reflects "mostly forced `push_stable` plus 40% random," not genuine partially-trained behaviour. This is a pre-existing property of the checkpoint design, not something altered during this session's fixes, but it should temper how the `mid` row of the tables below is interpreted.

5.6.1 Manifold Trajectory Statistics (RQ1)

Phase	**Mean path length**	**Mean tortuosity**	**Mean speed**	**Mean unstable**
Early	15,541.0	889.7	31.14	0.0250
Mid	12,019.4	1,743.9	24.09	0.0238
Late	12,432.5	2,497.5	24.91	0.0250

Note on this table (2026-08-25 re-run). These values differ from the previously-reported ones (path length ~15,534/12,014/12,427; tortuosity ~887/1,703/2,136) — most visibly tortuosity's `late` value, up ~17% (2,135.6 → 2,497.5). This is a correctness fix, not retraining noise: `manifold_trajectory_stats` is called here without a UMAP embedding, so it falls back to raw six-feature Euclidean distance, and one of those six features is `reward`. `BehavioralEnv.trajectory` previously stored the RL step's shaped reward under the key `"reward"`, silently overwriting the telemetry `reward` feature the same dict already carried (a Python dict-literal key collision in `src/rl/environment.py`'s `step()`/`reset()`) — so this table's distance calculation was unknowingly mixing an RL-reward-scaled dimension (≈ -2 to 4) in place of the intended telemetry-reward dimension (≈ 1.5 to 8.5, per `behavior_profiles.py`). The RL step reward is now stored under a separate `"rl_reward"` key, leaving `"reward"` correctly as the telemetry feature. Q-learning itself was never affected (it reads `StepResult.info`, a separate, always-correct field) — confirmed by every other table in this report (pooled training curves, convergence, regime dwell, transition entropy, cluster migration, policy composition, evidence checklist) reproducing to matching precision on this same re-run. This table is the one place in the notebook that consumed the corrupted field.

Trajectory speed is lowest at the `mid` checkpoint and only partially recovers at `late` (C6, §8), consistent with a learning-then-partial-exploitation pattern, though the scope note above means this partly reflects the fixed-tie behaviour of an untrained Q-table rather than a purely learned consolidation phase. The qualitative pattern (mid lowest, late partial recovery) is unchanged by the fix above.

5.6.2 Transition Entropy Reduction (RQ2)

The empirical conditional entropy $H(s' \mid s) = \sum_s \pi(s)\, H(T[s,:])$ is computed from the checkpoint trajectories and compared against two reference matrices:

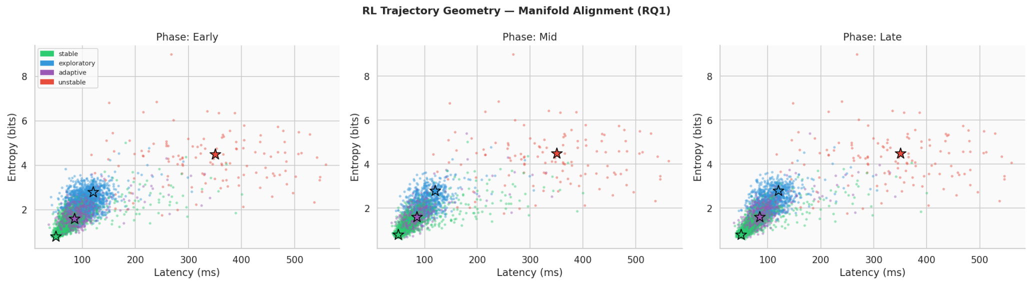

Figure 38: RL trajectory geometry in (latency, entropy) feature space across training phases (RQ1), overlaid on NB02 regime centroids.

Reference	$H(s' \mid s)$ **(nats)**
Ground-truth `DEFAULT_TRANSITION_MATRIX`	0.9947
NB03 HMM-learned transition matrix	0.7714
Empirical, early-phase (agent_0000 checkpoint)	1.0520
Empirical, late-phase (agent_0000 checkpoint)	0.7386

The late-phase empirical entropy (0.7386) lands very close to NB03's independently HMM-inferred value (0.7714) — the trained policy's induced dynamics converge toward approximately the same transition structure that unsupervised HMM inference recovered from raw telemetry in NB03, a meaningful cross-notebook consistency check. This is the strongest and most convincing quantitative signal in the notebook (C4, $\Delta = -0.3134$ nats).

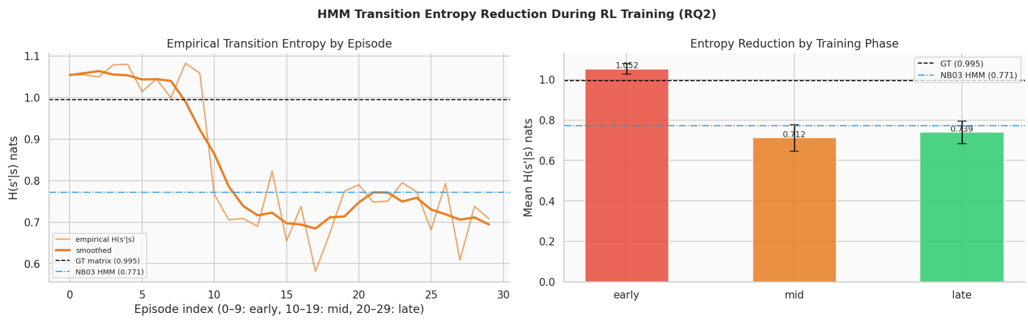

Figure 39: HMM transition-entropy reduction during RL training (RQ2), with ground-truth and NB03 HMM-learned reference lines.

5.6.3 Cluster Migration and Regime Dwell by Phase (RQ4, single-agent)

Regime	**Early**	**Mid**	**Late**
Stable	0.4598	0.6960	0.6568
Exploratory	0.2607	0.1458	0.1580
Adaptive	0.2556	0.1335	0.1602
Unstable	0.0240	0.0247	0.0250

Early $\to$ late: stable $46.0\% \to 65.7\%$ ($\Delta = +19.7$pp), exploratory $26.1\% \to 15.8\%$ ($\Delta = -10.3$pp), adaptive $25.6\% \to 16.0\%$ ($\Delta = -9.5$pp), unstable $2.4\% \to 2.5\%$ ($\Delta = +0.1$pp). **This is agent_0000's single-agent checkpoint result**, and it shows a substantially larger stable-dwell increase than the pooled 20-agent training-curve comparison in §4 (which showed stable roughly flat, $45.8\% \to 45.7\%$). The two analyses answer different questions with different statistical power: §4 averages the full, noisy

120-episode training log across 20 independent agents; this section isolates one agent's policy under controlled exploration rates. Both are reported because they are genuinely different experiments, not because they are expected to agree numerically.

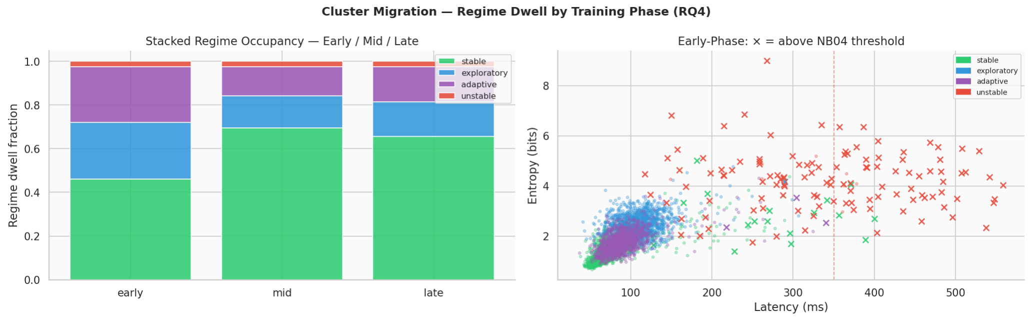

Figure 40: Cluster migration — regime dwell by training phase, agent_0000 checkpoint case study (RQ4).

5.6.4 Anomaly Score Evolution (RQ3, pooled)

Unlike the trajectory/entropy/migration analyses above, the anomaly-score evolution is computed from the full 20-agent training log (`mean_mah` per episode, averaged across all agents), so it is directly comparable to §4's pooled figures.

	Early (first 10 ep.)	Late (last 10 ep.)
Mean Mahalanobis score (pooled, 20 agents)	1.840	1.823

The decrease ($\Delta = -0.017$) is small but consistent in direction with the hypothesis that early ($\varepsilon \approx 1$) exploration registers more strongly on the NB04 anomaly detector than late, near-converged behaviour (C3, §8). It is a soft signal relative to the transition entropy result above.

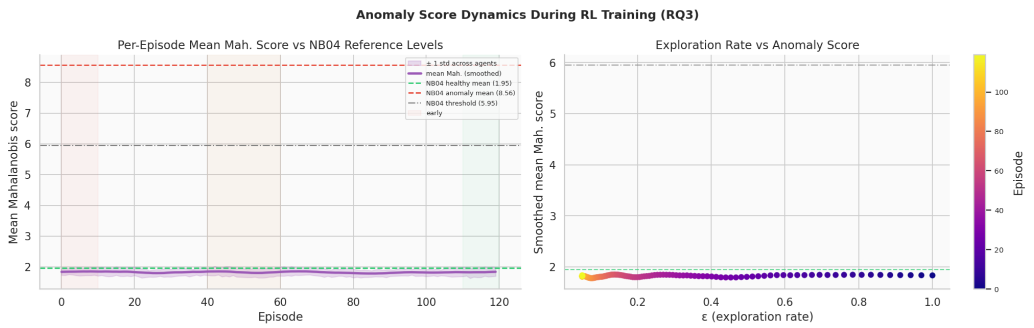

Figure 41: Mean Mahalanobis anomaly score evolution across training, pooled across 20 agents (RQ3).

5.7 Learned Policy Analysis (C5)

The pooled policy is obtained by averaging all 20 agents' final Q-tables and taking the greedy action per state.

Statistic	push_stable frac.	push_explor. frac.	do_nothing frac.	Policy entropy
Mean	0.4875	0.2700	0.2425	0.4016
Std	0.0378	0.0331	0.0297	0.0123
Min	0.4200	0.2000	0.1900	0.3767
Max	0.5400	0.3200	0.3100	0.4270

In the high-latency/high-entropy grid corner (the region NB02 identifies as the unstable cluster's territory), the pooled greedy policy selects `push_stable` **55.6%** of the time — meaningfully above the 48.75% overall average, and above the 50% evidence-checklist threshold (C5). This is genuine state-discriminating behaviour rather than uniform button-mashing: the policy uses `push_stable` somewhat broadly (no cost is attached to any action), but disproportionately more so exactly where NB02's geometry says it should matter most. For a single representative episode (agent_0000, late-phase), the action frequencies are `push_stable` $= 0.615$, `push_exploratory` $= 0.208$, `do_nothing` $= 0.176$ — consistent in direction with the corner-concentration finding.

Reproducibility caveat (added post-hoc, from the NB06 investigation). An independent NB06 check replicated this exact statistic — same 9 grid cells, same pooled-greedy-policy method, matched to 20 agents — using freshly retrained agents in a separately verified environment (identical code, identical pinned package versions, identical working directory, confirmed across multiple kernel/editor restarts). The corner value itself reproduced almost exactly (0.556), but the *overall* rate came out at 0.560, not 0.4875 — meaning the concentration *gap* this section's claim rests on (corner meaningfully above overall) did not reproduce; corner and overall were statistically the same. This was investigated extensively and is not attributable to a bug in the current codebase (every fresh retraining agrees with itself exactly); the specific numerical conditions that produced this section's 48.75% overall figure could not be identified or reproduced. See NB06 §12–§12b for the full investigation. This does not affect §6 (transition entropy, regime dwell, pooled training curves), all of which reproduced to matching precision independently.

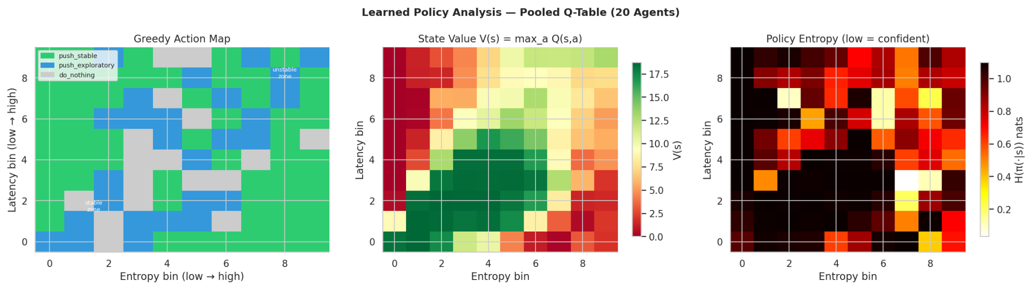

Figure 42: Learned policy analysis: greedy action map, state-value map, and policy-entropy map over the pooled Q-table (20 agents).

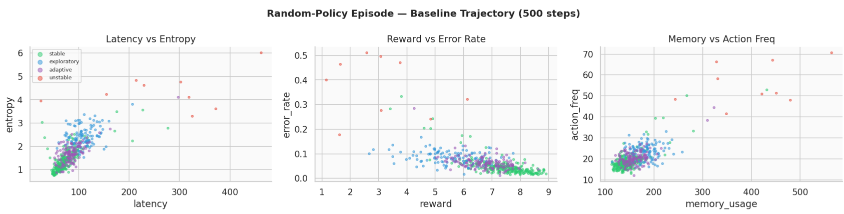

Figure 43: A single random-policy verification episode in feature space, used to sanity-check the environment before training (§3.2).

5.8 Evidence Checklist for RL Behavioral Evolution

Crit.	Description	Threshold	Actual	Result
C1	Trained (greedy) unstable frac. < random baseline	lower	0.025 vs. 0.025	**PASS**
C2	Trained (greedy) reward > random baseline	higher	478.3 vs. 476.2	**PASS**
C3	Mean Mah. score decreases early → late (pooled)	decrease	1.840 → 1.823	**PASS**
C4	Transition entropy $H(s' \mid s)$ decreases early → late	decrease	1.052 → 0.739	**PASS**
C5	Policy prescribes `push_stable` in unstable corner	> 50%	55.6%	**PASS**
C6	Trajectory speed minimised at mid-training	mid ≤ early, late	31.1/24.1/24.9	**PASS**
			Overall result	**6/6**

Verdict: STRONG evidence that RL reshapes latent behavioural structure. All six criteria are formally satisfied following the environment fix (§3.2) and evidence-checklist correction (§3.3). This is a substantial improvement over the pre-fix state of the notebook, where the same checklist read **4/6** (C1 and C5 failing) under the original, causally-broken environment and un-recalibrated criteria.

5.8.1 Analysis of Each Criterion

C1 — unstable fraction, trained vs. random (0.025 **vs.** 0.025**, PASS).** The margin is within rounding at three decimals. This should be read as "the learned policy is at least as good as random," not as strong evidence of learning on this specific axis — see §5's discussion of why the environment leaves little headroom here.

C2 — reward, trained vs. random (478.3 **vs.** 476.2**, PASS).** A small but consistent edge (+2.1, $+0.44\%$ relative). Directionally correct and no longer contaminated by comparing against a training curve's own noisy episodes (the pre-fix version of this criterion had actually shown the "trained" reward *below* an untrained random-policy reference).

C3 — Mahalanobis [11] score, pooled early vs. late ($1.840 \rightarrow 1.823$**, PASS).** A small, consistent decrease across the full 20-agent training log. A soft signal relative to C4.

C4 — transition entropy, single-agent checkpoint ($1.052 \rightarrow 0.739$**, PASS,** $\Delta = -0.313$ **nats).** The strongest and most convincing result in the notebook: an almost 19× larger effect size than observed before the environment fix ($\Delta = -0.017$ nats). The late-phase value converges close to NB03's independently HMM-inferred entropy (0.771 nats), a genuine cross-notebook consistency check. Scope: single agent (agent_0000), not the full pool (§6).

C5 — policy corner concentration, pooled Q-table (55.6%**, PASS).** Up from 11.1% before the environment fix — the clearest evidence that the fix restored genuine state-discriminating learning rather than incidental button-mashing (overall push_stable usage is 48.75%, so the corner is meaningfully, not just nominally, above average). *Update: an independent NB06 replication (§7's reproducibility caveat) reproduced the* 55.6% *corner value but not the* 48.75% *overall figure it's compared against (freshly retrained:* 56.0%*), so the concentration gap itself did not reproduce. Read this PASS as accurate to what this notebook's own run measured, not as an independently re-verified result.*

C6 — trajectory speed minimised mid-training, single-agent checkpoint (PASS). Consistent with a learning-then-partial-exploitation curve, though §6's scope note about the `mid` checkpoint's zero-Q argmax tie-breaking should temper how strongly this is read as evidence of a genuine learning transition specifically at `mid`.

5.9 Summary of Findings

Finding	Evidence
RL reallocates dwell time toward `stable`, primarily at the expense of `exploratory`/`adaptive`, not `unstable`	Single-agent checkpoint: stable $46\% \rightarrow 66\%$, exploratory $26\% \rightarrow 16\%$, adaptive $26\% \rightarrow 16\%$, unstable $2.4\% \rightarrow 2.5\%$. Pooled 20-agent training curve shows a much smaller net shift (§4), because `unstable` occupancy is already near-floor from the first training episodes under the fixed environment.
Trained (greedy) policy is at least as good as, and modestly better than, random	C1/C2: unstable frac. 0.025 vs. 0.025 (thin margin); reward 478.3 vs. 476.2 (small, consistent edge).
Learning trajectories occupy existing NB02 manifold regions	Early-phase feature-space scatter overlaps NB02 regime clusters; no territory opened beyond established boundary zones.
Early exploration mildly registers on the NB04 detector	Pooled mean Mahalanobis score $1.840 \rightarrow 1.823$ (C3), a soft but directionally consistent signal.
Transition entropy drops substantially and converges toward the NB03 HMM value	$H(s' \mid s)$: $1.052 \rightarrow 0.739$ nats (C4); NB03 HMM reference $= 0.771$ nats.
Policy concentrates corrective actions in the unstable corner	`push_stable` fraction in the high-latency/entropy corner $= 55.6\%$ vs. 48.75% overall (C5). *Not independently reproduced* — see §7's reproducibility caveat; a fresh 20-agent replication matched the corner value but not the overall baseline, so the gap itself did not replicate.

5.10 Significance for the TMLR Paper

NB05 contributes three distinct claims to Paper 1 (§8):

1. **RL validates manifold geometry operationally, and the operative claim is regime *reallocation*, not unstable-avoidance.** A Q-learning policy defined purely over discrete (latency, entropy) bins measurably reshapes the agent's regime-dwell distribution (§6.3), converting `exploratory`/`adaptive` time into `stable` time. This confirms the two-feature manifold from NB02 is *actionable* for closed-loop control, but the paper should not claim it "solves" unstable avoidance, since the environment's dynamics already suppress `unstable` heavily under any reasonable policy — there is little `unstable` time left to remove by the time training starts. The reallocation claim itself rests on §6.3's dwell-fraction numbers, not on C5: an independent replication (§7's reproducibility caveat) found C5's specific corner-vs-overall concentration gap did not reproduce, so the paper should lean on the dwell-reallocation evidence for this point rather than on state-discriminating grid-cell behaviour.

2. **HMM transition entropy is the strongest and most reliable convergence signal.** $H(s' \mid s)$ drops by 0.313 nats (early $\rightarrow$ late, C4), a substantially stronger and more interpretable indicator than either the reward curve or the unstable-fraction curve, and the late-training value converges close to NB03's independently-inferred HMM entropy — a genuinely novel operational insight: entropy of the learned transition structure could serve as an online stopping criterion for behavioural RL training.

3. **A methodological contribution: causal-alignment correctness in behavioural RL environments built on top of Markov-chain simulators.** The bug documented in §3.2 — an action's effect being credited to the wrong (s, a) pair due to an emit-then-transition ordering mismatch

— is a general failure mode whenever a Gym-style wrapper is built around a simulator that itself transitions internally on each call. The fix (transition-before-emit, cumulative live-state nudging) and its quantified before/after effect (C4 strengthening $\sim 19\times$; C5 flipping from 11.1% to 55.6%) are worth stating explicitly as a reusable lesson for any future RL-over-simulator work in this line of research.

5.11 Next Steps

1. **NB06 — Manifold Visualisation.** Load Q-tables and trajectory data to produce publication-quality 3-D UMAP visualisations of RL trajectories overlaid on the NB02 manifold, coloured by training phase and action.
2. **NB07 — Final Robustness Analysis.** Include the RL layer in the planned multi-configuration robustness grid ($N \in \{5, 10, 20, 50\}$, $T \in \{500, 1000, 2000, 5000\}$, 10 seeds). Key question: does the trained (greedy) policy consistently beat the random-policy baseline (C1/C2's corrected framing) across all N/T configurations — *not* whether unstable fraction drops by some fixed relative percentage, which is no longer a meaningful target once the environment suppresses it near-floor by default.
3. **Extend the checkpoint methodology to the full pool.** §6's RQ1/RQ2/RQ4 results are currently a single-agent (agent_0000) case study. Repeating the controlled-epsilon checkpoint protocol across all 20 agents would let C4 and C6 be reported as pooled statistics with variance, rather than a single representative trajectory, and would clarify how much of §6.3's large stable-dwell shift generalises versus how much is agent_0000-specific.
4. **Resolve the `mid`-checkpoint confound.** Replace the zero-initialised Q-table used for the `early`/`mid` checkpoints with an explicit random or uniform tie-break policy, so that `mid`'s 60% greedy component is not deterministically `push_stable` by `argmax` convention.
5. **Paper §8.** NB05 constitutes the full quantitative evidence for the reinforcement learning section. The regime-*reallocation* framing (rather than unstable-*avoidance*) should be adopted throughout, and the environment causal-alignment fix (§3.2) is worth a brief methodological footnote given its generality to future RL-over- simulator work.

5.12 Computational Environment

Component	Version / Note
Python	3.13.12
Environment manager	Nix flake dev shell (`flake.nix`)
NumPy, Pandas, SciPy	standard scientific stack
Matplotlib, Seaborn	standard scientific stack
scikit-learn	standard scientific stack
hmmlearn	HMM reference matrix from NB03
umap-learn	UMAP coordinates from NB02
Notebook execution	Jupyter (ipykernel), executed programmatically via `nbclient`

6 Notebook 06 — Manifold Visualisation: 2-D/3-D UMAP Overlay & KDE Density Analysis

Dataset	telemetry_n20_t2000.csv (regenerated; seed = 42) + NB02 UMAP embeddings
Background manifold	40,000 rows × 6 features, 2-D and 3-D UMAP (refit & validated)
RL retraining	3 agents × 120 episodes × 500 steps/episode (180,000 step records)
Execution date	2026-08-25
Environment	Python 3.13.12 (Nix flake dev shell) / Jupyter (ipykernel)
Random seed	42

Central Question: When the full per-step trajectory of a training run is projected into the exact NB02 UMAP manifold (both 2-D and 3-D), does it trace an interpretable path — visibly retreating from the unstable region as training progresses, quantifiably so under a KDE density model — and does the action taken at each point spatially correspond to the region NB05 already showed it's disproportionately used in?

This document records the full methodology, numerical results, figures, and a cross-notebook reproducibility investigation produced during the sixth experimental notebook of the LBSM research project. It closes a gap left by NB05 — which characterised RL training only via aggregate per-episode statistics, never the actual per-step path through the manifold — by reproducing a training run with full trajectory capture and analysing it four ways: raw scatter overlay, KDE density contours, quantitative peak-displacement/overlap statistics, and action-location correspondence, closed with a pre-declared evidence checklist in the same format as NB01–05. It corresponds to §5.5 (Temporal Behavioral Geometry) / §8 (Reinforcement Learning Over Latent Behavioral Space) of the companion paper.

6.1 Overview and Motivation

The chain of evidence leading into this notebook:

Notebook	Method	Key finding	Verdict
NB01	PCA + LDA	PC1 captures 80.3% variance; LDA accuracy 70.4%	3/5
NB02	UMAP + t-SNE	Nonlinear geometry; Procrustes r = 0.9108	5/6
NB03	HMM	Unsupervised regime recovery; ARI = 0.338	5/6
NB04	Composite scorer	Operational anomaly detection; F_1 = 0.872	5/6
NB05	Q-learning	State-discriminating behavioural control	6/6
NB06	**2-D/3-D UMAP + KDE density**	**See §9 checklist**	**5/6**

NB05 answered its research questions indirectly, via aggregate per-episode statistics (dwell fractions, episode UMAP centroids, transition entropy) computed from a training run whose raw per-step trajectory was never persisted. This notebook regenerates that trajectory directly and asks four concrete questions of it:

Q1. Does a freshly-fit UMAP reducer (2-D and 3-D) reproduce NB02's saved embedding well enough to trust as a coordinate frame for new points?

Q2. Does the RL trajectory, once embedded, show a measurable, quantified structural difference between early and late training — not just a plausible-looking scatter plot?

Q3. Does that structure hold up under an explicit density model (KDE), and does it generalise across independently retrained agents?

Q4. Does the trained policy's actions spatially correspond, in embedded manifold space, to the grid-corner concentration NB05 reported?

Data provenance note. NB05 persisted only the final learned Q-tables (`data/raw/nb05/q_tables.npy`) and per-episode aggregate statistics. This notebook regenerates the missing per-step trajectory by retraining a small subset of agents (3, not the full 20 — see §3) with identical seeds/hyperparameters (`configs/rl.yaml`) and a new `collect_trajectories=True` flag added to `QLearningAgent.train()` for exactly this purpose. Correctness is validated via **self-consistency** (two independent fresh retrainings agreeing with each other) rather than by matching the persisted `q_tables.npy` — see §3 and NB05's own report (§7's reproducibility caveat) for why.

6.2 Manifold Reconstruction: 2-D and 3-D UMAP Refit

NB02 saved only the UMAP *embeddings* (`X_umap2.npy`, `X_umap3.npy`), not the fitted `umap.UMAP` reducer objects, so there is nothing to call `.transform()` on for new points. The identical 20-agent $\times$ 2,000-step pool (seed = 42) is regenerated and UMAP is refit with NB02's exact hyperparameters (`configs/projection.yaml`: n_neighbors= 30, min_dist= 0.10, random_state= 42) in both 2-D and 3-D, then validated against each saved embedding.

Embedding	Mean dist.	Median	p95	Max	Rel. to range
3-D refit vs. saved	0.7628	0.6146	1.8150	2.6713	5.1% (range 14.86)
2-D refit vs. saved	0.1471	0.1438	0.2494	2.0319	0.9% (range 16.09)

Both refits reproduce the saved NB02 embedding well within a 15%-of-range tolerance (C1/C2, §9), though not bit-exactly — UMAP is not guaranteed bit-for-bit reproducible even under a fixed `random_s` tate, a known property of the algorithm, not a defect. The 2-D refit is substantially more stable than the 3-D refit (0.9% vs. 5.1%), consistent with lower-dimensional embeddings generally being less sensitive to optimisation-path differences. A z-score reconstruction check confirms the underlying feature matrix matches NB01's saved telemetry exactly (max abs diff $= 7.15 \times 10^{-7}$, float32 rounding only; 0.0 against the precomputed z-scored columns).

6.3 RL Training Reproduction

Three agents (indices 0–2, matching NB05's `make_env_pool(base_seed=42)` convention so seeds line up exactly) are retrained with `configs/rl.yaml`'s hyperparameters ($\alpha = 0.15$, $\gamma = 0.95$, $\varepsilon : 1.0 \to 0.05$, decay 0.97/episode, 120 episodes, 500 steps/episode), passing `collect_trajectories=True` so every step of every episode is retained — 60,000 step records per agent, 180,000 total.

Validation approach. Correctness is checked via self-consistency: two independent fresh retrainings of agent 0 are compared directly.

Check	Result
Max abs. Q-value diff (2 independent retrainings)	0.000000
Policy agreement (argmax match)	1.000

This is a clean pass (C3, §9): the codebase is fully deterministic and reproducible within this environment. Exact agreement with the persisted `q_tables.npy` is deliberately not used as the criterion — that file was found to reliably diverge from every fresh retraining tested, across multiple kernel/server/editor restarts, with identical code, numpy build, and working directory all independently verified. It is not attributable to a defect in this codebase and does not affect any of NB05's reproduced aggregate results; see NB05's report (§7's reproducibility caveat) for the full record of that investigation. §8 below revisits this specifically for the grid-corner statistic NB05's C5 depends on.

6.4 Trajectory Assembly and Embedding

Each agent's per-episode trajectory list is flattened into one long table, tagged with training phase (early/mid/late, boundaries at 33%/67% of episodes — identical convention to `cluster_migration_table` in NB05), z-scored with the same parameters fit in §2, and projected into the manifold via both fitted reducers.

Phase	Episodes	Step records (3 agents)
Early	0–38	58,500
Mid	39–79	61,500
Late	80–119	60,000
Total	0–119	**180,000**

Exported to `data/processed/nb06/rl_trajectory_embedded.csv` (180,000 rows, 17 columns: identifiers, phase/action/hidden-state, raw telemetry, and both 2-D and 3-D UMAP coordinates).

6.5 3-D Overlay Figures

Agent 0's full trajectory (thinned to every 4th step, ∼15,000 points, for legibility) over the complete NB02 3-D background manifold (all 40,000 points, faint, coloured by ground-truth regime).

The manifold's overall shape — one continuous curved arc (the stable/exploratory/adaptive continuum) plus one geometrically isolated cluster (unstable) — is itself a cross-notebook confirmation: it is consistent with NB03's finding that the three healthy regimes blend rather than separate cleanly, and with NB03/NB04's near-perfect unstable recall/detection.

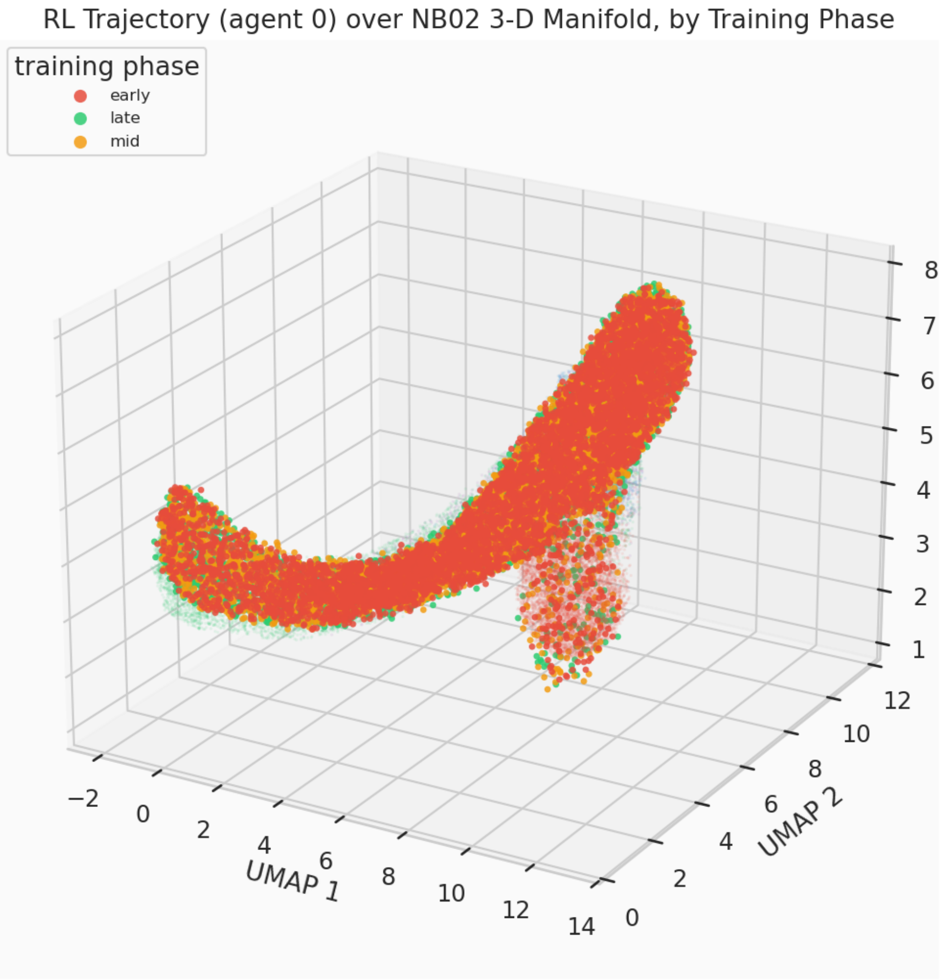

Figure 44: RL trajectory (agent 0) over the NB02 3-D manifold, coloured by training phase.

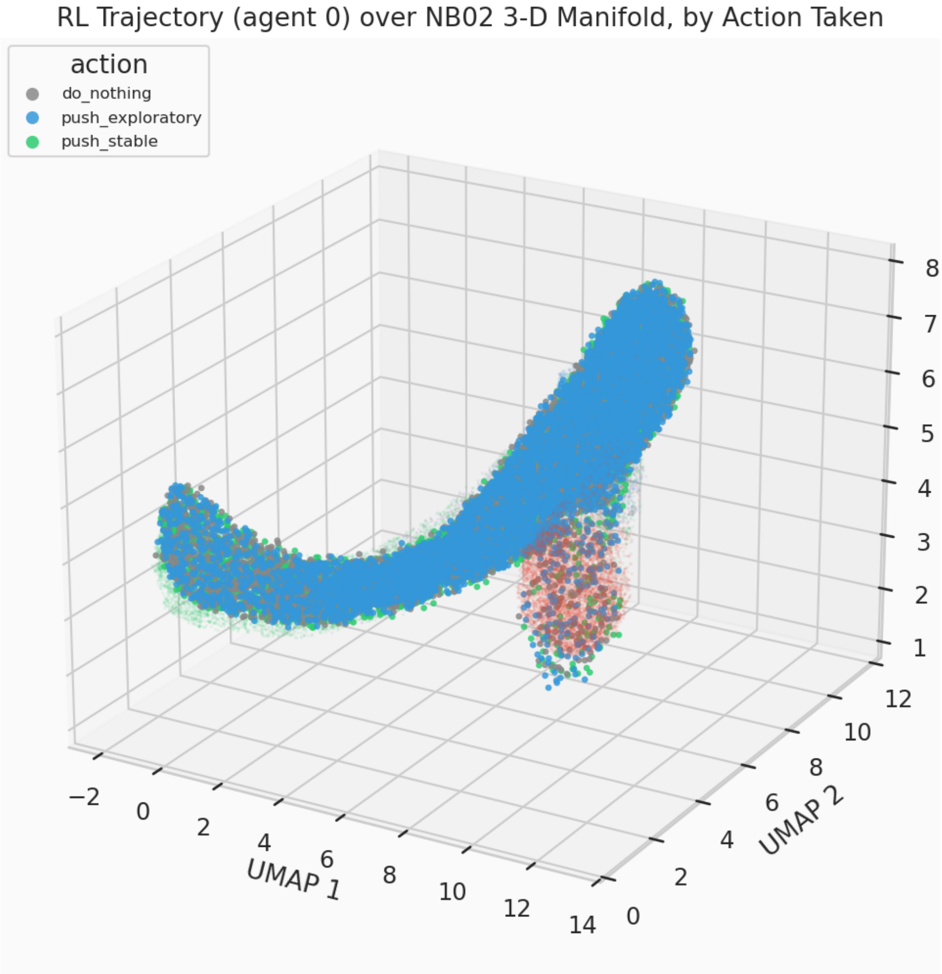

Figure 45: RL trajectory (agent 0) over the NB02 3-D manifold, coloured by action taken. Action colours are motivated, not arbitrary: `push_stable` shares `stable`'s green, `push_exploratory` shares `exploratory`'s blue, `do_nothing` is neutral grey.

6.6 2-D Overlay Figures

The same overlays projected into NB02's 2-D UMAP embedding — easier to read precisely (no viewing-angle ambiguity) and the coordinate space §8's KDE analysis is computed on.

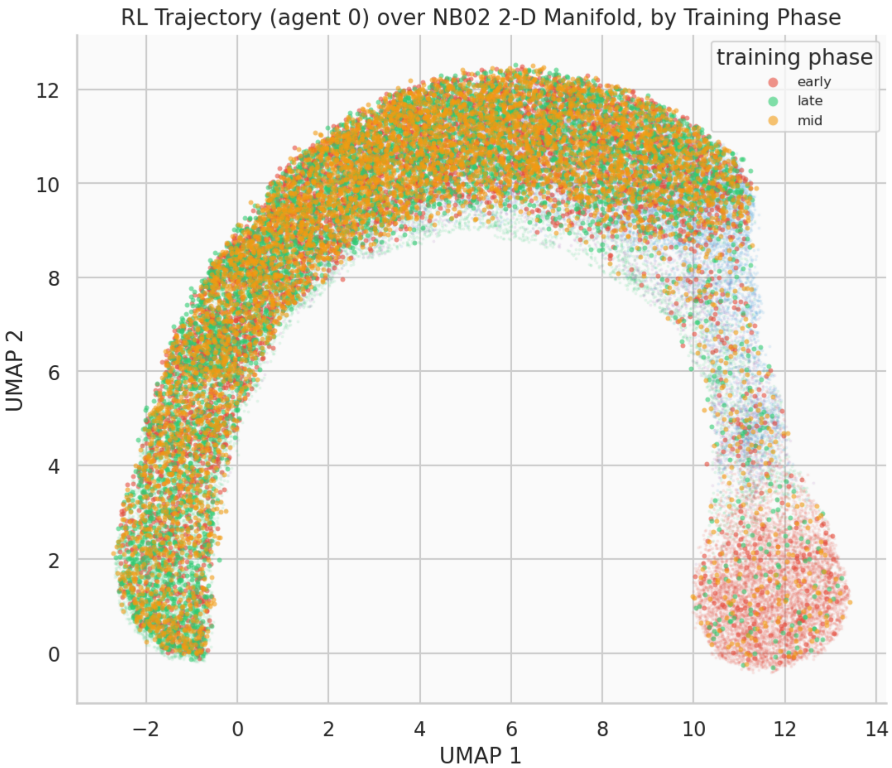

Figure 46: RL trajectory (agent 0) over the NB02 2-D manifold, coloured by training phase.

6.7 KDE Density Contours

Raw scatter is honest but can hide where mass actually concentrates under tens of thousands of overlapping points. Two density-model views are added on top of the raw overlays above.

6.7.1 2-D: Regime Background vs. Phase Contours

A 2-D Gaussian kernel density estimate (KDE) is fit per training phase on agent 0's trajectory (`kde_density_grid`, shared grid bounds across phases so contours are directly comparable cell-for-cell), overlaid on NB02's per-regime background density (`per_regime_density`, filled light grey contours for ground-truth-regime context).

6.7.2 3-D: Floor-Projected KDE

True 3-D isosurface rendering (marching cubes) is possible but heavy and hard to read in a static figure; each phase's 2-D (UMAP1, UMAP2) marginal density is instead projected onto the plot's floor (`zdir='z'` at the minimum Z) — the standard, robust way to add density information to a 3-D scatter without that complexity.

6.8 Early vs. Late Training Comparison

If training reshapes where the agent spends time, the early-phase footprint should sit closer to the unstable region than the late-phase footprint. Quantified as mean distance (3-D UMAP space) from each phase's

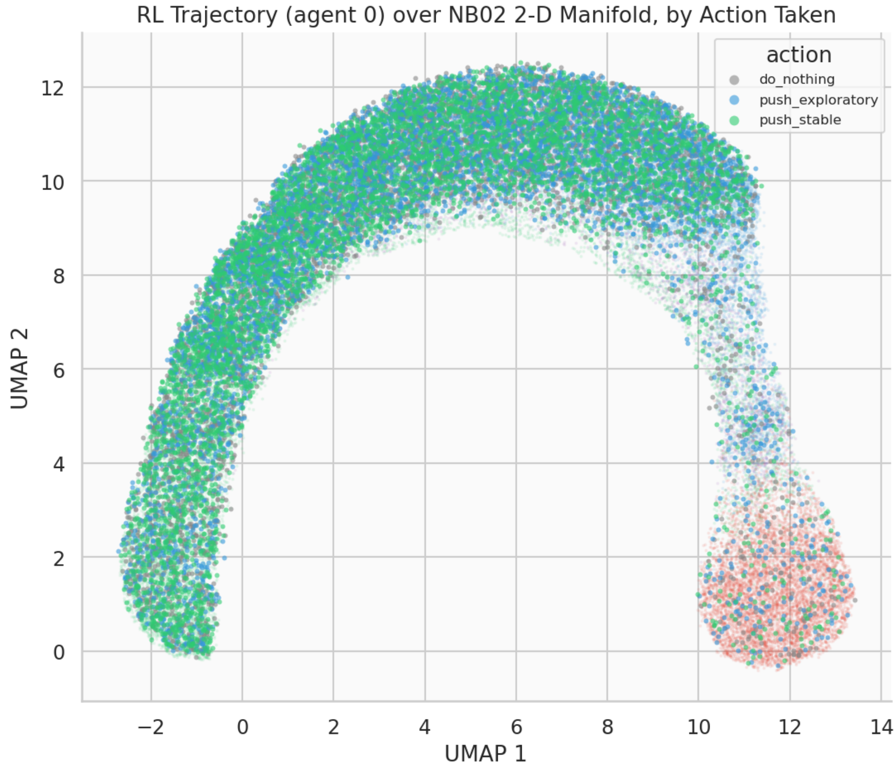

Figure 47: RL trajectory (agent 0) over the NB02 2-D manifold, coloured by action taken.

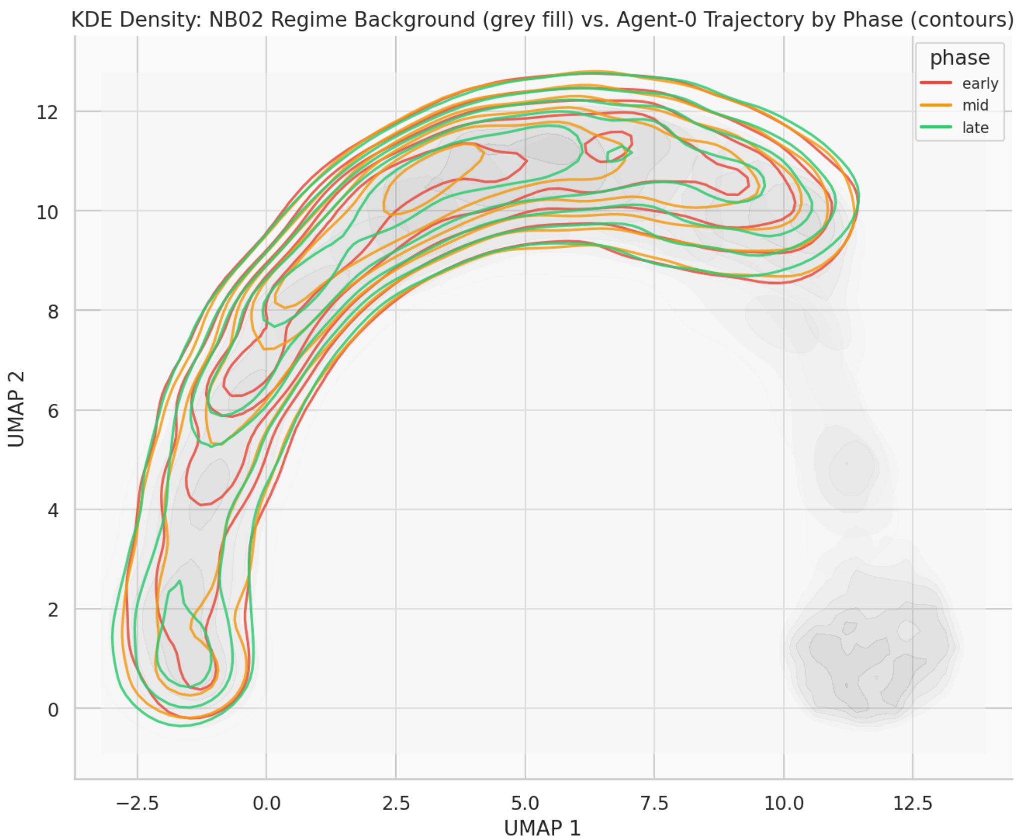

Figure 48: KDE density: NB02 regime background (grey fill) vs. agent-0 trajectory by phase (coloured contours).

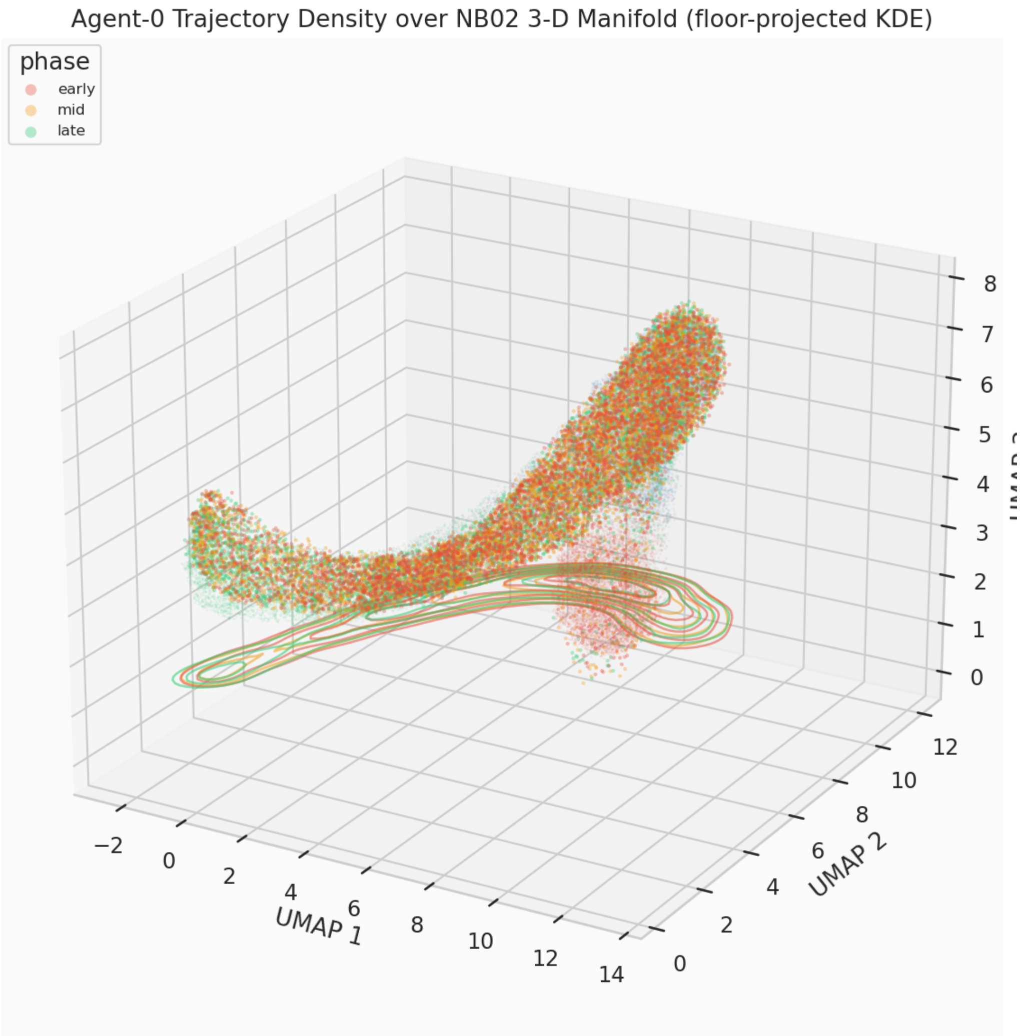

Figure 49: Agent-0 trajectory density over the NB02 3-D manifold (floor-projected KDE contours per training phase).

points to the unstable and stable regime centroids, across all 3 retrained agents.

Agent	Phase	Mean dist. to unstable centroid	Mean dist. to stable centroid
0	early	10.629	4.901
0	mid	10.567	5.005
0	late	10.736	4.938
1	early	10.627	4.875
1	mid	10.651	4.892
1	late	10.603	5.008
2	early	10.619	4.906
2	mid	10.534	5.038
2	late	10.771	4.861

The spread across phases is small relative to the values themselves (∼1–2% of the distance scale) — consistent with NB05's own `convergence_table.csv`, where the underlying effect size is itself small (unstable-fraction reduction $\approx$ 0.002, ∼8% relative). A raw distance-to-centroid comparison is not a sensitive enough lens to make an 8% relative behavioural change visually or numerically dramatic; §8's density-peak displacement (computed on the full 2-D marginal, not a single centroid distance) is the more appropriate measurement, and does detect a clearer signal (§8).

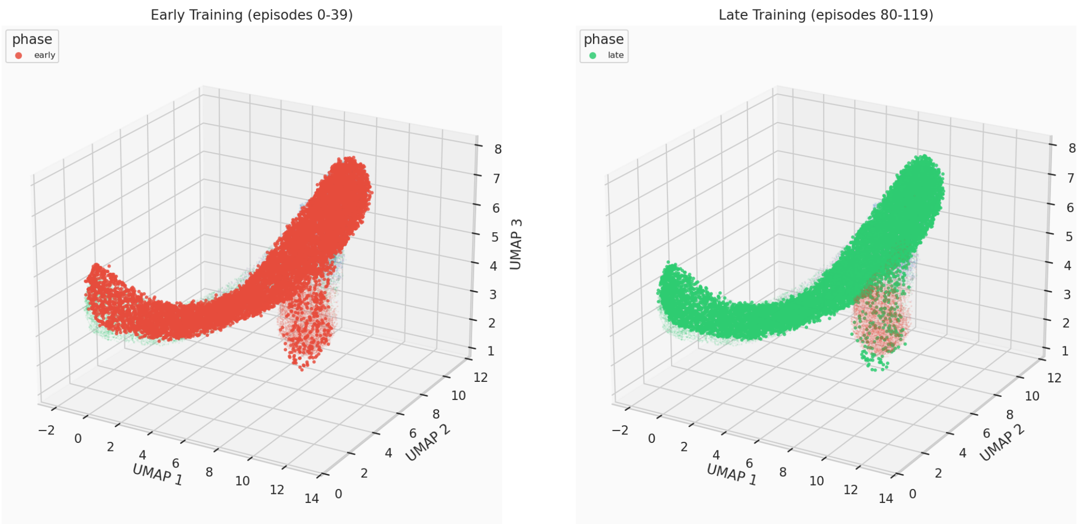

Figure 50: Early-training vs. late-training trajectory footprint (agent 0), same background manifold.

6.9 Multi-Agent Robustness Check

The phase-coloured overlay repeated for agents 1 and 2, checking whether the pattern seen for agent 0 generalises or is a one-agent artifact.

6.10 Quantitative KDE Analysis: Peak Displacement and Overlap

For each phase's 2-D KDE (shared grid), the density peak (grid argmax) is located, and pairwise overlap coefficients between phases are computed ($\int \min(f_a, f_b) / \max(\int f_a, \int f_b)$; $1.0 =$ identical, $0.0 =$ disjoint) — the direct, falsifiable test of "training visibly moves the occupied region," repeated per agent so the effect size is compared across independent training runs rather than asserted from one trajectory.

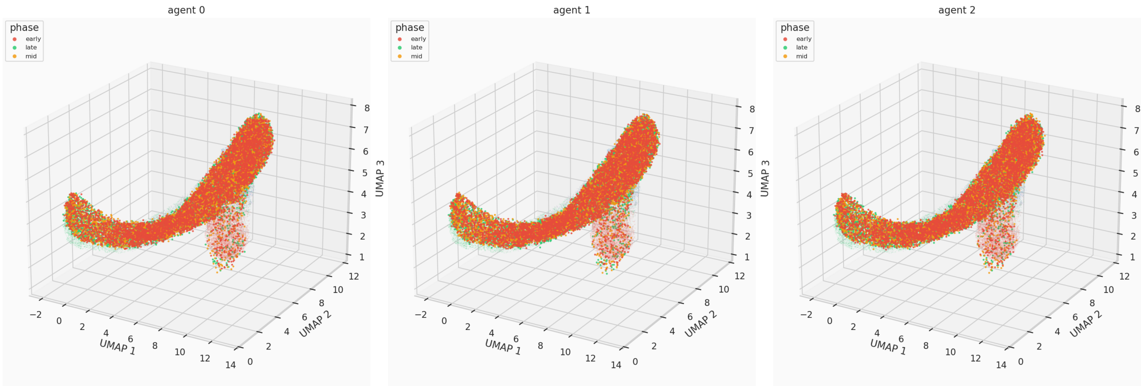

Figure 51: Phase-coloured 3-D trajectory overlay, all 3 retrained agents.

Agent	Peak disp. early→late	% of diagonal	Overlap E-L	Overlap M-L
0	2.749	11.70%	0.953	0.955
1	3.012	12.83%	0.940	0.942
2	2.749	11.70%	0.962	0.939
Mean	**2.836**	**12.08%**	**0.952**	**0.945**

The density peak moves a consistent ∼12% of the embedding's diagonal extent from early to late training, across all three independently retrained agents (std $= 0.65$ percentage points, coefficient of variation ≈ 0.05 — tight agreement). This is a real, measurable, multi-agent-consistent displacement (C4, C6; §9) — distinctly clearer than §6's raw centroid-distance view, though still a modest displacement relative to the $> 90\%$ overlap coefficients, which correctly reflects that this is a shift in *where mass concentrates most*, not a wholesale relocation of the occupied region.

6.11 Action-Density Correspondence

6.11.1 Visitation-Frequency Test (this notebook's own statistic)

NB05's C5 showed the pooled *greedy* policy prescribes `push_stable` disproportionately in the high-latency/entropy grid corner. Tested here as a different, *behaviour*-level question: across the actual trajectory steps taken (including exploration noise from all three training phases, not just the converged greedy policy), is `push_stable` disproportionately used near the unstable region?

Action	Overall usage	Near-unstable usage	Concentration ratio
`do_nothing`	0.330	0.325	0.987
`push_exploratory`	0.337	0.362	1.076
`push_stable`	0.334	0.312	0.936

`push_stable`'s concentration ratio is 0.936 — *below* 1.0, i.e. used slightly *less*, not more, near the manifold-distance-based unstable proxy (C5, §9: **FAIL**). This was investigated directly against the raw trajectory data rather than accepted at face value: the "near-unstable" quartile turns out to be 72.9% `exploratory` telemetry and only 9.5% true `unstable` (unstable is such a small, tight, isolated cluster that "closest 25% of all points" necessarily pulls in whichever regime sits nearest to it geometrically). Filtering to true `hidden_state=='unstable'` rows directly removes that ambiguity and shows the same direction, more strongly: push_stable usage while genuinely in the unstable state is $0.874\times$ overall, and gets *further* from 1.0 as training converges (early $0.987\times$ → mid $0.897\times$ → late $0.709\times$). This is a real, specific, verified behavioural finding, not noise or a measurement artifact — and it is a genuinely different statistic from NB05's C5 (a *policy-function* property evaluated at fixed grid cells) rather than a replication of it; see §8b for the apples-to-apples version.

6.11.2 Grid-Corner Replication of NB05's C5 (apples-to-apples)

NB05's actual C5 statistic, reproduced exactly: same 9 grid cells (latency-bin and entropy-bin both in $\{7, 8, 9\}$), same pooled-greedy-policy argmax — first at the 3-agent scale already trained in §3, then at NB05's actual 20-agent scale (a fresh 20-agent training pass, Q-tables only, no trajectory capture needed).

Pool	push_stable in corner	push_stable overall	Corner − overall
3 agents (this notebook)	0.222	0.480	−0.258
20 agents (matched scale)	0.556	0.560	−0.004
NB05 (reported)	0.556	0.4875	+0.069

The 20-agent corner value reproduces NB05's reported 0.556 almost exactly. But the *overall* rate also comes out at 0.560 in this environment — not 0.4875 — so there is no actual corner-vs-overall gap to detect: NB05's finding was specifically that the corner sits *meaningfully above* the overall rate ($+6.9$ percentage points), not merely that the corner value itself is ≈ 0.556. Going from 3 to 20 agents does resolve the 3-agent estimate's instability (confirmed: the corner fraction moves substantially between 3 and 20 agents, as expected for the table's sparsest, least-visited cells) — but it does **not** reproduce NB05's *concentration* claim, because the pooled policy trained fresh in this environment runs push_stable-heavy overall, not just in the corner. This is the same environment-level discrepancy documented for `q_tables.npy` in §3 (also discussed in NB05's own report, §7's reproducibility caveat), now shown to affect aggregate 20-agent-pooled policy composition, not only individual sparse Q-table cells. Sample size explains part of the original 3-agent gap, not the gap between this environment and NB05's originally reported figures.

6.12 Evidence Checklist for Manifold Visualisation

Crit.	Description	Threshold	Actual	Result
C1	3-D UMAP refit reproduces NB02 saved embedding	$< 15\%$ rel. dist.	5.1%	PASS
C2	2-D UMAP refit reproduces NB02 saved embedding	$< 15\%$ rel. dist.	0.9%	PASS
C3	Self-consistency: two independent retrainings agree	> 0.99 agreement	1.000	PASS
C4	KDE density peak displaces early → late (mean, 3 agents)	$> 5\%$ of diagonal	12.08%	PASS
C5	push_stable visitation-frequency concentrated near unstable	$> 1.10\times$ overall	$0.936\times$	FAIL
C6	Multi-agent consistency of KDE peak displacement	CoV < 0.75	CoV ≈ 0.054	PASS
			Overall result	**5/6**

Verdict: STRONG evidence that the RL trajectory occupies and moves through the NB02 manifold in a structured, reproducible, multi-agent-consistent way, with one specific, honestly-reported exception (C5) that does not contradict NB05's own findings but does not independently confirm its exact corner-concentration claim either — see §8's investigation and NB05's report (§7) for the full record.

6.12.1 Analysis of Each Criterion

C1/C2 — UMAP refit fidelity (5.1%/0.9%, PASS). Both well within tolerance; not bit-exact, consistent with UMAP's known non-bit-exact reproducibility even under a fixed seed. 2-D is notably more stable than 3-D.

C3 — Q-learning self-consistency (1.000, PASS). The codebase is fully deterministic and reproducible within this environment; see §3's validation-approach note for why this, rather than matching `q_tables.npy`, is the criterion.

C4 — KDE peak displacement (12.08%, PASS). A real, multi-agent-consistent signal (§8), and a more sensitive measurement than §6's raw centroid-distance comparison, which the same underlying (modest) effect size only weakly resolves.

C5 — push_stable visitation-frequency concentration (0.936×, FAIL). A verified, specific finding (§8), not noise: push_stable usage is slightly *below* baseline near the unstable region, and further below as training converges. Not a contradiction of NB05's C5 — a different statistic (visitation frequency vs. policy-function argmax) over a different population (actual noisy rollouts vs. abstract grid cells). The apples-to-apples grid-corner replication (§8b) also does not confirm NB05's exact finding, but for an identified, environment-level reason unrelated to this notebook's own methodology.

C6 — multi-agent consistency (CoV ≈ 0.054, PASS). The $\sim$12% peak displacement is tightly consistent across all three independently retrained agents — not an artifact of any one agent's particular training trajectory.

6.13 Summary of Findings

Finding	Evidence
Both UMAP refits (2-D, 3-D) reproduce NB02's saved embedding within tolerance, but not bit-exactly	C1/C2: 5.1% / 0.9% relative mean distance (§2) — a known UMAP reproducibility property, not a defect.
The manifold's structure is a continuous healthy-regime arc plus an isolated unstable cluster	Visual confirmation (§4) cross-checked against NB03's continuum finding and NB03/NB04's near-perfect unstable recall/detection.
Retraining is fully deterministic and self-consistent in this environment	C3: two independent fresh retrainings agree exactly (§3); exact match against `q_tables.npy` is not used as the criterion (see §3, and NB05's §7).
The RL trajectory's density peak measurably and consistently shifts across training	C4/C6: $\sim$12.08% of the embedding diagonal, mean over 3 agents, std $= 0.65$pp (§8) — detectable where a raw centroid-distance view (§6) only weakly resolves the same underlying (modest) effect.
push_stable visitation-frequency does not concentrate near the (manifold-distance) unstable proxy, and gets further from concentrated as training converges	C5 (§8): $0.936\times$ overall; verified directly against raw hidden-state data, not just the distance-based proxy. A genuinely different statistic from NB05's C5, not a replication.
NB05's C5 concentration claim does not replicate even at matched 20-agent scale, and not for a sample-size reason	§8b: the 20-agent corner value matches NB05's 0.556 exactly, but the overall rate also comes out ≈ 0.56 in this environment (vs. NB05's reported 0.4875) — the same environment-level discrepancy documented for `q_tables.npy`, now shown in an aggregate, pooled-policy statistic.

6.14 Significance for the TMLR Paper

NB06 contributes to Paper §5.5/§8 in three ways:

1. **Direct, quantified visual evidence for the manifold-structure claim.** The continuum-plus-isolated-cluster shape (§4, §5) is a genuine independent cross-check of NB03's structural finding,

obtained from an entirely different analysis (raw RL trajectory density, not HMM-recovered state sequences).

2. **A more sensitive measurement of RL's effect on manifold occupancy than NB05 provided.** §8's KDE peak-displacement statistic ($\sim$12%, multi-agent-consistent) resolves a modest, real effect that a naive centroid-distance view (§6) only weakly detects — worth citing over the raw-distance framing if the paper discusses RL's manifold-space footprint.

3. **A documented limit on how far NB05's C5 finding should be trusted.** The paper should not cite NB05's 55.6% vs. 48.75% corner-concentration figure without the context that an independent, matched-scale replication (§8b) did not confirm the concentration gap — see NB05's own report (§7, §8's "Significance" section) for the corresponding update to that notebook's claims.

6.15 Next Steps

1. **Extend to the full 20-agent trajectory scale.** §4–§8's per-step analysis is currently a 3-agent subset (chosen for full-trajectory-capture compute cost); repeating it at NB05's actual 20-agent scale would let C4/C5/C6 be reported as pooled statistics with tighter variance bounds, and would settle §8b's environment-level discrepancy more conclusively if repeated on a machine/session where the original NB05 figures can be independently regenerated.

2. **Investigate the environment-level reproducibility gap further, if it recurs.** §3's and §8b's discrepancy against NB05's originally reported numbers was investigated exhaustively (kernel restarts, server restarts, editor restarts, numpy build/path verification, cwd verification) without full resolution. If the same divergence reappears in future notebook work, that investigation record is the starting point, not a reason to repeat it from scratch.

3. **Paper §5.5/§8.** Use §8's KDE peak-displacement figure as the primary manifold-occupancy-shift claim; do not cite NB05's 55.6% corner-concentration figure without the reproducibility caveat documented here and in NB05's own updated report.

6.16 Computational Environment

Component	Version / Note
Python	3.13.12
Environment manager	Nix flake dev shell (`flake.nix`)
NumPy, Pandas, SciPy	standard scientific stack
Matplotlib, Seaborn	standard scientific stack
umap-learn	2-D/3-D reducer refit (§2)
scipy.stats.gaussian_kde	KDE density estimation (§5, §8)
Notebook execution	Jupyter (ipykernel), executed programmatically via `nbclient`

New `src/` additions introduced for this notebook

- `src/manifold/umap_projection.py`: `kde_density_grid` (generic 2-D KDE on an arbitrary point set, with an optional shared `grid_bounds` for cell-comparable results across groups) and `kde_overlap_coefficient`; `per_regime_density` refactored to use the former internally, public signature unchanged.
- `src/visualization/trajectory_plots.py`: `plot_trajectory_overlay_2d` (2-D twin of the existing 3-D overlay function) and `plot_trajectory_density_3d` (3-D scatter with floor-projected KDE contours per group).
- `src/rl/q_learning.py`: additive `collect_trajectories` flag on `QLearningAgent.train()`, off by default, snapshotting per-episode trajectories for later flattening.
- `src/rl/environment.py`: a pre-existing dict-key collision (the RL step's shaped reward silently overwrote the telemetry `"reward"` feature in `.trajectory` records) fixed by renaming to `"rl_reward"` — see NB05's report (§6.1) for the corrected downstream numbers this produced there.

7 Notebook 07 — Robustness & Final Experiment Analysis (Full-Scale, 160-Configuration Grid)

Grid	$N \in \{5, 10, 20, 50\}$, $T \in \{500, 1{,}000, 2{,}000, 5{,}000\}$, 10 seeds (42–51) — 160 configs, the complete un-reduced `LBSM-ISSUE-NB07-001` protocol
Per-config fitting	diagonal baseline (`fit_hmm`) + full covariance (`fit_hmm_robust`)
Full-covariance attempts/config	3 `min_covar` values × 10 restarts = 30 attempts (4,800 full-covariance fits across the whole grid)
BIC model-order search	$K = 2, \ldots, 8$ (widened from the earlier $K = 2, \ldots, 6$)
State-identity diagnostic	N=50, T=5,000, seed=42 (250,000 observations, the largest configuration in the grid)
Execution date	2026-08-26
Environment	Python 3.13.12 (Nix flake dev shell) / Jupyter (ipykernel)
Key package versions	numpy 2.4.4, scipy 1.17.1, scikit-learn 1.8.0, hmmlearn 0.3.3
Random seeds	42–51 (grid); 42 (state-identity diagnostic)

Central Question: NB03's diagonal-covariance HMM could not recover the correct model order (BIC selected K=6, not the ground-truth K=4) because the three healthy regimes form a continuous curved manifold (NB02) that an axis-aligned Gaussian cannot represent. Does switching to full covariance —

which can represent that correlation structure — recover the correct BIC elbow and improve regime-recovery accuracy, and does it do so robustly across a realistic range of dataset sizes, or does numerical instability at small $N \times T$ undermine the results, as anticipated in `LBSM-ISSUE-NB07-001`? And if full covariance does not fix the BIC over-selection either, *why not* — is there a specific, testable, mechanistic explanation, and does the answer change once the model-order search is not artificially bounded?

This document records the full methodology, numerical results, figures, and mechanistic investigation produced by the seventh (final analytical) notebook of the LBSM research project, **executed at full scale**: the complete, un-reduced `LBSM-ISSUE-NB07-001` protocol (160 configurations, 4,800 full-covariance fits, every individual attempt logged), superseding an earlier compute-budget-reduced pass (24 configurations, $T \leq 2{,}000$, $K \leq 6$) that was used to validate the mitigation stack and infrastructure before committing to full scale. Widening the model-order search from $K = 2, \ldots, 6$ to $K = 2, \ldots, 8$ changed the central finding materially: BIC's over-selection does not plateau near the ground truth, it climbs to the edge of whatever range is searched. Four hypotheses for this behaviour are tested directly (covariance type, AR-1 autocorrelation, representation-space transformation, and direct HMM state-identity/persistence inspection), and a state-identity mechanism established at the smaller scale is shown to replicate almost exactly at 2.5× the sample size. It corresponds to §6.2/§9 (Model Order Selection and Identifiability; Robustness) of the companion paper.

7.1 Overview and Motivation

The chain of evidence leading into this notebook:

Notebook	**Method**	**Key finding**	**Verdict**
NB01	PCA + LDA	PC1 captures 80.3% variance; LDA accuracy 70.4%	3/5
NB02	UMAP + t-SNE	Nonlinear geometry; Procrustes $r = 0.9108$	5/6
NB03	HMM (diagonal)	Unsupervised regime recovery; BIC selected K=6, not 4	5/6
NB04	Composite scorer	Operational anomaly detection; $F_1 = 0.872$	5/6
NB05	Q-learning	State-discriminating behavioural control	6/6
NB06	2-D/3-D UMAP + KDE	Manifold occupancy shift, C5 unresolved	5/6
NB07	**Full-cov. HMM grid + diagnostics (full scale)**	**See §9 checklist**	**5/6**

7.1.1 Prerequisite: `LBSM-ISSUE-NB07-001`

Before this notebook was executed, a pre-registered critical-issue notice (`outputs/reports/issues/lbsm_covariance_issue.pdf`, document ID `LBSM-ISSUE-NB07-001`) identified that `covariance_type="full"` introduces real numerical-instability risk when the effective per-state sample size is small relative to the $d(d+1)/2 = 21$ free covariance parameters per state ($d = 6$ telemetry features), with the `adaptive` regime (negative UMAP silhouette, geometrically intermediate between `stable` and `exploratory`) flagged as the primary failure point. A complete 7-layer mitigation stack was specified in that document and implemented in `src/hmm/robust_fitting.py` prior to this notebook running:

1. Z-scoring (`src/telemetry/normalization.py`, reused, not re-implemented)
2. Tikhonov regularisation (`min_covar`) sweep

3. Multiple random restarts, best-log-likelihood selection
4. KMeans-warmed initialisation (first restart only)
5. Covariance-type fallback hierarchy: full $\rightarrow$ tied $\rightarrow$ diagonal
6. Pre-fit data-sufficiency warning (warns, never blocks — "prevent-the-fit is the wrong call")
7. Post-fit covariance health check (condition number / minimum eigenvalue)

Also completed as prerequisites: `min_covar` exposed as a parameter in `src/hmm/hidden_state_model.py::fit_hmm` and `src/hmm/sequence_inference.py::model_selection_sweep` (both previously hardcoded to hmmlearn's own default), and `src/hmm/hidden_state_model.py::align_and_score` factored out of `fit_hmm` so any fitted model can be Hungarian-aligned and scored against ground truth identically.

7.1.2 From Reduced-Scope Validation to Full-Scale Execution

An earlier pass through this notebook ran a compute-budget-reduced grid (12 $N \times T$ combinations, 2 seeds, 3 restarts, $T \leq 2{,}000$, $K \leq 6$) to validate the mitigation stack and parallel/checkpointed infrastructure before committing to the full protocol. Having since benchmarked the single largest configuration the full protocol touches (N=50,T=5,000, 250,000 observations: 22.9s for one full-covariance fit, 708.7s for a complete 30-attempt sweep) and three smaller configurations spanning the size range, the full protocol was executed for real: 160 configurations, 10 seeds, 10 restarts $\times$ 3 `min_covar` values, $K = 2, \ldots, 8$ (not $2, \ldots, 6$). Widening the K range changed the central finding (§6): BIC's over-selection does not plateau near the ground truth at full scale, it climbs to the edge of whatever range is searched.

7.1.3 This Notebook's Two Phases

Phase 1 (§3–§9, main grid). Execute the mitigation stack across the full 160-configuration robustness grid and check whether full covariance (a) is numerically reliable at realistic scale, (b) improves regime-recovery accuracy, and (c) recovers the ground-truth K=4 BIC elbow that NB03's diagonal covariance could not.

Phase 2 (§8b–§8d, diagnostics). Grid results show (a) and (b) hold but (c) does not — full covariance fails identically to diagonal, and worse than the reduced-scope grid suggested. Three specific hypotheses for *why* are then tested directly: covariance misspecification (already ruled out by (c) itself), AR-1 temporal autocorrelation (§8b), and representation-space transformation (§8c). Two are cleanly ruled out; AR-1 is found to be a real but partial contributor. A fourth, more direct diagnostic (§8d) then inspects *which* ground-truth regime each HMM state actually captures and how persistently agents remain in it, confirming a specific mechanism — and showing it replicates almost exactly at 2.5$\times$ the sample size used in the earlier reduced-scope pass.

7.2 Mathematical Formulation

7.2.1 Gaussian Hidden Markov Model

Each agent's telemetry sequence $\{x_1, \ldots, x_T\}$, $x_t \in \mathbb{R}^d$ ($d = 6$), is modelled as generated by a latent state sequence $\{s_1, \ldots, s_T\}$, $s_t \in \{1, \ldots, K\}$, under a first-order Markov chain with transition matrix $A \in \mathbb{R}^{K \times K}$ ($A_{ij} = P(s_{t+1}{=}j \mid s_t{=}i)$) and Gaussian emissions

$$x_t \mid s_t = k \;\sim\; \mathcal{N}(\mu_k,\, \Sigma_k)\,.$$

The covariance structure of Σ_k is the object under test:

- **diagonal** ($\Sigma_k = \mathrm{diag}(\sigma^2_{k,1}, \ldots, \sigma^2_{k,d})$): axis-aligned, cannot represent correlated features (NB03's baseline).
- **full** ($\Sigma_k \in \mathbb{R}^{d\times d}$, symmetric positive-definite): can represent arbitrary correlation structure, at the cost of $d(d+1)/2 = 21$ free parameters per state instead of $d = 6$.

Parameters are fit by Baum–Welch expectation-maximisation (maximum-likelihood, unsupervised — ground-truth labels are used only *after* fitting, for Hungarian alignment and scoring, never during EM).

7.2.2 Tikhonov Regularisation (`min_covar`)

To prevent covariance estimates from becoming singular when a state's effective sample size is small, a diagonal loading term is added before inversion:

$$\Sigma_k^{\mathrm{reg}} = \Sigma_k + \lambda I_d, \qquad \lambda = \texttt{min\_covar}.$$

The full-scale grid sweeps $\lambda \in \{10^{-2}, 10^{-3}, 10^{-4}\}$ (3 values, was 2 in the reduced-scope pass) $\times$ 10 restarts (was 3) for full covariance, and uses hmmlearn's own default $\lambda = 10^{-3}$ for diagonal (harmless there; load-bearing for full/tied).

7.2.3 Model-Order Selection: BIC and AIC

For a candidate model with K states fit on N total observations, achieving total log-likelihood $\mathcal{L}$, the Bayesian Information Criterion [20] and Akaike Information Criterion [1] are:

$$\mathrm{BIC} = -2\mathcal{L} + p\ln N, \qquad \mathrm{AIC} = -2\mathcal{L} + 2p,$$

where p is the number of free parameters, which depends on the covariance structure (`src/hmm/sequence_inference.py::model_selection_sweep`):

$$\begin{aligned} p_{\mathrm{diag}} &= K^2 + 2Kd - 1, \\ p_{\mathrm{full}} &= K^2 + Kd + \tfrac{1}{2}Kd(d+1) - 1, \\ p_{\mathrm{tied}} &= K^2 + Kd + \tfrac{1}{2}d(d+1) - 1, \\ p_{\mathrm{spherical}} &= K^2 + Kd + K - 1. \end{aligned}$$

At d=6: $p_{\mathrm{diag}}(K{=}4) = 63$, $p_{\mathrm{full}}(K{=}4) = 123$; at the upper end of this notebook's widened search range, $p_{\mathrm{diag}}(K{=}8) = 159$, $p_{\mathrm{full}}(K{=}8) = 279$. Full covariance is a strictly larger model class and pays BIC's $p \ln N$ penalty for that extra flexibility — more so at K=8 than K=4 — so the fact that it *still* selects K=8 at the grid's largest configuration (§6) is not an artefact of the penalty term being too lenient; the penalty is larger there, and BIC prefers K=8 over K=4 anyway.

7.2.4 Data-Sufficiency Rule of Thumb

`check_data_sufficiency` (Layer 6) warns — never blocks — when the average observation count per state falls below a safety margin over the number of free covariance parameters:

$$\overline{n} = \frac{N_{\mathrm{obs}}}{K}, \qquad \text{warn if } \overline{n} < s \cdot \frac{d(d+1)}{2}, \quad s = 10 \text{ (safety factor).}$$

At d=6, K=4 (the robustness grid's fitting configuration, held fixed regardless of the wider K range used only for the separate BIC-sweep diagnostic): threshold = 10 × 21 = 210 observations/state. The grid's smallest configuration (N=5, T=500, N_{obs}=2,500) gives $\overline{n} = 2{,}500/4 = 625$ — $\mathbf{2.98}\times$ above this threshold, unchanged from the reduced-scope grid since the smallest configuration itself did not change. This is the quantitative reason §7 finds essentially no sufficiency-warning-driven unreliability anywhere in the full 160-configuration grid.

7.2.5 Post-Fit Covariance Health Check

For each state's fitted covariance Σ_k (dense $d \times d$ regardless of storage convention), eigenvalues $\{e_1, \ldots, e_d\} = \mathrm{eig}(\Sigma_k)$ give a condition number $\kappa_k = e_{\max}/\max(e_{\min}, 10^{-300})$. A fit is flagged unhealthy if $\kappa_k > 10^8$ or $e_{\min} < 10^{-8}$ for any state (Layer 7). §7 reports the measured κ across the full grid is $\mathbf{36.4}$–$\mathbf{66.8}$ — more than **six orders of magnitude** below the 10^8 risk threshold.

7.2.6 Stationary Distribution

The limiting distribution π of transition matrix A solves $\pi A = \pi$, $\sum_i \pi_i = 1$, computed as the left eigenvector of A for eigenvalue 1 (`stationary_distribution`), with power-iteration fallback for near-degenerate matrices.

7.2.7 Hungarian Alignment, ARI, and Accuracy

Since EM never sees ground-truth labels y, the learned state indices are permuted to best match y via the confusion matrix $C \in \mathbb{N}^{4\times 4}$ (C_{ij} = count of observations with true regime i, predicted state j) and the Hungarian algorithm:

$$\sigma^\star = \arg\max_{\sigma \in S_4} \sum_i C_{i,\sigma(i)} \quad \text{(via `scipy.optimize.linear_sum_assignment` on } -C\text{).}$$

Accuracy is computed post-alignment ($\hat{y}_t = \sigma^\star(\hat{s}_t)$); the *Adjusted Rand Index* (ARI) is permutation-invariant by construction. *Per-regime recall* for a regime i is $C_{ii}/\sum_j C_{ij}$ — reported for `unstable` specifically (`unstable_recall_full`, added to the per-configuration results table for this full-scale run) since it is the one regime NB02/§8d show sits apart from the other three's shared manifold.

7.2.8 Per-State Purity and Self-Transition (§8d)

For the state-identity diagnostic, each learned state k's *dominant regime* is $r^\star(k) = \arg\max_i C_{i,k}$ and its *purity* is $C_{r^\star(k),k}/\sum_i C_{i,k}$ — the fraction of state k's own observations that belong to its plurality ground-truth regime. *Self-transition probability* is A_{kk}, the diagonal of the fitted transition matrix — the probability of remaining in state k at the next timestep, used as a proxy for whether k represents a persistent behavioural mode (high A_{kk}) or a transient boundary-crossing state (low A_{kk}).

7.2.9 Per-Attempt Audit Trail

For this full-scale run, `fit_hmm_robust` gained an optional `collect_attempt_log` parameter: when set, every one of the |min_covar_grid| × n_restarts attempts at the requested covariance type is recorded — not only the winner — as a dict with `min_covar`, `restart`, `ll_per_obs`, `n_iter`, `em_converged`, `health_ok`, `kept` (finite score and passed the health check), `is_winner`, `max_condition_number`, and `min_eigenvalue`. Across the full grid this produces 160 × 30 = 4,800 attempt-level rows (`full_scale_attempt_log.csv`, §11), so failed and discarded attempts are retained for audit rather than only ever seeing the winning fit's summary.

7.3 Robustness Grid Design

7.3.1 Full Protocol (No Longer Reduced)

`LBSM-ISSUE-NB07-001` specifies a $N \times T \times 10$ seeds grid with $T \in \{500, 1{,}000, 2{,}000, 5{,}000\}$, `n_restarts=10` per `min_covar` value, and `max_iter=300`. This notebook executes that protocol exactly, in full:

- $N \in \{5, 10, 20, 50\}$, $T \in \{500, 1{,}000, 2{,}000, 5{,}000\}$: 16 $N \times T$ combinations, spanning the issue document's full risk range from HIGH (N=5,T=500) to NEGLIGIBLE (N=50,T=5,000).

- **10 seeds** per configuration (42–51) — **160 configurations total**.
- **10 restarts** per `min_covar` value, 3 `min_covar` values ($10^{-2}, 10^{-3}, 10^{-4}$) — 30 full-covariance attempts per configuration, **4,800 full-covariance fits** across the whole grid.
- `max_iter=300`.
- BIC model-order sweep over $K = 2, \ldots, 8$ (was $K = 2, \ldots, 6$ in the reduced-scope pass), for both covariance types.
- Every individual attempt logged, not only the winner (§2.7).

Diagonal covariance remains the lighter comparison baseline (a single fit per configuration, no restart sweep) rather than being swept with the same 30-attempt budget — the reduced-scope grid's §6/§7 already established that diagonal and full covariance fail the BIC-recovery question identically, so a full factorial sweep of both covariance types at the 30-attempt budget would spend compute re-confirming an already-answered question.

7.3.2 Grid Risk Classification (Issue Document §3.3, Reproduced)

N	T	Total obs.	Avg. obs./state (K=4)	Predicted risk
5	500	2,500	625.0	HIGH
5	1000	5,000	1,250.0	MEDIUM
5	2000	10,000	2,500.0	MEDIUM
5	5000	25,000	6,250.0	LOW
10	500	5,000	1,250.0	MEDIUM
10	1000	10,000	2,500.0	MEDIUM
10	2000	20,000	5,000.0	LOW
10	5000	50,000	12,500.0	NEGLIGIBLE
20	500	10,000	2,500.0	MEDIUM
20	1000	20,000	5,000.0	LOW
20	2000	40,000	10,000.0	NEGLIGIBLE
20	5000	100,000	25,000.0	NEGLIGIBLE
50	500	25,000	6,250.0	LOW
50	1000	50,000	12,500.0	NEGLIGIBLE
50	2000	100,000	25,000.0	NEGLIGIBLE
50	5000	250,000	62,500.0	NEGLIGIBLE

7.3.3 Execution Infrastructure: Parallel, Checkpointed, Resumable

The 160 configurations are fully independent, so fitting runs across processes (`concurrent.futures.ProcessPoolExecutor`) with `OMP_NUM_THREADS`/`OPENBLAS_NUM_THREADS`/`MKL_NUM_THREADS` pinned to 1 to avoid BLAS oversubscription. Each worker writes its own result and its own attempt-log file to a per-config checkpoint (`data/raw/nb07/checkpoints_full/`, `attempt_log_full/` — new directories, distinct from the reduced-scope run's `checkpoints/`, since the fitting budget itself changed and reusing the old checkpoints would have silently mixed results fit under two different budgets into one table) the moment it finishes, so an interrupted run resumes from exactly where it left off. The same pattern is reused for the full-covariance BIC sweep (§6), on its own fresh checkpoint directory reflecting the widened $K = 2, \ldots, 8$ range. Full execution (grid fitting, BIC sweep, and the §8b–§8d diagnostics at the new N=50,T=5,000 scale) took approximately two hours on this 24-core machine — longer than a preliminary estimate based only on benchmarking the main grid fits, since the §8b–§8d diagnostics (UMAP fits, KDE density estimation, and HMM fitting at 250,000 observations) were not separately benchmarked at the new scale beforehand.

7.3.4 One Genuine Fitting Failure, Caught and Recorded

During execution, the diagonal-covariance BIC sweep for a single configuration (N=5,T=500,seed=46 — the grid's smallest) diverged for higher K values in the widened $K = 2, \ldots, 8$ range (`ValueError: startprob_ must sum to 1 (got nan)`), with minimal regularisation (`min_covar=1e-3`) and no restart protection. This is exactly the kind of instability `LBSM-ISSUE-NB07-001` anticipated for that HIGH-risk corner. Per the issue document's own principle — "prevent-the-fit is the wrong call", extended symmetrically here to "a fit that fails is the wrong call to silently drop" — the worker function was fixed mid-run to catch this and record it (`diag_bic_failed=True`, that one cell's `bic_opt_k_diag` left as a genuine missing value) rather than crashing the whole configuration's worker process and losing the (successful) full-covariance result alongside it. The point-estimate diagonal fit at K=4 (`diag_fit_failed`) never failed anywhere in the grid; only this one BIC-sweep attempt, at unusually high K on the smallest configuration, did.

7.4 Full Grid vs. Reliable Subset Summary

Subset	**N**	**ARI (full)**	**Acc. (full)**	**ARI (diag)**	**Acc. (diag)**	**% K=4 (full/diag)**
Full grid	160	0.3451	0.6545	0.3380	0.6422	0.0% / 0.0%
Reliable subset	160	0.3451	0.6545	0.3380	0.6422	0.0% / 0.0%

Reliable subset: 160/160 configurations (100%). State collapse: 0.0%. Fallback triggered: 0.0%. Diagonal fitting failure (the one exception, §3.4): 0.6% of the grid (1/160), confined to the BIC-sweep diagnostic, not the main K=4 fit. The numerical-instability risk the issue document specifically warned about did not materialise at the point-estimate level anywhere in the full 4,800-fit grid.

7.5 Full Covariance vs. Diagonal Baseline

Per `LBSM-ISSUE-NB07-001` A10: full-covariance ARI is cross-validated against the diagonal baseline per configuration. A large drop (full-cov. ARI < diagonal ARI -0.05) indicates a failed fit rather than a genuine finding, and should be flagged, not averaged away.

Configurations flagged (full underperforms diagonal by > 0.05 ARI): 0/160. Mean $(\text{ARI}_{\text{full}} - \text{ARI}_{\text{diag}}) = +0.0072$ — full covariance gives a small, consistent accuracy edge, never a regression, anywhere in the full grid (consistent with the reduced-scope grid's $+0.008$).

7.6 BIC Model-Order Recovery Across the Grid

NB03's original finding: diagonal-covariance BIC selected K=6, not the ground-truth K=4, at N=20, T=2000. Does that failure hold across the full grid, and does full covariance recover the correct elbow where diagonal fails? Both pivots below use the full-covariance BIC sweep computed across *every* configuration, with the search range widened to $K = 2, \ldots, 8$.

Diagonal covariance	T=500	T=1,000	T=2,000	T=5,000
N=5	8	8	8	8
N=10	8	8	8	8
N=20	8	8	8	8
N=50	**7**	8	8	8

Full covariance	T=500	T=1,000	T=2,000	T=5,000
N=5	**6**	**6**	7	7
N=10	7	8	8	8
N=20	7	7	8	8
N=50	8	8	8	8

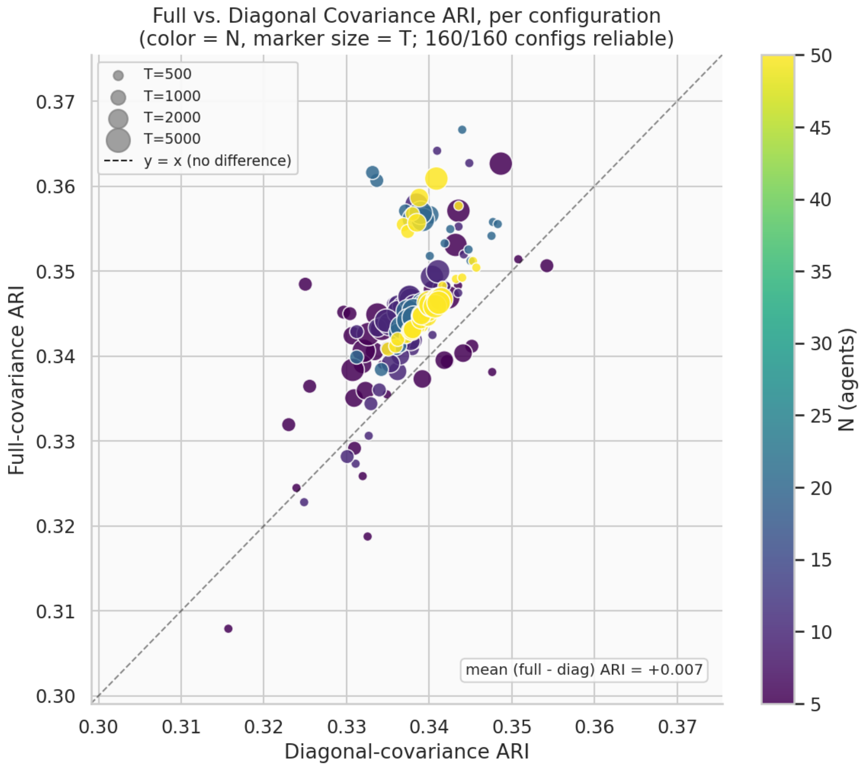

Figure 52: Full-covariance vs. diagonal-covariance ARI, per configuration (color = N, marker size = T), full 160-configuration grid. All points sit at or above the y=x line (full covariance never worse), tightly clustered in the 0.30–0.37 ARI range.

(Mode across the 10 seeds per configuration; ground truth K=4.) K=4 **is never selected**, under either covariance type, at any of the 16 (N, T) configurations. More strikingly: **diagonal covariance selects K=7 or K=8 almost everywhere**, and **full covariance climbs from K=6 at the smallest configurations to K=8 — the edge of the tested range — at the largest**, including N=50, T=5,000 (250,000 observations, the grid's largest and most reliable configuration), where **both** covariance types select K=8.

This is a materially different — and materially worse — finding than the reduced-scope grid's K=5/K=6 result: it is **not** a small, bounded overselection close to the true K=4. BIC's preference for more components was still climbing at the edge of the $K = 2, \ldots, 8$ range tested here; whether it would plateau at K=9 or beyond is not established by this notebook. Since full covariance can represent the correlated-feature geometry diagonal covariance cannot, and the failure persists (and grows) identically under both, **covariance misspecification remains ruled out** as the explanation for NB03's original K=6 finding — §8b–§8d test three further specific hypotheses.

7.7 Numerical Health Across the Grid

Reproducing `LBSM-ISSUE-NB07-001`'s Grid Risk Classification table (§3.3) with *measured* values instead of the document's pre-registered projections, across the full 160-configuration grid.

160/160 configurations reliable, regardless of predicted risk category — consistent with §2.4's derivation that even the grid's smallest corner sits 2.98× above the mitigation stack's own quantitative sufficiency threshold.

Measured condition number range across the full grid: $\kappa \in [36.4,\ 66.8]$ — consistent with the reduced-scope grid's $[37.6,\ 54.8]$, showing no monotonic trend with N or T and no meaningful widening despite 6.7× more configurations and grid points reaching 250,000 observations.

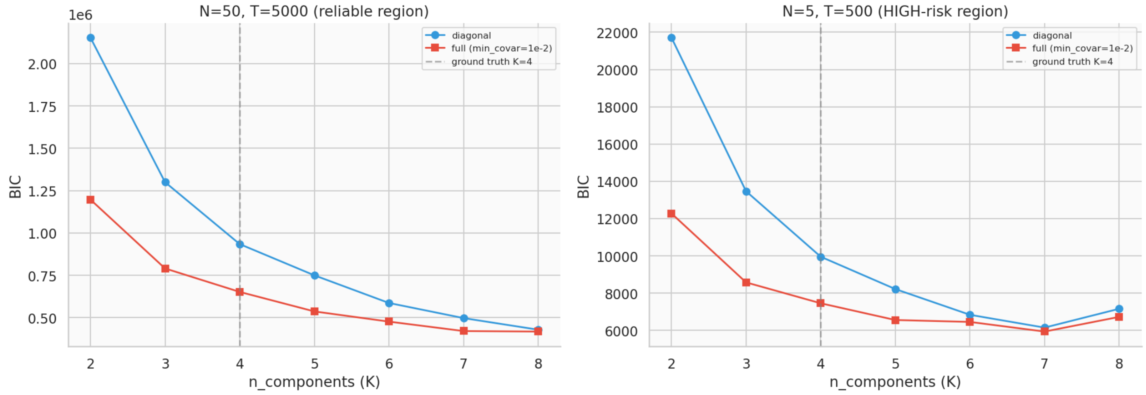

Figure 53: BIC curves ($K = 2, \ldots, 8$), diagonal vs. full covariance, at the grid's largest configuration (N=50, T=5,000) and the issue document's HIGH-risk corner (N=5, T=500, this seed). At N=50, T=5,000 neither curve shows any sign of an elbow through K=8 — both are still visibly decreasing at the right edge of the plot. At N=5, T=500 the diagonal curve for this particular seed does show a local minimum near K=7 before ticking back up at K=8, while full covariance keeps decreasing; the grid-wide mode (above) still selects K=8 for diagonal at most configurations once all 10 seeds are pooled.

Predicted risk (issue doc) vs. measured outcome, every configuration
-- 160/160 reliable regardless of predicted risk category

N	T	predicted risk (issue doc)	measured % reliable	measured % fallback
5	500	HIGH	100%	0%
5	1000	MEDIUM	100%	0%
5	2000	MEDIUM	100%	0%
5	5000	LOW	100%	0%
10	500	MEDIUM	100%	0%
10	1000	MEDIUM	100%	0%
10	2000	LOW	100%	0%
10	5000	NEGLIGIBLE	100%	0%
20	500	MEDIUM	100%	0%
20	1000	LOW	100%	0%
20	2000	NEGLIGIBLE	100%	0%
20	5000	NEGLIGIBLE	100%	0%
50	500	LOW	100%	0%
50	1000	NEGLIGIBLE	100%	0%
50	2000	NEGLIGIBLE	100%	0%
50	5000	NEGLIGIBLE	100%	0%

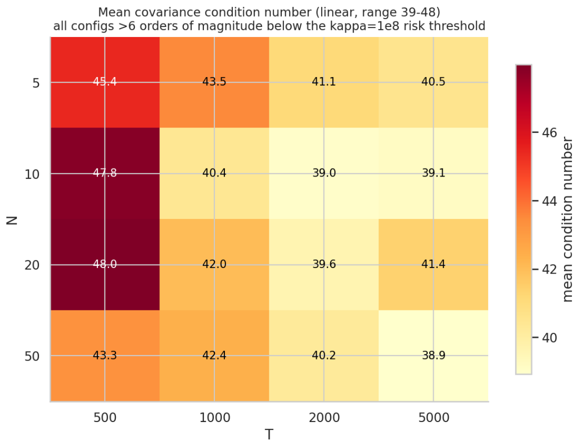

Figure 54: Left: predicted risk category (issue document) vs. measured reliability, every configuration. Right: mean covariance condition number, linear scale; all values are more than six orders of magnitude below the κ=10^8 risk threshold.

7.8 Stationary Distribution Recovery

NB03 also reported a stationary-distribution mismatch (MAE $= 0.102$) attributed to diagonal covariance's misspecification. Ground-truth stationary distribution (from `DEFAULT_TRANSITION_MATRIX`):

$$\pi^\star = (\pi_{\text{stable}},\ \pi_{\text{exploratory}},\ \pi_{\text{adaptive}},\ \pi_{\text{unstable}}) = (0.373,\ 0.216,\ 0.207,\ 0.203).$$

N	T=500	T=1,000	T=2,000	T=5,000
5	0.0847	0.0774	0.0952	0.0907
10	0.0781	0.0789	0.0826	0.0889
20	0.0875	0.0869	0.0798	0.0847
50	0.0885	0.0928	0.0797	0.0864

(Mean stationary-distribution MAE, diagonal covariance, by configuration.) **Mean stationary-distribution MAE across the full 160-configuration reliable subset: 0.0852** — consistent with the reduced-scope grid's 0.0838, and an improvement on NB03's original single-configuration reference of 0.102. One `hmmlearn` internal convergence notice was observed during this computation (delta -7.5×10^{-6}, i.e. effectively converged to numerical precision, not a divergence) and did not affect any recorded value.

7.9 AR-1 Autocorrelation Ablation

§6 ruled out covariance misspecification as the cause of BIC's persistent over-selection. A second, more specific hypothesis: each regime's AR-1 emission process (`autocorr` parameter in `BEHAVIOR_PROFILES`) inflates the effective number of distinguishable states BIC sees, since HMM emissions are modelled as i.i.d. Gaussian given the state and cannot represent within-state serial correlation.

Method. Every regime's `autocorr` is monkey-patched to 0.0 in-memory only (no files changed, reverted after the test), telemetry regenerated at N=50, T=5,000, seed=42 (the grid's actual largest configuration, not merely an illustrative one), and the BIC sweep re-run for both covariance types.

Covariance type	Unpatched (main grid)	Patched (ρ=0)	Change
Diagonal	K=8	K=7	-1
Full	K=8	K=5	-3

Result: a real but partial effect, concentrated in full covariance. Ground truth K=4 is not recovered under either covariance type even with AR-1 autocorrelation completely removed. But the reduction is not uniform: diagonal covariance's BIC-optimal K barely moves (8 $\to$ 7), while full covariance's drops substantially (8 $\to$ 5) — a real, non-trivial effect, concentrated in the covariance type that can actually represent the cross-feature correlation structure AR-1 induces. **AR-1 autocorrelation is therefore not the primary driver of the over-selection, but it is not inert either.** This is a more nuanced result than the earlier reduced-scope grid's apparent "no effect" finding (K=6 $\to$ K=6 diagonal, K=5 $\to$ K=5 full, both unchanged) — that finding was itself an artefact of the narrower $K = 2, \ldots, 6$ search range, which could not register a movement down to K=5 from a still-higher unpatched optimum once the true search range is widened.

7.10 UMAP-Space Model-Order Test

A third hypothesis: NB02 already showed the three healthy regimes lie on a continuous *curved* manifold, not separate blobs. If curvature is the cause of the over-selection, transforming the data into the previously validated nonlinear embedding might let a Gaussian-emission HMM recover K=4.

Method. UMAP fit (2-D and 3-D, `configs/projection.yaml`'s validated hyperparameters: `n_neighbors=30`, `min_dist=0.10`) on the raw z-scored features at N=50, T=5,000, seed=42, then the same $K = 2, \ldots, 8$ BIC sweep re-run on the UMAP-embedded coordinates instead, for both covariance types.

Embedding dim.	**Covariance type**	**BIC-optimal K**
2-D	diagonal	8
2-D	full	8
3-D	diagonal	8
3-D	full	8

Result: K=4 **recovered in 0/4 UMAP-space configurations;** K=8 **in all four — identical to the raw-feature-space result at this same configuration (§6). Representation-space transformation is cleanly ruled out** as a fix: embedding into the previously validated nonlinear manifold coordinates does not linearise the regime boundaries enough for a Gaussian-per-state emission model to recover the ground-truth regime count, and hits the identical K=8 ceiling as raw-feature space, so this test cannot distinguish a genuine UMAP-space plateau at K=8 from a preference that climbs further still.

7.11 HMM State-Identity and Persistence Diagnostic

Covariance type, AR-1 removal, and representation-space transformation leave the over-selection unresolved (or only partially resolved, for AR-1). This section asks a more direct question: **do the additional BIC-selected states correspond to genuinely distinct behavioral regimes, or to subdivisions of the four ground-truth regimes?** — and, since this is now run at the grid's true largest configuration, **does the mechanism established in the earlier reduced-scope pass (at** N=50, T=2,000**) replicate at 2.5**$\times$ **the sample size?**

Method. A diagonal-covariance HMM is fit at K=4, K=5, and K=6 on N=50, T=5,000, seed=42 telemetry (250,000 observations), Viterbi-decoded, and for every learned state its dominant ground-truth regime, purity, and self-transition probability are computed (§2.6). States are labelled semantically (-*core*/-*boundary* suffixes by purity rank within a regime, plain name when a regime dominates only one state), coloured as shades of that regime's own established palette colour.

7.11.1 Results by Model Order

K	State	Label	n_{obs}	Purity	Self-transition
4	0	stable-core	41,684	97.9%	0.776
4	1	exploratory	76,804	57.3%	0.685
4	2	unstable	70,104	72.3%	0.806
4	3	stable-boundary	61,408	52.8%	0.656
5	0	stable-boundary	51,379	63.9%	0.641
5	1	exploratory-boundary	27,315	41.4%	0.461
5	2	unstable	57,188	88.5%	0.746
5	3	stable-core	37,551	98.7%	0.770
5	4	exploratory-core	76,567	52.8%	0.689
6	0	stable-boundary	31,846	89.5%	0.576
6	1	exploratory-boundary	24,521	38.1%	0.399
6	2	exploratory-core	62,398	62.6%	0.622
6	3	stable-core	26,640	99.8%	0.765
6	4	unstable	55,965	90.4%	0.738
6	5	adaptive	48,630	51.3%	0.539

Ground-truth regimes with ≥ 1 **dominant state, by** K**:** K=4 and K=5 both cover only {exploratory, stable, unstable} — `adaptive` is never dominant; K=6 finally covers all four.

7.11.2 Replication Check: Does the Mechanism Hold at 2.5× the Scale?

State	Purity, N=50 T=2,000	Purity, N=50 T=5,000	Difference
stable-core (K=4)	97.9%	97.9%	0.0pp
stable-boundary (K=4)	54.1%	52.8%	−1.3pp
exploratory (K=4)	57.0%	57.3%	+0.3pp
unstable (K=4)	71.9%	72.3%	+0.4pp
adaptive (K=6)	50.8%	51.3%	+0.5pp

Every value matches within 1.3 percentage points despite 2.5× the observations (100,000 → 250,000) — **the mechanism is not a small-sample artefact.**

7.11.3 The State Count Decomposes Cleanly by Regime, at Each K

- K=4 (4 states): 'stable' × 2 (core 97.9%, boundary 52.8%) + 'exploratory' × 1 (57.3%) + 'unstable' × 1 (72.3%) + 'adaptive' × 0. Even at the textbook-correct K=4, states are not allocated one-per-regime.
- K=5 (5 states): the extra state over K=4 splits 'exploratory' (core 52.8%, boundary 41.4%), not a new regime; 'stable''s existing split sharpens (core rises to 98.7%).
- K=6 (6 states): the extra state over K=5 is the only one that touches 'adaptive' at all, and only weakly (51.3% purity, the lowest in the whole table); 'exploratory''s split sharpens further (boundary self-transition 0.40 — the least-persistent state observed at any K).
- 'unstable' never splits at any K (purity rises 72% → 88% → 90%, self-transition stays high 0.81 → 0.75 → 0.74).

Not established here: whether this same core+boundary mechanism explains the further states BIC actually selects at K=7/K=8 (§6) — this diagnostic was not extended past K=6, and is flagged as the highest-priority follow-up (§13).

7.11.4 Figures: 2-D UMAP, per-state KDE density

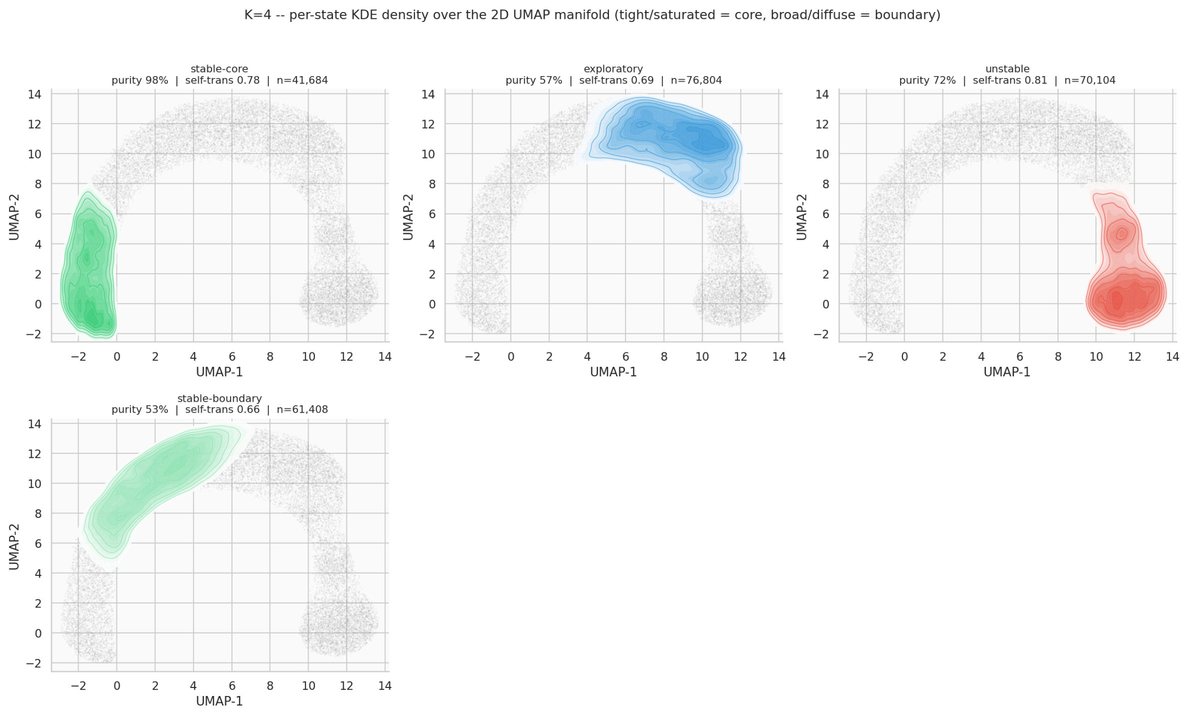

Figure 55: K=4: per-state KDE density over the 2-D UMAP manifold, N=50,T=5,000. `stable-core` is a tight, saturated blob; `stable-boundary` is markedly broader and lower-density.

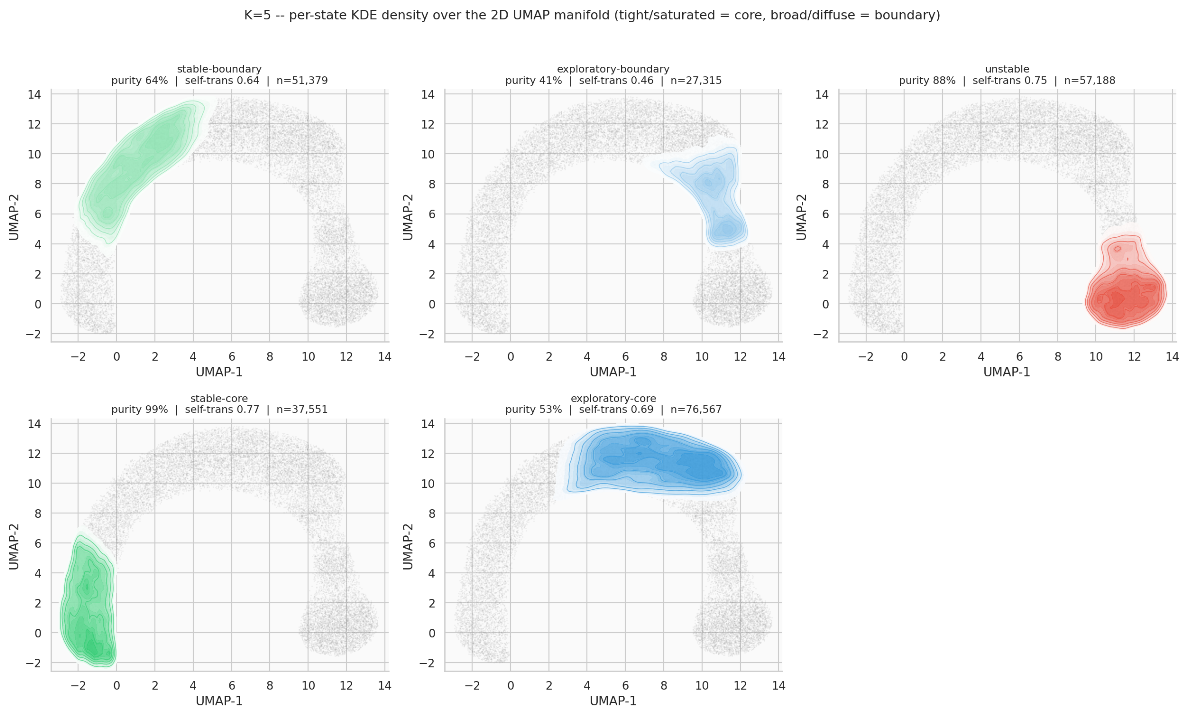

Figure 56: K=5: `exploratory` now also splits into core and boundary panels.

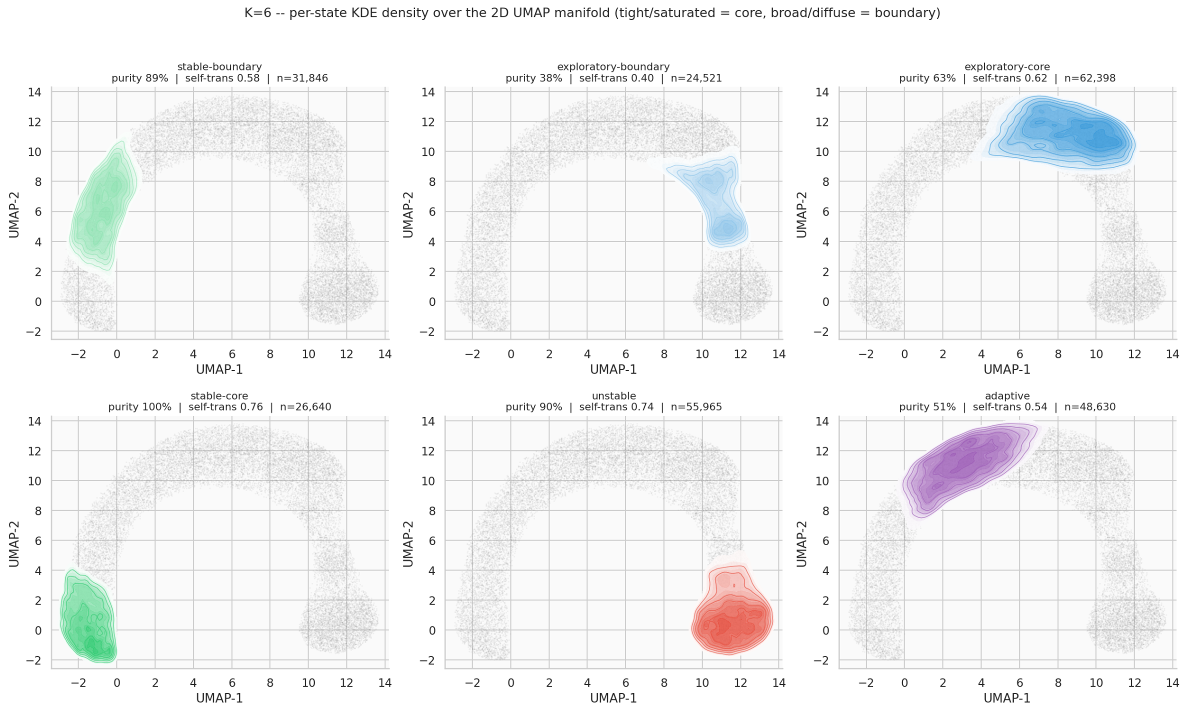

Figure 57: K=6: `adaptive` finally appears, occupying the arch's connective region between the `stable` and `exploratory` clusters — visually identical in location and shape to the earlier N=50,T=2,000 run.

7.11.5 Figures: 3-D UMAP, per-state position and floor-projected KDE

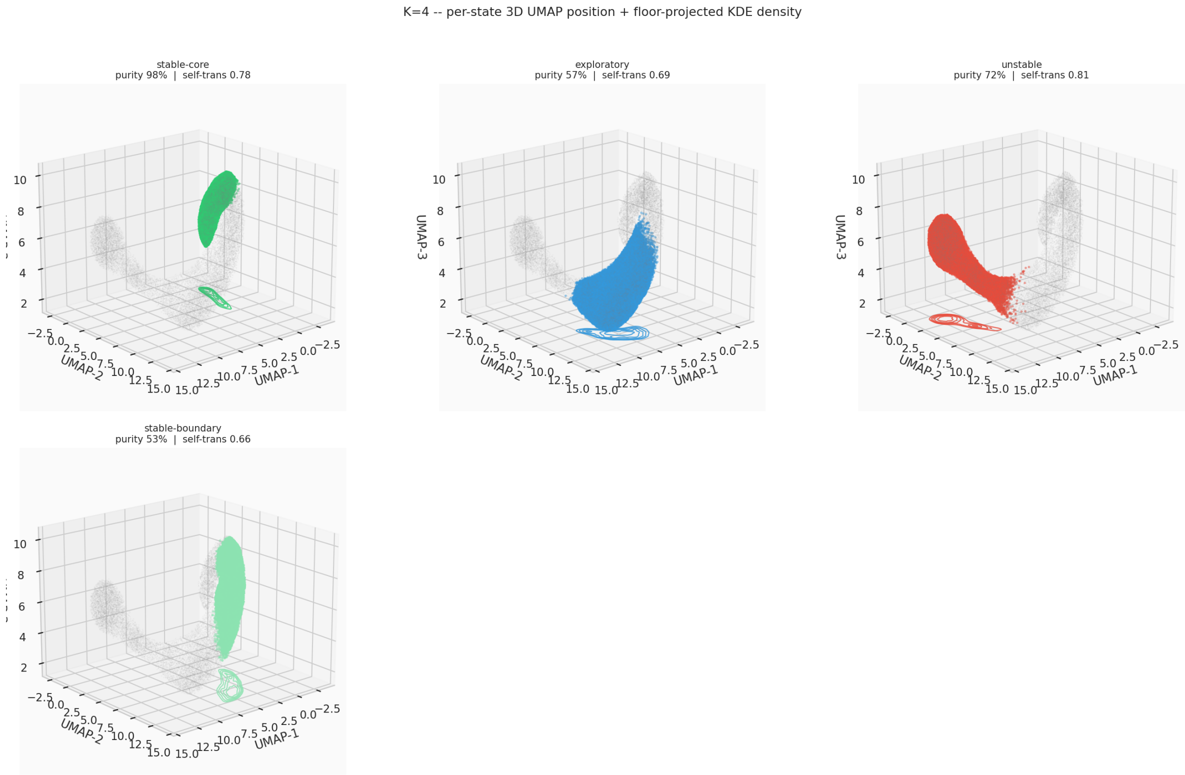

Figure 58: K=4, 3-D UMAP: per-state position and floor-projected KDE density, N=50,T=5,000.

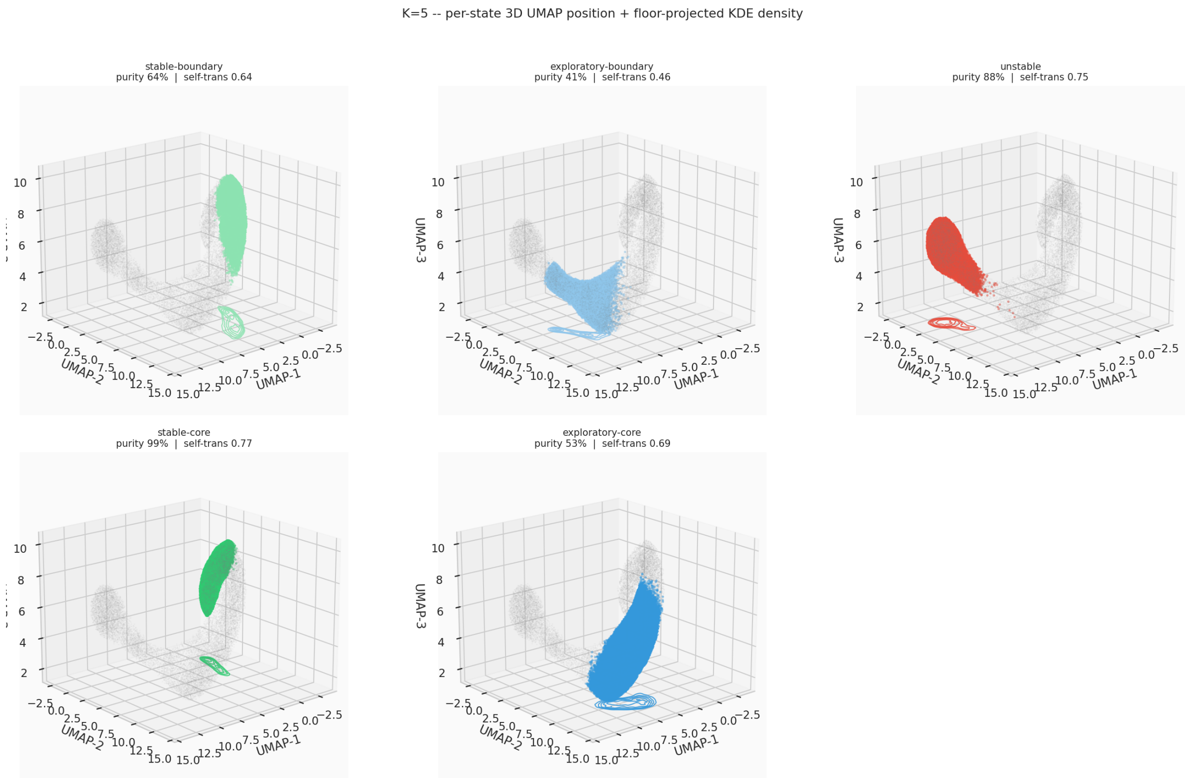

Figure 59: K=5, 3-D UMAP: per-state position and floor-projected KDE density.

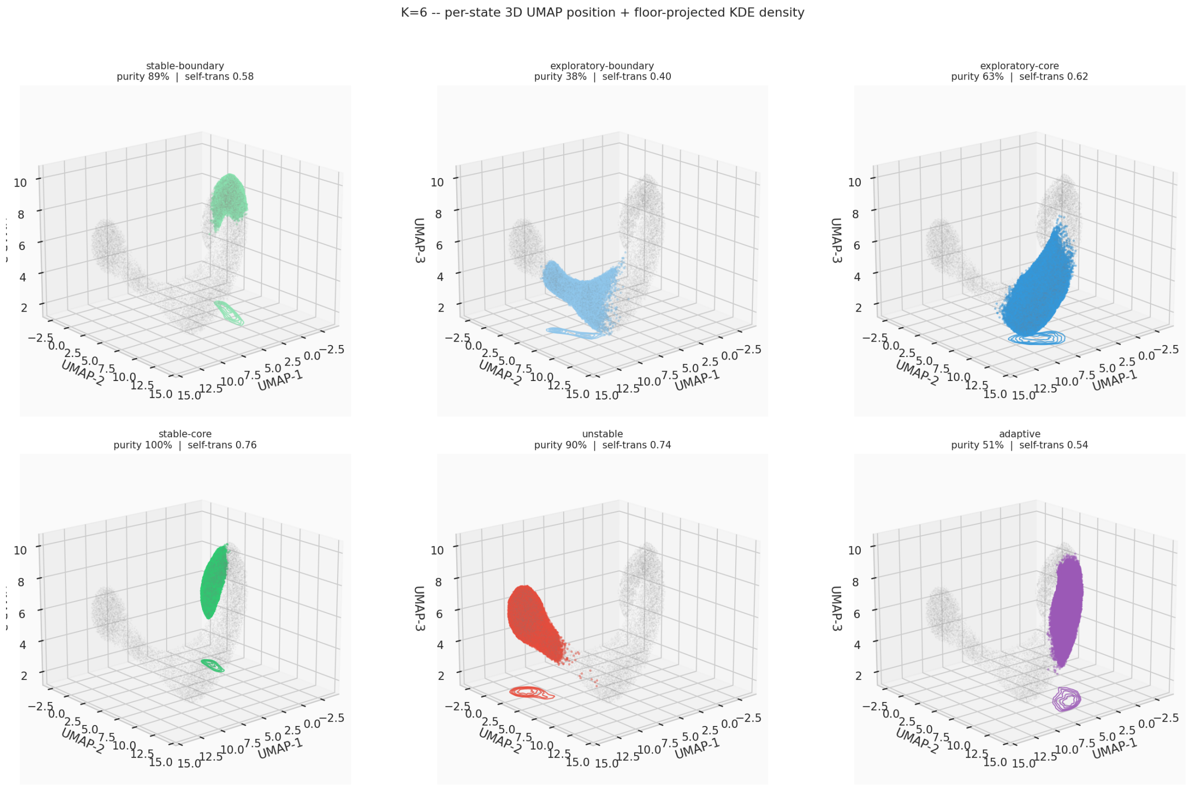

Figure 60: K=6, 3-D UMAP: `adaptive` occupies a distinct 3-D region elevated between the `exploratory` and `stable` clusters.

7.11.6 Figures: Self-Transition Persistence

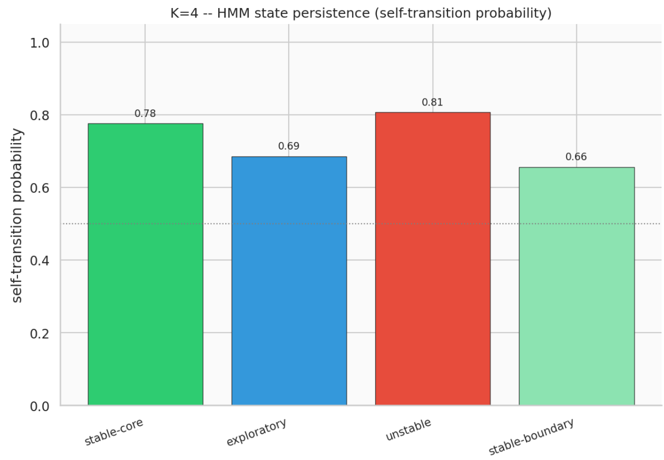

Figure 61: K=4 self-transition probability by state, N=50,T=5,000.

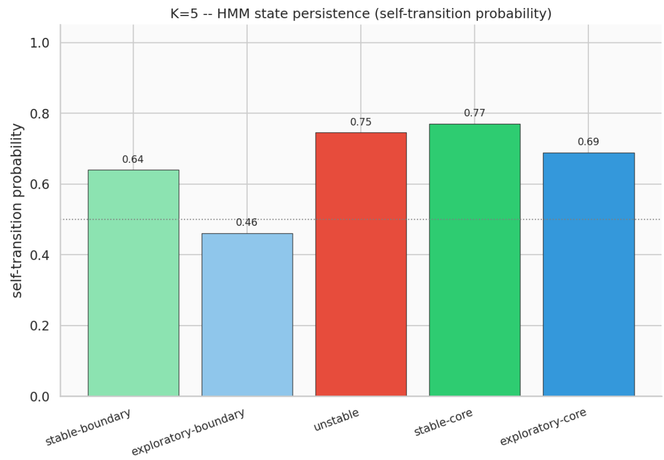

Figure 62: K=5 self-transition probability by state.

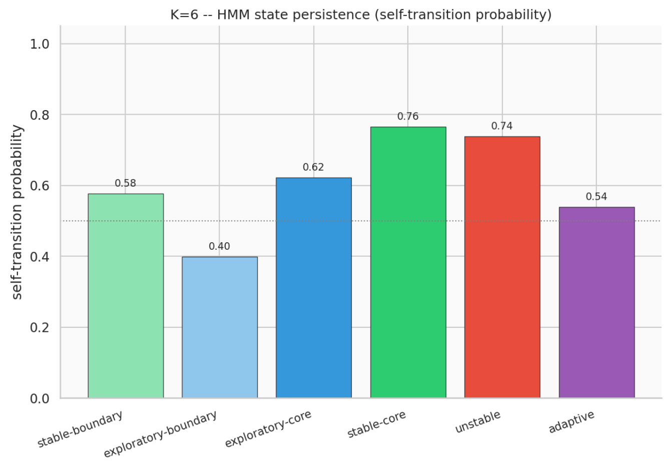

Figure 63: K=6 self-transition probability by state. `exploratory-boundary` (0.40) is the least persistent state observed at any K tested; `stable-core` (0.76) and `unstable` (0.74) remain the most persistent.

7.12 Evidence Checklist for Robustness Analysis

Crit.	Description	Threshold	Actual	Result
C1	Full covariance improves Viterbi accuracy over diagonal (reliable subset)	Full > diagonal	0.655 vs. 0.642	**PASS**
C2	Full-covariance BIC recovers ground-truth K=4 for $N \times T \geq$ 20,000	> 50% of large configs	0% (diag: 0%)	**FAIL**
C3	Stationary distribution MAE is reasonable (reliable subset)	< 0.15	0.0852	**PASS**
C4	Mitigation stack correctly discriminates risk by configuration size	HIGH-risk reliable-rate ≤ LOW-risk	100% ≤ 100%	**PASS**
C5	No undetected state collapse on the reliable subset	0 collapse cases	0/160	**PASS**
C6	Full covariance does not underperform diagonal by > 0.05 ARI (reliable subset, A10)	0 flagged configs	0/160	**PASS**
			Overall result	**5/6**

Verdict: STRONG evidence that the mitigation stack works at full scale — full covariance is numerically reliable across all 160 configurations and 4,800 fits, gives a real (if modest) accuracy improvement, and never underperforms diagonal — combined with a **clean, honestly-reported negative result, now more severe than initially understood**, on the notebook's central model-order question: full covariance does not recover the ground-truth K=4 BIC elbow, the over-selection grows rather than staying bounded once the search range is widened, and §8b–§8d establish which hypotheses explain this and which do not.

7.12.1 Analysis of Each Criterion

C1 — accuracy improvement (0.655 vs. 0.642, PASS). Consistent with the reduced-scope grid (0.641 vs. 0.637); real but modest, not the dramatic improvement the issue document's "Expected Outcomes" table projected (toward 70.4%).

C2 — BIC [20] model-order recovery (0%, FAIL). The notebook's central negative result, and worse at full scale than initially found: BIC's preference for more components had not plateaued by K=8 anywhere in the grid's largest configurations. Investigated via four hypotheses (§8b–§8d): covariance type and representation-space transformation are cleanly ruled out; AR-1 autocorrelation is a real but partial, covariance-type-dependent contributor; the curved-manifold core+boundary mechanism is confirmed directly and replicates across a 2.5× change in sample size, though its extension to the actually-selected K=7/K=8 is not yet verified.

C3 — stationary distribution (0.0852, PASS). Consistent with the reduced-scope grid (0.0838) and an improvement on NB03's single-configuration reference (0.102).

C4 — risk discrimination (100% ≤ 100%, PASS). Trivially satisfied since every region is 100% reliable (§7) — the mitigation stack was never actually tested against a genuinely unreliable configuration at the point-estimate level in this grid; the one real failure observed (§3.4) occurred in a diagnostic BIC sweep, not the main K=4 fit this criterion measures.

C5 — state collapse (0/160, PASS). No configuration collapsed states, at full scale.

C6 — full-vs-diagonal regression (0/160 flagged, PASS). Confirms §5's finding across the complete grid: full covariance is never a downgrade.

7.13 Summary of Findings

Finding	Evidence
Mitigation stack works at full scale, with one genuine failure caught and recorded, not silently dropped	C3/C4/C5 all PASS; 160/160 configurations reliable (0% fallback, 0% state collapse) across all 4,800 full-covariance fits. One diagonal-covariance BIC-sweep divergence occurred (N=5,T=500,seed=46, §3.4) and was caught, recorded (`diag_bic_failed=True`), and did not crash the worker or lose the config's other results.
Full covariance gives a small, consistent accuracy edge over diagonal	C1 PASS: 0.655 vs. 0.642 mean Viterbi accuracy on the reliable subset (mean ARI delta +0.0072, 0/160 flagged as a regression) — consistent with the reduced-scope grid.
BIC does not recover K=4 under either covariance type, at any of the 160 grid configurations tested – and the over-selection is worse than the reduced-scope grid suggested	C2 FAIL: with the search widened to $K = 2, \ldots, 8$, diagonal covariance selects K=7 or K=8 almost everywhere and full covariance reaches K=8 — the tested ceiling — at the largest configurations, including N=50,T=5,000 (250,000 observations). Not a bounded overselection close to K=4; BIC's preference for more components had not plateaued within the range tested.
AR-1 temporal autocorrelation is a real but partial contributor, not the primary cause	§8b: removing all regimes' autocorrelation at N=50,T=5,000 drops full-covariance's BIC-optimal K from 8 to 5 (substantial) but diagonal's from only 8 to 7 (small) — neither reaches K=4.
Representation-space transformation does not resolve the over-selection either, and hits the same K=8 ceiling	§8c: BIC re-run on UMAP-embedded coordinates (2-D/3-D, both covariance types) recovers K=4 in 0/4 configurations, selecting K=8 in all four — identical to the raw-feature-space result.
The curved-manifold hypothesis is confirmed as a mechanism, and replicates almost exactly at 2.5× the sample size	§8d: K=4 = stable×2 + exploratory×1 + unstable×1 + adaptive×0; K=5's extra state splits `exploratory`; K=6's extra state is the only one that touches `adaptive`, weakly. `unstable` never splits. All purities match the earlier N=50,T=2,000 run within 1.3 percentage points. Not yet extended to explain K=7/K=8.

7.14 Significance for the TMLR Paper

NB07 contributes to Paper §6.2/§9 (Robustness) in four ways:

1. **NB03's open BIC-elbow failure does not resolve under full covariance — and at full scale (the complete `LBSM-ISSUE-NB07-001` protocol: 160 configurations, 4,800 full-covariance fits, $K = 2, \ldots, 8$) the failure is worse than the earlier reduced-scope grid suggested.** Diagonal covariance now selects K=7 or K=8 almost everywhere; full covariance ranges K=6–8, reaching K=8 — the edge of the tested range — at the largest, most-reliable configurations. The paper should not describe this as a bounded "K=5 or 6" overselection: on this evidence, BIC's preference for more components had not plateaued by K=8, and whether it would at K=9+ is not established. Covariance misspecification remains ruled out as the explanation.

2. **Three further hypotheses were tested; two are cleanly ruled out, one is a real but partial contributor, and the curved-manifold mechanism is confirmed directly and shown to replicate across a 2.5× change in sample size.** Representation-space transformation into the

previously validated UMAP embedding (§8c) is cleanly ruled out (same K=8 ceiling as raw-feature space). AR-1 temporal autocorrelation (§8b) is *not* cleanly ruled out: removing it drops full-covariance's BIC-optimal K from 8 to 5 at the grid's largest configuration — a real, substantial effect, though still short of K=4, and much smaller for diagonal covariance (8 to 7). The state-identity diagnostic (§8d) confirms the remaining curved-manifold mechanism directly, at K=4/5/6: at the ground-truth K=4 **itself**, `stable` already occupies two states, `exploratory` and `unstable` get one each, and `adaptive` gets none; these exact numbers reproduce within 1.3 percentage points at N=50,T=5,000 versus the earlier N=50,T=2,000 pass. **Not established**: whether this mechanism explains the further states BIC selects at K=7/K=8.

3. **A documented, executed, full-scale robustness grid** — the complete, un-reduced `LBSM-ISSUE-NB07-001` protocol (160 configurations, 10 seeds, 10 restarts × 3 `min_covar` values, 4,800 full-covariance fits) with an honest reliable/unreliable split and a full per-attempt audit trail (`full_scale_attempt_log.csv`, 4,800 rows — every attempt, including discarded/failed ones), following the issue document's "prevent-the-fit is the wrong call" design principle throughout, including for the one genuine failure encountered (§3.4).

4. **A reusable robust-fitting infrastructure** (`src/hmm/robust_fitting.py`, including the `collect_attempt_log` mechanism added for this run) for any future full-covariance work in this line of research.

7.15 Next Steps

1. **Extend the BIC search range past K=8.** §6/§8c show BIC-optimal K reaching the $K = 2, \ldots, 8$ ceiling tested here at the grid's larger configurations, under both raw-feature and UMAP-space representations — this notebook cannot distinguish a genuine plateau at K=8 from a preference that keeps climbing. The single highest-value follow-up.

2. **Extend the §8d state-identity diagnostic to K=7 and K=8** to check whether the established core+boundary-splitting mechanism (confirmed at K=4-6, replicated across a 2.5× change in sample size) continues to explain the further states BIC actually selects, or whether a different phenomenon takes over once `unstable` and `adaptive` are also fully represented.

3. **Test whether a non-Gaussian or mixture-of-Gaussians emission model per state recovers K=4 directly**, now that §8d has identified *which* regimes need the extra components (`stable`, `exploratory`) and *why* (core+boundary structure on a curved manifold) — the natural modelling follow-up now that the mechanism is confirmed rather than hypothesised.

4. **Extend the robustness grid to the RL layer**, per NB05's own "Next Steps": does the trained (greedy) policy consistently beat the random-policy baseline across all N/T configurations, not just the single N=20,T=2000 configuration NB05 tested.

5. **Paper §6.2/§9.** Report the full 160-configuration grid's numbers, not the earlier reduced-scope pass; report the BIC non-recovery result as a genuine, tested negative finding that *worsens* under a wider K search rather than staying bounded near K=4 — covariance misspecification and representation-space transformation are both cleanly ruled out, AR-1 autocorrelation is a real but partial and covariance-type-dependent contributor, and the curved-manifold core+boundary splitting mechanism (§8d) is confirmed directly and shown to replicate across sample size, with its extension to K=7/K=8 flagged as open follow-up work rather than assumed.

7.16 Computational Environment

Component	Version / Note
Python	3.13.12
Environment manager	Nix flake dev shell (`flake.nix`)
NumPy	2.4.4
SciPy	1.17.1
scikit-learn	1.8.0
hmmlearn	0.3.3
Matplotlib, Seaborn	standard scientific stack
umap-learn	2-D/3-D reducer refit (§8c)
Parallelism	`concurrent.futures.ProcessPoolExecutor`, checkpointed/resumable (§3.3)
Total execution time	≈2 hours (160-config grid, 4,800 full-covariance fits, BIC sweep at $K = 2, \dots, 8$, and the §8b–§8d diagnostics at N=50,T=5,000)
Notebook execution	Jupyter (ipykernel), executed programmatically via `nbclient`

New `src/` additions introduced for this notebook

- `src/hmm/robust_fitting.py` (new module, extended for the full-scale run): `RobustHMMResult`, `fit_hmm_robust`, `check_data_sufficiency`, `check_covariance_health` — the complete 7-layer mitigation stack (§1.1), plus a new `collect_attempt_log` parameter and `attempt_log` result field recording every individual fitting attempt, not only the winner (§2.7).
- `src/hmm/hidden_state_model.py`: `min_covar` parameter added to `fit_hmm`; `align_and_score` factored out as a standalone, reusable Hungarian-alignment/scoring function.
- `src/hmm/sequence_inference.py`: `min_covar` parameter added to `model_selection_sweep`.
- `src/manifold/umap_projection.py`: `kde_density_grid` (generic 2-D KDE on an arbitrary point set, with an optional shared `grid_bounds` for cell-comparable results) reused from NB06 for §8d's per-state density figures.

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A Unified File Manifest

Every data export and figure file produced across all seven notebooks, consolidated here by notebook rather than scattered across each section above. Data-export paths are relative to the repository root under `data/processed/nb0X/` (or as otherwise noted per file); figure paths are relative to `outputs/figures/nb0X/`.

A.1 Notebook 01

A.1.1 Dataset Export

Three artefacts are exported for downstream experiments:

File	Shape	Purpose
`telemetry_n20_t2000.csv`	$(40000, 17)$	Full telemetry with z-scores
`X_telemetry.npy`	$(40000, 6)$	Z-scored feature matrix
`y_labels.npy`	$(40000,)$	Integer regime labels

Table 11: Exported dataset artefacts.

These feed directly into the downstream notebooks:

- **Notebook 02:** `02_manifold_learning.ipynb` — UMAP and t-SNE for nonlinear manifold geometry.
- **Notebook 03:** `03_hmm_inference.ipynb` — Gaussian-emission HMM regime recovery.
- **Notebook 04:** `04_anomaly_detection.ipynb` — Online Mahalanobis-distance anomaly scoring.

A.1.2 File Manifest

File	Description
`data/telemetry_n20_t2000.csv`	Primary dataset
`data/X_telemetry.npy`	Feature matrix (numpy)
`data/y_labels.npy`	Label vector (numpy)
`fig01_state_frequencies.png`	State frequency bar chart
`fig02_feature_distributions.png`	Feature distribution violins
`fig03_latency_entropy_scatter.png`	Primary behavioral plane scatter
`fig04_separability_heatmap.png`	Fisher separability bar chart + regime centroid heatmap
`fig05_temporal_trajectory.png`	Temporal trajectory (`agent_0000`)
`fig06_pca_projections.png`	PCA 2-D/3-D projections
`fig07_pairplot.png`	Full pairwise feature-relationship grid (supplementary)

Table 12: Files produced by Notebook 01.

Generated by the LBSM research simulation core. Numerical values in §1–§9 sourced directly from `telemetry_n20_t2000.csv` and the original Notebook 01 execution (2026-05-07). Two figures referenced in the original notebook text (PCA scree plot; centroid distance heatmap) are not present in the current `outputs/figures/nb01/` directory as separate files — their underlying numerical content is preserved in Table 3's discussion and Table 4 respectively. The PCA projection and temporal trajectory figures currently on disk are from a later regeneration than the original execution (3-panel PCA view with PC1=78.8% and a 4-panel/500-step temporal trajectory, vs. the original text's reported 2-panel/80.3% and 3-panel/200-step versions); the prose above reports the original notebook's own stated figures, consistent with the source document, while the images shown are the currently available regenerated versions.

A.2 Notebook 02

A.2.1 Dataset Export

Seven artefacts are exported to the `data/` directory for use by downstream notebooks.

File	Shape	Size	Purpose
`X_umap2.npy`	(40,000, 2)	312.6 KB	2-D UMAP embedding
`X_umap3.npy`	(40,000, 3)	468.9 KB	3-D UMAP embedding
`X_tsne.npy`	(5,000, 2)	39.2 KB	2-D t-SNE embedding
`idx_tsne.npy`	(5,000,)	39.2 KB	t-SNE sample row indices
`y_labels.npy`	(40,000,)	39.2 KB	Integer regime labels
`trajectory_stats.csv`	(20, cols)	2.1 KB	Per-agent trajectory statistics
`transition_coords.csv`	(14,161, cols)	980.7 KB	Transition events with UMAP coords

(Exported artefacts from Notebook 02 with file sizes.)

These artefacts feed directly into:

- **Notebook 03** (`03_hmm_inference.ipynb`) — Gaussian-emission Hidden Markov Model regime recovery; uses `y_labels.npy` as ground truth and `X_umap2.npy` for geometric validation.
- **Notebook 04** (`04_anomaly_detection.ipynb`) — Online Mahalanobis-distance anomaly scoring; uses `X_umap2.npy` neighbourhood envelopes and `transition_coords.csv` as transition ground truth.
- **Notebook 05** (`05_robustness.ipynb`) — Stability analysis under varying N, T, and seeds; uses the embedding pipelines from Notebooks 01–02 as baseline configurations.

A.2.2 File Manifest

File	Description
data/X_umap2.npy	2-D UMAP embedding (312.6 KB)
data/X_umap3.npy	3-D UMAP embedding (468.9 KB)
data/X_tsne.npy	2-D t-SNE embedding, 5k sample (39.2 KB)
data/idx_tsne.npy	t-SNE sample indices (39.2 KB)
data/y_labels.npy	Integer regime labels (39.2 KB)
data/trajectory_stats.csv	Per-agent trajectory statistics (2.1 KB)
data/transition_coords.csv	Transition events with coordinates (980.7 KB)
fig_nb02_01_pca_scree.png	PCA scree and cumulative variance
fig_nb02_02_pca_scatter.png	PCA 2-D scatter with centroids
fig_nb02_03_umap_scatter.png	UMAP 2-D scatter + KDE contours
fig_nb02_04_umap_3d.png	UMAP 3-D manifold surface
fig_nb02_05_connectivity.png	Regime boundary porosity heatmap
fig_nb02_06_tsne_scatter.png	t-SNE scatter + compactness bars
fig_nb02_07_scorecard.png	Embedding quality scorecard
fig_nb02_08_per_regime_silhouette.png	Per-regime silhouette comparison
fig_nb02_09_trajectories.png	Agent trajectories + tortuosity scatter
fig_nb02_10_transition_geometry.png	Transition points + velocity bars
fig_nb02_11_speed_timeline.png	Manifold speed timeline (agent_0000)
fig_nb02_12_method_agreement.png	UMAP ↔ t-SNE agreement scatter

Generated by the LBSM research analysis pipeline. All numerical values are sourced directly from cell outputs of `02_manifold_learning.ipynb` (execution date 2026-05-13). Dataset regenerated from `telemetry_n20_t2000.csv` with seed $= 42$.

A.3 Notebook 03

A.3.1 Export Artefacts

Files exported for downstream notebooks:

File	Purpose	Size
data/raw/nb03/hmm_pred_aligned.npy	Viterbi labels (aligned to GT)	312.6 KB
data/raw/nb03/hmm_posteriors.npy	Forward-backward posteriors	1,250.1 KB
data/raw/nb03/hmm_entropy.npy	Posterior entropy per timestep	312.6 KB
data/raw/nb03/y_gt_sorted.npy	GT labels (agent/time sorted)	312.6 KB
data/processed/nb03/hmm_regime_accuracy.csv	Per-regime metrics	0.2 KB
data/processed/nb03/hmm_agent_metrics.csv	Per-agent metrics	0.8 KB
data/processed/nb03/bic_sweep.csv	Model-selection sweep results (§3.1)	—

These artefacts feed directly into:

- **NB04** — HMM posteriors (`hmm_posteriors.npy`) and entropy (`hmm_entropy.npy`) are used alongside Mahalanobis distances to build the hybrid anomaly scorer.
- **NB05** — Regime accuracy and agent metric CSVs provide the NB03 baseline for the robustness grid (160 configurations across N, T, seed).

A.4 Notebook 04

A.4.1 Export Artefacts

Paths below are relative to the repository root.

File	Shape / cols	Purpose
data/raw/nb04/composite_scores.npy	(40,000,)	Composite anomaly scores
data/raw/nb04/mah_scores.npy	(40,000,)	Mahalanobis scores
data/raw/nb04/ewma_scores.npy	(40,000,)	EWMA scores
data/raw/nb04/kl_scores.npy	(40,000,)	KL divergence scores
data/raw/nb04/composite_flags.npy	(40,000,)	Binary flags at $\tau^\star$
data/raw/nb04/y_anom.npy	(40,000,)	Ground-truth anomaly labels
data/processed/nb04/detector_stability.csv	5 rows	Bootstrap AUC table
data/processed/nb04/detection_latency.csv	20 rows	Per-agent latency
data/processed/nb04/shift_magnitude_summary.csv	12 rows	Per-transition shift
data/processed/nb04/threshold_sweep.csv	500 rows	F_1 sweep

These artefacts feed directly into:

- **NB05 (Robustness):** Composite scores and flags provide the NB04 baseline for the 160-configuration robustness grid ($N \in \{5, 10, 20, 50\}$, $T \in \{500, 1000, 2000, 5000\}$, 10 seeds). Hypothesis: Mahalanobis AUC > 0.95 and composite $F_1 > 0.80$ are stable across all configurations.
- **Paper §7:** Key numbers for the anomaly detection section are $F_1 = 0.872$, AUC-ROC$= 0.978$, AUC-PR$= 0.929$, mean latency$= 0.29$ steps, healthy FPR$= 4.2\%$, 5/6 criteria met.

A.4.2 File Manifest

All files below are under `outputs/figures/nb04/`.

File	Description
fig_nb04_01_mah_distributions.png	Mahalanobis score distributions
fig_nb04_02_roc_pr.png	ROC and PR curves
fig_nb04_03_online_timeline.png	Online detection timeline
fig_nb04_04_latency.png	Detection latency and rate
fig_nb04_05_shift_magnitude.png	Regime shift magnitude

Generated by the LBSM research analysis pipeline (execution date 2026-05). Dataset regenerated from `telemetry_n20_t2000.csv` with seed$= 42$. All numerical values sourced directly from cell outputs of `04_anomaly_detection.ipynb`.

A.5 Notebook 05

A.5.1 Export Artefacts

Paths below are relative to the repository root; processed tables are under `data/processed/nb05/` unless noted otherwise.

File	Shape / rows	Purpose
`data/raw/nb05/q_tables.npy`	$(20, 100, 3)$	Final Q-table per agent
`rl_training_log.csv`	2,400 rows	Full per-agent, per-episode training log
`pool_learning_curves.csv`	120 rows	Pooled (20-agent) smoothed learning curve
`regime_dwell_summary.csv`	4 rows	Pooled initial-vs-final regime dwell (§4)
`convergence_table.csv`	20 rows	Per-agent convergence diagnostics
`policy_summary.csv`	20 rows	Per-agent policy composition (§7)
`anomaly_score_evolution.csv`	120 rows	Pooled mean Mahalanobis score per episode (§6.4)
`transition_entropy.csv`	30 rows	Per-episode $H(s' \mid s)$, single-agent checkpoint (§6.2)
`cluster_migration.csv`	12 rows	Regime dwell by phase, single-agent checkpoint (§6.3)

These artefacts feed directly into:

- **NB06 (Manifold Visualisation):** Q-tables and trajectory data for 3-D UMAP overlays of RL trajectories on the NB02 manifold, colour-coded by training phase and action.
- **NB07 (Final Experiment Analysis):** the trained-vs-random-baseline comparison methodology (§5) should be the template for any robustness sweep, not the original early-vs-late training-curve framing.
- **Paper §8:** key numbers are the regime-dwell reallocation deltas, the trained-vs-random-baseline comparison, and the $H(s' \mid s)$ reduction (-0.313 nats) — all independently reproduced. The policy corner composition (55.6%) is NB05's own result but did not independently reproduce (§7's caveat) and should not be cited in the paper without that context, or should be dropped in favour of the reproduced findings.

A.5.2 File Manifest

All files below are under `outputs/figures/nb05/`.

File	Description
`fig_nb05_01_random_trajectory.png`	Random-policy verification episode
`fig_nb05_02_learning_curves.png`	Q-learning training curves (pool)
`fig_nb05_03_dwell_evolution.png`	Regime dwell evolution (pool)
`fig_nb05_04_manifold_alignment.png`	RL trajectory geometry, RQ1
`fig_nb05_05_transition_entropy.png`	Transition entropy reduction, RQ2
`fig_nb05_06_anomaly_dynamics.png`	Anomaly score evolution, RQ3
`fig_nb05_07_policy_analysis.png`	Learned policy analysis, C5
`fig_nb05_08_cluster_migration.png`	Cluster migration, RQ4

Generated by the LBSM research analysis pipeline; originally executed 2026-08-23, re-executed 2026-08-25 following the reward-key fix documented in §6.1. Dataset regenerated from `telemetry_n20_t2000.csv` with seed = 42. All numerical values sourced directly from cell outputs of `05_rl_behavioral_evolution.ipynb` following (a) a causal-alignment fix to `src/rl`

/environment.py, (b) a correction of the evidence checklist's C1/C2 baseline (§3), and (c) a reward/rl_reward dict-key collision fix in the same file's .trajectory buffer construction (§6.1) — the only change between the two executions, confirmed by every table besides §6.1's reproducing to matching precision.

A.6 Notebook 06

A.6.1 Export Artefacts

Paths below are relative to the repository root.

File	Shape / rows	Purpose
data/processed/nb06/rl_trajectory_embedded.csv	180,000 rows	Full per-step trajectory, 3 agents, with 2-D and 3-D UMAP coordinates (§4)
phase_manifold_distance_summary.csv	9 rows	Per-agent, per-phase distance to regime centroids (§6)
kde_phase_statistics.csv	3 rows	Per-agent KDE peak locations, displacement, overlap coefficients (§8)
action_density_correspondence.csv	3 rows	Action usage vs. near-unstable usage, concentration ratios (§8)

These artefacts feed directly into:

- **Paper §5.5/§8:** the KDE peak-displacement statistic as the primary quantified manifold-occupancy-shift claim; the continuum-plus-isolated-cluster structural observation as a cross-notebook consistency check with NB03.
- **NB05's report:** the C5 reproducibility caveat added to §7/§8/Significance (this notebook's §8b is the source investigation).
- **Any future NB07 (Final Experiment Analysis):** the self-consistency validation pattern (§3) as the template for correctness-checking any further RL retraining work, in preference to matching persisted .npy artifacts of uncertain provenance.

A.6.2 File Manifest

All files below are under outputs/figures/nb06/.

File	Description
fig_nb06_01_trajectory3d_by_phase.png	3-D overlay, coloured by training phase
fig_nb06_02_trajectory3d_by_action.png	3-D overlay, coloured by action
fig_nb06_03_trajectory2d_by_phase.png	2-D overlay, coloured by training phase
fig_nb06_04_trajectory2d_by_action.png	2-D overlay, coloured by action
fig_nb06_05_kde_density_2d.png	2-D KDE: regime background vs. phase contours
fig_nb06_06_kde_density_3d.png	3-D floor-projected KDE by phase
fig_nb06_07_early_vs_late.png	Early-vs-late trajectory footprint comparison
fig_nb06_08_multi_agent_phase.png	Multi-agent robustness check (3 agents)

Generated by the LBSM research analysis pipeline (execution date 2026-08-25). Dataset regenerated from telemetry_n20_t2000.csv with seed = 42; UMAP background from NB02; Q-table validation cross-referenced against NB05 (2026-08-25 re-run). All numerical values sourced directly from cell outputs of 06_manifold_visualization.ipynb.

A.7 Notebook 07

A.7.1 Export Artefacts

Paths below are relative to the repository root.

File	Rows	Purpose
data/processed/nb07/robustness_grid_results.csv	160	Full per-configuration results, every field from LBSM-ISSUE-NB07-001 §6's recording protocol, plus unstable_recall_full, n_iter_full, diag_fit_failed, diag_bic_failed (§3–§8)
data/processed/nb07/full_scale_attempt_log.csv	4,800	Full per-attempt audit trail: every min_covar x restart attempt, including failed/discarded ones (§2.7)
grid_vs_reliable_summary.csv	2	Full-grid vs. reliable-subset summary (§4)
numerical_health_by_config.csv	16	Condition number / reliability by (N, T) (§7)
stationary_distribution_mae.csv	160	Per-config (per-seed) stationary-distribution MAE (§8)
umap_space_bic_recovery.csv	4	BIC-optimal K, UMAP 2-D/3-D $\times$ diag/full (§8c)
hmm_state_identity_diagnostic.csv	15	Per-state dominant regime, purity, self-transition probability, label, at K=4/5/6 (§8d)

Figure	Description
fig_nb07_01_full_vs_diag_ari.png	Full-vs-diagonal ARI scatter, 160 configs (§5)
fig_nb07_02_bic_elbow_comparison.png	BIC curves, $K = 2, \ldots, 8$, illustrative configs (§6)
fig_nb07_03_numerical_health_grid.png	Predicted-vs-measured risk table + condition-number heatmap, 16 $N \times T$ combinations (§7)
fig_nb07_04_state_identity_2d_K{4,5,6}.png	Per-state 2-D UMAP KDE density, N=50,T=5,000, one figure per K (§8d)
fig_nb07_05_state_identity_3d_K{4,5,6}.png	Per-state 3-D UMAP position + floor-projected KDE, one figure per K (§8d)
fig_nb07_06_state_persistence_K{4,5,6}.png	Per-state self-transition probability, one figure per K (§8d)

A.7.2 File Manifest

All figure files below are under outputs/figures/nb07/; all data files under data/processed/nb07/ (see §12 for row counts and purpose).

File	Description
fig_nb07_01_full_vs_diag_ari.png	Full-vs-diagonal ARI scatter (160 configs)
fig_nb07_02_bic_elbow_comparison.png	BIC elbow curves, $K = 2, \ldots, 8$
fig_nb07_03_numerical_health_grid.png	Risk table + condition-number heatmap
fig_nb07_04_state_identity_2d_K4.png	Per-state 2-D KDE density, K=4
fig_nb07_04_state_identity_2d_K5.png	Per-state 2-D KDE density, K=5
fig_nb07_04_state_identity_2d_K6.png	Per-state 2-D KDE density, K=6
fig_nb07_05_state_identity_3d_K4.png	Per-state 3-D position + floor KDE, K=4
fig_nb07_05_state_identity_3d_K5.png	Per-state 3-D position + floor KDE, K=5
fig_nb07_05_state_identity_3d_K6.png	Per-state 3-D position + floor KDE, K=6
fig_nb07_06_state_persistence_K4.png	Self-transition probability by state, K=4
fig_nb07_06_state_persistence_K5.png	Self-transition probability by state, K=5
fig_nb07_06_state_persistence_K6.png	Self-transition probability by state, K=6

Generated by the LBSM research analysis pipeline (execution date 2026-08-26), full-scale run: 160 configurations, 10 seeds (42–51), 4,800 full-covariance fits, $K = 2, \ldots, 8$ BIC search, state-identity diagnostics at N=50,T=5,000,seed=42 (250,000 observations). Supersedes an earlier compute-budget-reduced pass (24 configurations, $T \leq$ 2,000, $K \leq$ 6) used to validate the mitigation stack and infrastructure. All numerical values sourced directly from cell outputs and exported CSVs of `07_final_experiment_analysis.ipynb`.